

Fire-Safety Personnel

80 practical questions for evidence-aware, privacy-preserving building support

80

QUESTIONS

8

OPERATIONAL CATEGORIES

6

SUBSTANTIVE FIELDS PER QUESTION

1 Current authority

Command, current emergency plan, signed owner decisions and competent records

2 Verified evidence

Mapped, commissioned, permitted and healthy sources with times and provenance

3 Bounded analysis

Lookup, observation, calculation, estimate, forecast or comparison labelled correctly

4 Safe failure

Unknown, qualified or local check required when evidence is stale, missing or conflicting

DECISION SUPPORT, NOT LIFE-SAFETY CONTROL

Current authoritative owners, competent professionals, the Responsible Person, the fire service and the current emergency plan always take precedence.

Never generate or expose a personal emergency evacuation plan. Never operate, reset or command a life-safety system.

Scope and operating assumptions

This catalogue frames practical questions that fire-safety personnel may ask of a future Talking Abacws service. Each record connects a decision to proposed evidence, current authoritative records, a named analytical operation, uncertainty and an explicit safe failure.

1 Proposed capability

Sensors, integrations and analytics remain proposed until deployment, asset mapping, commissioning, permission, clock alignment and current health are independently verified.

2 Authority first

Fire-service command and the current emergency plan lead during an incident. Otherwise, current signed Responsible Person decisions and competent findings outrank system records, telemetry and derived analysis.

3 Historical plan limit

The May 2021 floor plan is historical spatial reference only. It cannot establish current room use, access, services, hazards, routes, assets, closures or system configuration.

4 Privacy by design

Use aggregate, purpose-limited evidence. No identity inference, person tracking, movement profiling, speech recording, personal-plan extraction or inspection of research content is required.

5 Evidence discipline

Every source must expose spatial coverage, observed/effective/retrieved time, calibration or functional-check status, clock offset, uptime, gaps, drift, version, permissions and provenance.

6 Honest failure

When evidence is missing, stale, conflicting, unmapped or unauthorised, answer "unknown", "qualified" or "not demonstrated" and name the responsible local or owner-led check.

NO OPERATIONAL OVERRIDE

The service does not certify fire safety, approve work or impairments, operate alarms, doors, lifts, smoke control, suppression or emergency power, issue evacuation commands, or replace the Responsible Person, competent professionals, the fire service or the current emergency plan.

How to read every question card

01 Fire-safety personnel question

A natural, decision-centred question that a fire-safety professional could genuinely ask before, during or after routine operations.

02 Sensors and telemetry

Physically plausible proposed evidence with modality, asset/location coverage, timing, aggregation, measurement quality, health, gaps and limitations.

03 Authoritative supporting data

Current owners, precedence, effective/observed/retrieved times, permissions, provenance, staleness and explicit handling of conflicts.

04 Analysis required

The operation type - lookup, observation, calculation, estimate, forecast, comparison, diagnosis or recommendation - plus interval, minimum evidence, method, uncertainty and safe failure.

05 Answer boundary

What the service must not infer, disclose, promise, certify, authorise, control or present as a substitute for competent authority.

06 Why this is useful

The specific inspection, escalation, coordination, maintenance, planning or evidence action fire-safety personnel can take.

Evidence status used throughout

SUPPORTED

Required authoritative and qualified evidence is present for the narrow claim.

QUALIFIED

A partial answer is possible, but material gaps, assumptions or uncertainty are shown.

UNAVAILABLE

Evidence, permission or provenance is insufficient; a local or owner-led check is required.

Categories and evidence precedence

Eight operational categories

01	Compartmentation, doors, routes and fire-safety assets	FS-001 to FS-010
02	Alarms, detection, suppression and monitoring status	FS-011 to FS-020
03	Evacuation routes, exits, refuges and assembly points	FS-021 to FS-030
04	Occupancy, visitors and inclusive evacuation information	FS-031 to FS-040
05	Drills, training, wardens and communications	FS-041 to FS-050
06	Inspection, testing, maintenance and records	FS-051 to FS-060
07	Impairments, hot work, changes and temporary controls	FS-061 to FS-070
08	Incidents, fire-service liaison and post-event evidence	FS-071 to FS-080

Evidence precedence

1

Fire-service incident command and the current emergency plan

2

Current signed Responsible Person decisions, impairments and competent findings

3

Current authoritative asset, permit, schedule, training and maintenance systems

4

Commissioned, mapped, healthy and permitted telemetry or attended observations

5

Derived analysis with assumptions, coverage and uncertainty shown

RULE

Lower layers may corroborate higher layers. They must never overrule them. A conflict remains visible.
The May 2021 floor plan may support legacy labels only and cannot establish current conditions.

Effective time: when a rule, permit or plan applies

Observed time: when a condition was measured

Retrieved time: when the service fetched it

FS-001

FIRE-SAFETY PERSONNEL QUESTION

Which mapped fire doors should be checked next because closure, release, latching or swing-clear evidence is missing or contradictory?

SENSORS AND TELEMETRY

Proposed evidence. Door contacts on selected leaves; approved hold-open controller command/feedback; latch switches only where designed; optional overhead time-of-flight sensing confined to a marked swing or clear zone; and time-stamped attended observations. Coverage and timing. Only commissioned fire doors and explicitly monitored approach zones linked to a current door ID, side and compartment; uninstrumented doors remain unobserved. Event states within 5-10 seconds; clear-zone data reduced to a one-minute exception flag; manual reports retain source time and validity period, with no images or person traces in the answer. Quality and limits. Require current mapping, commissioning and latest permission; retain functional-check or calibration status, configuration, source-clock/UTC offset, heartbeat, uptime, maintenance windows, gaps and raw-to-summary provenance. Compare paired contacts or attended checks for stuck states or drift; never fill a live-state gap longer than two minutes. A contact does not prove latching, seals, fire resistance or unobstructed swing; time-of-flight sensing may miss low, reflective or screened objects; sensor silence is not proof of safety.

ANALYSIS REQUIRED

Operation and interval. Observation plus comparison; recommendation is limited to where a human check is warranted, with no causal diagnosis. Every 30 seconds during occupied periods and on demand before opening, events, handover or a route change. Minimum evidence and method. A current mapped door, healthy recent channel or attended observation, current impairment/defect status, owner-defined expected state and latest inspection reference. Reconcile observed and expected state, apply a short persistence window to remove normal transients, then rank exceptions by route or compartment dependency rather than by sensor confidence alone. Uncertainty and safe failure. Monitored-door coverage, clock offset, sensor health, data age, unobserved leaves, and the possibility of a hidden obstruction or failed latch. Return 'local condition not demonstrated', list the missing or conflicting evidence and identify the responsible local checker.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person or delegated fire-safety lead owns the fire-door register and expected states; Estates owns asset location; Security and maintenance owners hold current closures, defects and work orders. Live incident restrictions and signed impairment/defect decisions first; then the current approved register and competent inspection; then healthy permitted telemetry and attended reports. Times, access and provenance. Role-based fire-safety, Estates and Security access; suppress access-control events, restricted schedules and secure-route detail from wider audiences. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Two minutes for live state, the stated validity for an attended observation, and the current acceptance date for inspection evidence. Conflict. If telemetry, a current defect record and a local report disagree, show all three and require an attended check; never select the reassuring source. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer fire resistance, correct latching, seal condition, safe route availability or the cause of an exception from contact or proximity data. Do not disclose individual door-use events, identities, movement histories or security-sensitive door schedules. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

Fire-safety personnel can target a site check, remove an inappropriate hold-open or obstruction, raise a defect and amend the operational briefing without treating telemetry as acceptance evidence.

FS-002

FIRE-SAFETY PERSONNEL QUESTION

Which live or recently closed work packages need competent inspection before cover-up or handback because fire-stopping reinstatement is not evidenced at each penetration?

SENSORS AND TELEMETRY

Proposed evidence. No general-purpose sensor can verify concealed fire stopping. Proposed evidence is permit-linked QR/NFC scans at the work face, restricted close-out photographs, installer seal identifiers, ceiling/riser access events and work-package location records. Coverage and timing. Only authorised works intersecting current owner-approved compartment lines, protected shafts or risers; each penetration must have a unique location and work-package reference. Event-driven at inspection, cover-up and close-out; retain original capture times and file hashes; aggregate only to penetration-level evidence status, never worker activity. Quality and limits. Verify scanner/device clock against UTC at upload; retain capture device, uploader, file hash, editing history, access periods and missing-sequence gaps. Calibration is not applicable to photographs, but scale/reference markers and image quality checks are required. Do not infer continuity across unrecorded cover-up periods. Photographs and tags cannot reveal hidden voids, material suitability, depth, annular fill or workmanship outside the view; access events do not prove work quality.

ANALYSIS REQUIRED

Operation and interval. Lookup, spatial join and evidence-completeness comparison; recommendation only to inspect or withhold evidence close-out, not to diagnose integrity. Daily while work is live, at permit closure, before cover-up and before an affected area is returned to normal use. Minimum evidence and method. A current compartment map, geolocated permit/work order, unique penetration ID, complete installation record and competent close-out result. Intersect approved work extents with current compartment boundaries, verify required evidence fields and sequence, and flag locations with missing, stale or mismatched acceptance evidence. Uncertainty and safe failure. Drawing registration error, concealed scope, incomplete photographs, mistaken IDs, clock error and changes after inspection. Return 'reinstatement not demonstrated' and require competent physical inspection or owner-led evidence correction before relying on the close-out.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead or appointed fire engineer owns compartment information and acceptance criteria; Estates project managers own change control; the competent installer and inspector own installation and close-out evidence. A current signed competent acceptance or unresolved defect first; then approved change/permit records; then traceable close-out evidence. Telemetry may establish timing, not compliance. Times, access and provenance. Restricted to named project, fire-safety and assurance roles because images may expose secure spaces; preserve original evidence and derivative lineage. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale if the work scope, location or cover-up state changes; otherwise use the acceptance period set by the owner. Conflict. Any mismatch in penetration ID, coordinates, work scope, material record or competent sign-off remains open and is escalated; a closed work order cannot resolve it. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer fire-stopping integrity, material suitability or workmanship from a tag, photo, access event or closed permit. Do not expose worker identities, unrestricted construction images, secure service routes or named workmanship allegations beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps schedule inspection before ceilings close, handback occurs or dependent work proceeds, preventing an undocumented penetration from disappearing into the fabric.

FS-003

FIRE-SAFETY PERSONNEL QUESTION

Which compartment lines, ceiling voids, risers or shafts have the weakest current evidence after intrusive work, and where should the next competent inspection be targeted?

SENSORS AND TELEMETRY

Proposed evidence. No general-purpose sensor can establish compartment integrity. Use current compartment drawings, permit-linked work-zone records, unique penetration or opening IDs, ceiling/riser access events, restricted inspection photographs with scale references, installer records and competent inspection findings. Coverage and timing. Cover only owner-mapped compartment lines, shafts, risers and voids intersected by authorised intrusive work; mark concealed or uninspected lengths explicitly. Capture evidence before concealment, at inspection and close-out, preserving original observed times, file hashes and location accuracy. Quality and limits. Retain mapping version, registration error, device clock/UTC offset, upload gaps, editing history, permissions and source-to-answer provenance. Calibration applies only to measurement tools, not photographs. Tags, access events and absence of a work order cannot prove continuity, depth, material suitability, workmanship or hidden condition.

ANALYSIS REQUIRED

Operation and interval. Lookup, spatial join, coverage calculation and comparison; recommendation is limited to where competent inspection should be prioritised, with no integrity diagnosis. Run daily while intrusive work is active, before concealment or handback, and on demand after a change. Minimum evidence and method. Require a current compartment map, verified work extent, unique opening or penetration references, inspection status, concealment state and known uninspected coverage. Intersect work and access extents with current compartment lines, score evidence completeness and age, then rank gaps by impending concealment and dependency of protected routes or shafts. Uncertainty and safe failure. State drawing registration error, concealed scope, incomplete images, mistaken IDs, undocumented work and changes after inspection. Return "compartment evidence not demonstrated", identify the exact line or opening and require competent physical inspection or owner-led evidence correction.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person or delegated fire-safety lead owns the required assurance question; an appointed fire engineer or competent compartmentation specialist owns technical criteria and findings; Estates project owners hold change and work records. Current signed competent findings and unresolved defects outrank project status, photographs and derived coverage scores. Times, access and provenance. Use role-restricted access because records may reveal secure service spaces. Retain effective, observed and retrieved times, source ID, drawing/work-package version, signature or hash, and every spatial transformation. Evidence becomes stale when scope, concealment state, mapping or acceptance status changes. Conflict. A mismatch in line location, penetration ID, work scope, photograph, material record or competent sign-off remains open and triggers owner-led clarification or inspection. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer fire resistance, continuity, material suitability or workmanship from a map, tag, photograph, access event or apparently closed work package. Do not recommend destructive opening-up or assign workmanship blame. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

Enables fire-safety personnel to target scarce competent inspection before concealment or handback, rather than allowing weakly evidenced compartment lines to disappear into the building fabric.

FS-004

FIRE-SAFETY PERSONNEL QUESTION

In the latest authorised cause-and-effect test, which hold-open fire doors did not complete the expected release-close-latch sequence?

SENSORS AND TELEMETRY

Proposed evidence. Fire-alarm test event, hold-open controller command and feedback, local power state, door contact, latch-position feedback where designed, and a competent observer's pass/fail result. Coverage and timing. Every commissioned hold-open device included in the approved cause-and-effect test, linked to its door, release circuit, test zone and observed leaf. Capture all state changes at native event resolution; align controller and panel clocks before the test; retain a complete per-door trace plus the observer's source time. Quality and limits. Record functional-check dates, controller configuration, firmware, clock offset, heartbeat, gaps and provenance. Validate contacts against the observer before use; flag impossible state sequences, stuck values and timing drift. Do not reconstruct across any missing test event. Electrical release feedback does not prove unobstructed closure, latch engagement, seals or future performance; observer coverage may be incomplete.

ANALYSIS REQUIRED

Operation and interval. Event-sequence observation and comparison with approved cause-and-effect; fault localisation may be suggested, but causal diagnosis is not claimed. At every authorised test and after relevant maintenance or configuration change; summary on demand. Minimum evidence and method. Authorised test ID, approved sequence, mapped door/device, aligned clocks, complete command and endpoint trace, and competent observation. Reconstruct command-release-close-latch order, compare each transition and duration with owner-set tolerances, and distinguish no-command, no-release, no-movement and no-latch evidence gaps. Uncertainty and safe failure. Clock offset, contact reliability, observer coverage, transient obstruction and unobserved mechanical condition. Return 'sequence not demonstrated', preserve the trace and require competent retest or physical examination.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The life-safety system owner owns cause-and-effect and test scope; the fire-safety lead authorises the test; the competent maintainer and observer own endpoint results and defects. Signed test result and declared impairment first; then panel/controller logs; then healthy mapped telemetry. A local competent failure overrides a normal command log. Times, access and provenance. Test traces and detailed system logic are restricted to authorised fire-safety, Estates and specialist maintenance roles. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Valid only for the tested configuration and until a subsequent fault, change or impairment; live states stale after two minutes. Conflict. Show command, feedback and observation separately; any missing endpoint or disagreement is an incomplete/failed demonstration, not a pass. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer door integrity or future reliability from a successful cycle, and do not reset, release or test any door. Do not disclose detailed cause-and-effect logic or secure door locations to unauthorised users. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the maintainer a precise failed step and lets the fire-safety lead record an impairment or arrange local controls without treating controller output as a competent pass.

FS-005

FIRE-SAFETY PERSONNEL QUESTION

Which fire shutters or curtains need specialist attendance because command, travel, endpoint or witnessed-test evidence is incomplete or abnormal?

SENSORS AND TELEMETRY

Proposed evidence. Upper/lower limit switches, controller state, approved motor-current telemetry, obstruction/safety-edge input, local power status and witnessed functional-test outcome. Coverage and timing. Only registered shutters and curtains with commissioned interfaces, mapped to the opening or compartment they protect; unmonitored barriers remain inspection-only. State changes within seconds, daily heartbeat and full native-resolution trace during an authorised cycle; aggregate to one exception per asset while retaining raw test events. Quality and limits. Retain limit-switch functional checks, current-sensor calibration where used, configuration, source clock/UTC offset, uptime, gaps and maintenance history. Compare travel time and current signature with accepted baselines after servicing; flag drift, but never infer across a missing cycle. Limit switches may be misadjusted and motor current is indirect; completed travel does not show fabric, guides, seals, obstructions or hidden damage.

ANALYSIS REQUIRED

Operation and interval. State-machine observation, comparison and limited fault localisation; no causal diagnosis and no automatic recommendation to restore service. On each commanded movement, after every authorised test and in a daily exception summary. Minimum evidence and method. Mapped asset, expected resting state, command, terminal limits or approved equivalent, controller health and competent witnessed result. Validate order, completeness, duration and terminal state; compare current signature only with the same asset and configuration; separate command, motion and endpoint evidence. Uncertainty and safe failure. Sensor drift, missing events, mechanical condition outside the sensors, clock alignment and baseline changes after maintenance. Return 'position/readiness not demonstrated' and require specialist attendance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates or the life-safety system owner owns the register and expected state; the specialist maintainer owns test results, defects and accepted baselines; the fire-safety lead owns impairment status. Current impairment and signed competent test first; then controller logs and healthy telemetry. A physical defect report overrides a normal controller state. Times, access and provenance. Role-restricted because detailed barrier locations and control logic may be security-sensitive. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Two minutes for live position, 24 hours for heartbeat, and only until the next fault or configuration change for a test result. Conflict. Retain all conflicting states; classify as unverified and escalate to the system owner rather than choosing the controller or telemetry result. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer barrier integrity, smoke leakage, safe passage beneath it or the cause of abnormal movement; do not move, test, bypass or reset the asset. Do not disclose raw control logic, secure openings or fault histories beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps isolate whether evidence is missing at command, movement or endpoint and gives the specialist provider a usable trace for a safe attended investigation.

FS-006

FIRE-SAFETY PERSONNEL QUESTION

Which fire or smoke dampers lack current asset-specific evidence of location, test, position and accepted remedial close-out?

SENSORS AND TELEMETRY

Proposed evidence. Actuator position feedback and limit switches for motorised dampers; controller faults; test-mode pressure/airflow readings only where part of an approved method; tagged inspection and witnessed results for passive dampers. Coverage and timing. Only dampers in the current register, each linked to duct, wall/floor penetration, served zone and access point; passive or uninstrumented assets rely on competent inspection records. Continuous heartbeat for monitored actuators, event trace during authorised tests and event-driven inspection updates; no general inference between tests. Quality and limits. Retain actuator functional checks, pressure-sensor calibration where applicable, configuration, clock/UTC offset, uptime, gaps and test provenance. Compare feedback with witnessed position; flag drift or repeated non-terminal states. Do not fill any test gap. Actuator feedback does not prove blade, seal or penetration condition; duct pressure is indirect and can be affected by fan state, leakage and other dampers.

ANALYSIS REQUIRED

Operation and interval. Lookup plus evidence-completeness comparison; event-sequence observation for motorised tests; no diagnosis of damper integrity from indirect telemetry. Daily evidence summary, at each test, and before handback of affected works or systems. Minimum evidence and method. A current asset/location record, applicable test requirement, competent result, current defect/impairment status and complete telemetry if a live state is claimed. Join each damper to its latest accepted test and remedial actions, validate test scope and asset identity, and separate monitored position from physical inspection evidence. Uncertainty and safe failure. Hidden blade/seal condition, inaccessible assets, mapping errors, incomplete access, clock gaps and changes after testing. Return 'damper condition not demonstrated' and identify the required competent inspection or record correction.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The life-safety/HVAC system owner owns the register and cause-and-effect; the specialist contractor owns competent test and remediation records; the fire-safety lead owns acceptance and impairment status. Current competent test, defect and impairment decision first; then controller evidence; then permitted telemetry. A normal actuator signal cannot close an open defect. Times, access and provenance. Access is role-restricted because damper locations expose service routes and system topology. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live actuator state stale after two minutes; test evidence valid only for the tested configuration and owner-defined period. Conflict. Mismatch in asset ID, location, test result, actuator state or close-out remains unresolved until the owner accepts competent evidence. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer fire/smoke performance, compartment integrity or safe restoration from actuator or pressure data; do not command or cycle dampers. Do not expose service-route geometry or contractor details beyond operational need. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This lets fire-safety staff prioritise inaccessible, overdue or mismatched dampers and stop a close-out claim from relying on controller feedback alone.

FS-007

FIRE-SAFETY PERSONNEL QUESTION

Which extinguishers, hose reels or fire blankets need an attended check because presence, access, service status or physical condition is not demonstrated?

SENSORS AND TELEMETRY

Proposed evidence. Time-stamped inspection scans of asset IDs; optional cabinet door/tamper state; approved presence or weight/pressure telemetry only where supplied by the manufacturer; and attended defect reports. Coverage and timing. Only assets in the current register, mapped to a current location and inspection route; loose equipment without a reliable fixed-location sensor remains scan-and-inspect only. Inspection events at source time; optional cabinet states on change with daily heartbeat; aggregate to asset-level exception and route-level completion. Quality and limits. Retain scan device clock/UTC offset, inspector/source, asset tag history, optional sensor calibration/functional check, uptime, gaps and provenance. Detect duplicate/moved tags, flat-line cabinet states and location drift; never infer condition between inspections. Presence, weight or pressure does not prove suitability, serviceability, pin/seal condition, access, signage or correct mounting; a scan can be made without a full inspection.

ANALYSIS REQUIRED

Operation and interval. Lookup, observation and comparison with the current register and service schedule; recommendation only to inspect, relocate through authority or raise service. Daily exception list and before events, area handback or high-risk work. Minimum evidence and method. A current register entry, mapped location, latest competent inspection/service, current defect/work-order status and any telemetry health required for a live claim. Compare expected and observed asset/location/status, identify overdue or missing evidence, and avoid inferring condition from presence alone. Uncertainty and safe failure. Tag loss, asset swap, unrecorded relocation, missed inspection steps, sensor drift and unobserved access obstruction. Return 'equipment provision/condition not demonstrated' and require an attended inspection.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead or equipment owner owns the register and local provision; the competent service provider owns service records; Estates/Cleaning/Security hold relocation, access and obstruction reports. Current competent service/inspection and declared defect first; then register and work orders; then optional healthy telemetry and scans. Times, access and provenance. Asset-level status is role-based; photographs, named inspection notes and secure locations remain restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Optional live cabinet state stale after 24 hours; inspection/service evidence stale at the owner-set due date or immediately after relocation/damage. Conflict. A missing or contradictory asset ID, location or service result is treated as not verified; telemetry cannot overrule a competent defect. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not judge extinguisher type suitability, firefighting strategy, serviceability or legal sufficiency from a scan or presence signal. Do not expose secure asset locations or named inspection findings to unauthorised users. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This directs a targeted equipment round, service request or location correction before an asset is assumed available in an emergency plan.

FS-008

FIRE-SAFETY PERSONNEL QUESTION

Which protected riser, shaft or service-cupboard doors need action because they are open, unsecured, defective or affected by current work?

SENSORS AND TELEMETRY

Proposed evidence. Door contacts, approved lock/bolt status where available, local power or tamper state, permit-linked access events and attended observations; no continuous video. Coverage and timing. Only registered protected riser, shaft and service-cupboard doors mapped to current compartments and authorised work zones; unmonitored doors are explicit gaps. State changes within 10 seconds; access/permit events at source time; one-minute persistent-open summary and daily unverified-door list. Quality and limits. Retain contact/lock functional checks, configuration, source-clock/UTC offset, uptime, maintenance windows, gaps and source lineage. Compare expected closed state with attended checks; flag stuck contacts and repeated clock reversal. Do not fill gaps over two minutes. A closed or locked signal does not prove latching, fire rating, seal condition, internal housekeeping or authorised work completion.

ANALYSIS REQUIRED

Operation and interval. Observation plus comparison with expected state and active-work context; no identity or cause diagnosis. Every minute during active work, daily otherwise, and at permit suspension or close-out. Minimum evidence and method. Current mapped door, expected state, recent healthy signal or attended observation, active permit/defect/impairment status and responsible work owner. Correlate persistent-open or unsecured states with authorised work windows, but report unexplained overlap without inferring who opened the door. Uncertainty and safe failure. Unmonitored leaves, contact misalignment, clock offset, temporary access, hidden internal condition and stale permit status. Return 'protected opening not verified' and request the work owner or local checker to attend.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead owns protected-opening requirements; Estates owns doors and service spaces; Security owns authorised access records; project/maintenance owners hold permits, defects and handback. Live incident restrictions and active impairments first; then current permit/closure and competent findings; then healthy permitted telemetry. Times, access and provenance. Access-event detail and secure service locations are restricted to named operational roles. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Two minutes for live state; access context stale when the permit or work period changes; inspection evidence follows owner validity. Conflict. An active permit or physical defect overrides a normal closed signal; unexplained opening requires local verification, not attribution. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer unauthorised behaviour, fire-door performance or compartment integrity from access and contact data. Do not disclose individual access events, identities, secure service routes or restricted lock states beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This enables rapid coordination with live works, correction of an open protected door and accurate handover of an unresolved opening.

FS-009

FIRE-SAFETY PERSONNEL QUESTION

Where do fire-safety signs, plans or asset labels conflict with the current approved route, asset register or emergency information?

SENSORS AND TELEMETRY

Proposed evidence. Time-stamped mobile inspection scans or restricted photographs of signs and labels; QR/NFC asset IDs; digital-plan and signage-register versions. Continuous cameras are neither required nor appropriate. Coverage and timing. Only inspected locations and registered signs/labels linked to current route, asset or action-plan records; unseen or inaccessible signs remain unknown. Event-driven at installation, change and inspection; retain original capture time and version; aggregate to mismatch type and location, not image analytics alone. Quality and limits. Verify capture-device clock/UTC offset, image hash, sign/label ID, checker identity and source-version provenance. Calibration is not applicable, but image resolution, glare and OCR confidence must be recorded. Flag duplicate IDs and stale templates; never infer text not legible in the source. OCR may misread text; a photograph does not show visibility under all lighting or viewing conditions, physical mounting, accessibility or user understanding.

ANALYSIS REQUIRED

Operation and interval. Lookup, text comparison and spatial consistency check; recommendation is limited to inspect, replace or reconcile through the owner. After each approved change and in scheduled inspection summaries. Minimum evidence and method. A current approved source document, sign/label ID and location, legible captured evidence, assessment date and current change record. Compare exact controlled wording, symbol/version, destination and location against the approved source; flag uncertain OCR for human assessment rather than auto-correcting. Uncertainty and safe failure. Capture quality, OCR confidence, hidden signs, unrecorded relocation and current-plan uncertainty. Return 'sign/label agreement not demonstrated' and require attended verification.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead owns emergency information content; Estates/Space Management owns current locations and layouts; the asset/system owner owns identifiers; Communications may own approved public wording. Current approved plan/instruction and signed change decision first; then signage/asset registers and competent inspection; OCR or scan output is supporting evidence only. Times, access and provenance. Detailed emergency plans, secure asset locations and raw images are role-restricted; public output uses only approved wording. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale after a route, asset, naming or emergency-plan change; otherwise by the inspection schedule. Conflict. Any text, version, location or identifier mismatch remains open for owner decision; analytics cannot choose which document is authoritative. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not publish or rewrite emergency instructions, infer compliance from OCR, or declare a route correct because signs agree with each other. Do not expose secure drawings, equipment locations or unapproved emergency wording. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports a controlled replacement list and prevents staff, occupants or responders receiving contradictory route or asset information.

FS-010

FIRE-SAFETY PERSONNEL QUESTION

Which approved changes since the accepted baseline require a compartment, protected-route or fire-safety asset check before the changed area is relied upon?

SENSORS AND TELEMETRY

Proposed evidence. No sensor can establish the full effect of building change. Proposed evidence is change-control records, permits, work orders, approved drawings/models, asset-register deltas, commissioning results and restricted site observations. Coverage and timing. All approved, emergency and discovered changes intersecting current compartments, protected routes, fire-safety assets or emergency information; include temporary changes and rollback status. Event-driven at approval, start, inspection, commissioning and handback; retain each version and effective interval; aggregate to unresolved impact by zone and dependency. Quality and limits. Require signed/versioned files, stable asset/location IDs, uploader/source, clock/UTC offset, file hashes, supersession links and transformation lineage. Calibration applies only to any supporting measurements. Flag unversioned drawings, orphaned assets, overlapping effective dates and missing handback gaps. Records may omit informal or emergency changes; model geometry may be inaccurate; a work-order closure does not prove physical implementation or fire-safety acceptance.

ANALYSIS REQUIRED

Operation and interval. Lookup, version comparison, spatial join and evidence-gap recommendation; no causal diagnosis of physical condition. Daily for live works, at every approval/handback and before fire-risk reassessment or major event. Minimum evidence and method. Accepted baseline, signed change record, current geometry/asset IDs, commissioning or inspection evidence, effective dates and unresolved defects/impairments. Diff versions, intersect change extents with fire-safety dependencies, verify required owner acceptances and classify each impact as confirmed, possible or unknown. Uncertainty and safe failure. Unrecorded work, drawing registration, temporary states, incomplete commissioning, stale asset mapping and conflicting effective dates. Return 'change impact not established', list affected dependencies and require competent assessment or site verification.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The change owner and Estates project manager own scope and implementation; the fire-safety lead or competent fire engineer owns fire-safety impact and acceptance; asset/system owners own commissioning and register updates. Current emergency restrictions and signed competent findings first; then approved change/commissioning records; then asset-register and telemetry corroboration. Historical plans do not establish current condition. Times, access and provenance. Role-restricted project, drawing and secure-system access; outputs expose only the impact needed for the decision. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale when scope, sequence, temporary state or handback changes; baseline valid only when explicitly accepted. Conflict. Conflicting drawings, registers, physical observations or dates are retained as an unresolved configuration issue and escalated to the owners. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not treat a design approval, work-order closure, digital model or sensor agreement as proof that the physical fire-safety outcome is acceptable. Do not expose restricted drawings, contractor identities or secure asset topology outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the fire-safety lead a bounded change list for inspection, risk reassessment, register update and briefing before the accepted baseline is relied upon.

FS-011

FIRE-SAFETY PERSONNEL QUESTION

What alarm, fault, disablement, power and communications state is currently evidenced at each fire-alarm panel, repeater and monitored zone?

SENSORS AND TELEMETRY

Proposed evidence. Read-only fire-alarm panel/repeater event export, loop and power-supply status, network/communication heartbeat and an attended panel observation; access must use a commissioned, manufacturer-supported interface. Coverage and timing. Only panels, repeaters and zones in the current system register, with explicit indication of interfaces or panels that are not integrated. Native event timestamps with read-only polling no faster than the approved interface permits; present current state and a 24-hour event summary without suppressing repeated faults. Quality and limits. Record interface commissioning, firmware/configuration, panel/source clock and UTC offset, polling success, heartbeat, uptime, gaps and raw event provenance. Compare panel and gateway sequence numbers for dropped events; detect clock jumps and repeated stale states. Never infer state during an outage. A read-only export can lag, omit local indications or use vendor-specific codes; a normal panel does not prove detector, sounder, interface or building safety.

ANALYSIS REQUIRED

Operation and interval. Lookup and observation of reported states, followed by comparison across panel, integration and impairment records; no fault diagnosis from codes alone. Every 30 seconds where approved, on demand at handover, and immediately after a reported alarm, fault, test or disablement. Minimum evidence and method. Current panel identity/configuration, healthy read-only link, aligned clock, complete recent event sequence, current impairment record and an attended panel source when material. Normalise vendor codes without changing meaning, reconcile active and restored events, and report alarm, fault, disablement and communication states with source and age. Uncertainty and safe failure. Vendor-code mapping, event loss, clock offset, local indications not exported, stale integrations and unregistered configuration change. Return 'panel/zone state not established', name the missing source and direct the user to the authorised panel observer or system owner.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-alarm system owner and Responsible Person own operational status and impairment decisions; the specialist maintainer owns diagnostic interpretation; Security/control-room staff own attended observations. Live fire-service command and current panel indication first; then signed impairment/maintenance records; then complete read-only export. A local panel observation overrides a stale integration. Times, access and provenance. Detailed zone, device and cause-and-effect data is restricted to authorised fire-safety, Estates, Security and maintainer roles. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Thirty seconds for current state or the approved polling interval, whichever is slower; immediately stale after a communication loss or panel change. Conflict. Show panel, integration and attended observations separately. Any missed poll, unknown code or disagreement is 'state not established' and requires local panel confirmation. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not interpret a normal state as certification, suppress or acknowledge an event, or advise occupants how to act during an alarm. Do not expose sensitive zone maps, device addresses, access credentials or named operator actions outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This provides a traceable handover view and identifies exactly which panel or zone needs local confirmation instead of relying on a green dashboard.

FS-012

FIRE-SAFETY PERSONNEL QUESTION

Which detectors, call points, loops or network segments need competent follow-up because faults, isolations, lost communication or abnormal diagnostics affect coverage?

SENSORS AND TELEMETRY

Proposed evidence. Read-only device/loop status, fault and isolation events, approved analogue-value or contamination indicators, network-node heartbeat, power status and attended test results. Coverage and timing. Only registered devices and loops supported by the approved interface; each record must map to current device ID, zone, loop/address and physical location. Native event reporting; diagnostic values at the manufacturer's supported interval, not arbitrary high-rate polling; daily exception aggregation with raw values retained for authorised assessment. Quality and limits. Retain device commissioning/test date, diagnostic reference range, configuration, firmware, source clock/UTC offset, heartbeat, uptime, gaps and provenance. Compare each device only with approved limits and its own baseline; flag drift or flat-line behaviour, but never interpolate through a communication gap. Contamination or analogue values are vendor-specific and do not establish sensitivity, siting, physical condition or fire presence; network silence may be an interface fault rather than device failure.

ANALYSIS REQUIRED

Operation and interval. Observation and comparison with approved device/loop criteria; recommendation to investigate; causal diagnosis is explicitly not claimed. On every event, every five minutes for outstanding communication exceptions, and in daily maintenance triage. Minimum evidence and method. Current device register, complete recent events, healthy panel/interface link, current isolation/defect record, approved diagnostic interpretation and latest competent test where available. Group exceptions by common loop/network dependency, distinguish active fault, isolation, missing communication and abnormal diagnostic, and rank by affected coverage rather than raw count. Uncertainty and safe failure. Mapping errors, vendor thresholds, clock alignment, shared-loop effects, missing events, sensor drift and unobserved physical obstruction/contamination. Return 'device/coverage state not demonstrated' and require competent panel and field investigation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-alarm system owner owns device inventory and isolations; the specialist maintainer owns diagnostic meaning and competent test; the fire-safety lead owns operational impairment and follow-up priority. Current alarm/incident command and signed isolation/defect records first; then competent test; then complete diagnostic telemetry. Times, access and provenance. Device-level topology and addresses are role-restricted; public output is limited to approved operational messages. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live fault/isolation state stale after one polling cycle plus 30 seconds; diagnostics stale at the owner-set interval or after maintenance/configuration change. Conflict. Do not merge conflicting IDs or silently clear an exception. A competent failed test or active isolation remains open even if diagnostics appear normal. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer detector sensitivity, fire risk, exact cause of a fault or safe coverage from analogue values or heartbeat. Do not disclose device addresses, restricted locations or named operator actions without permission. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This lets maintenance investigate the likely common dependency first and lets fire-safety staff understand the affected coverage without claiming a diagnosis.

FS-013

FIRE-SAFETY PERSONNEL QUESTION

Which detectors need attended siting or contamination checks because temporary works, dust, heat, airflow or partitions have changed their surroundings?

SENSORS AND TELEMETRY

Proposed evidence. Work permits, temporary-partition and equipment records, local inspection reports, and - only in approved works zones - dust/particle, temperature or humidity sensors positioned near the affected detector, not inside it. Coverage and timing. Only detectors within the current mapped extent of live works, temporary layouts or known contamination sources; no building-wide inference from sparse environmental sensing. Environmental samples at 1-5 minute intervals, aggregated to hourly range and threshold duration; work changes and inspections are event-driven with effective times. Quality and limits. Retain environmental-sensor calibration, mounting location/height, source clock/UTC offset, uptime, gaps and provenance. Compare with a co-located reference or pre-work baseline for drift; flag sensor relocation. Do not bridge gaps over 15 minutes or extrapolate beyond the works zone. Dust/temperature/humidity near a detector does not measure detector sensitivity, obstruction, airflow, contamination inside the chamber or compliance of siting.

ANALYSIS REQUIRED

Operation and interval. Spatial lookup, observation and comparison; recommendation is to obtain competent siting/contamination assessment, not to diagnose detector performance. Hourly during relevant works, at each layout/process change, and before removing temporary controls. Minimum evidence and method. A current detector/location record, live work extent, intended environment/owner criteria, recent healthy local evidence or attended observation, and current impairment status. Intersect works and temporary changes with detector locations, compare observed environmental range/duration with owner criteria, and classify only the need for assessment. Uncertainty and safe failure. Sensor placement, sparse coverage, airflow, localised dust, baseline drift, unrecorded work and clock/data gaps. Return 'detector environment not established' and request an attended check; do not imply normal operation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-alarm system owner and competent designer/maintainer own detector siting and environmental limits; project/permit owners own temporary conditions; the fire-safety lead owns operational assessment. Current impairment, permit conditions and competent inspection first; then approved design information; then healthy local environmental evidence. Times, access and provenance. Works, hazard and secure-location data is restricted to authorised project and fire-safety roles; no worker tracking. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Environmental evidence stale after 15 minutes during active work and immediately after layout/process change; inspection valid only for the observed state. Conflict. Any permit or attended report indicating adverse conditions overrides apparently normal local sensing. Conflicts trigger a physical check. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer detector sensitivity, contamination level, compliance, fire probability or the need to isolate a detector from environmental sensors alone. Do not expose worker identity, work content beyond need, or raw environmental traces tied to individuals. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This focuses competent checks on detectors affected by dust, partitions, heat or airflow changes before temporary conditions are allowed to continue.

FS-014

FIRE-SAFETY PERSONNEL QUESTION

Which sounder, voice-alarm or visual-alarm areas lack current evidence of circuit health and effective sampled notification?

SENSORS AND TELEMETRY

Proposed evidence. Panel/circuit supervision, amplifier and battery status, speaker/strobe loop events, and competent sampled tests using calibrated sound-level equipment, visual checks or approved intelligibility methods; no continuous microphone recording. Coverage and timing. Every registered notification circuit plus the exact sampled spaces and test positions; unsampled spaces remain unverified and must not inherit a nearby result automatically. Continuous supervised-state events; sampled measurements only during authorised tests with source time, position, operating condition and test signal; aggregate by circuit and sampled area. Quality and limits. Retain meter calibration certificate, microphone/photometer setup, test position, panel and recorder clock/UTC offset, configuration, heartbeat, gaps and provenance. Recheck after room, ceiling, furnishing or system change; never use an expired calibration or interpolate between sample points. Circuit health does not prove audibility, intelligibility or visibility; sampled measurements vary with occupancy, noise, doors, finishes and position; visual alarms may not serve every need.

ANALYSIS REQUIRED

Operation and interval. Observation and comparison with approved test criteria; calculation of sampled coverage; no inference to untested locations and no diagnosis of user comprehension. On every fault and after each authorised test; coverage assessment after room or system changes. Minimum evidence and method. Current circuit map, healthy supervised state, calibrated test equipment, documented sample positions/conditions, accepted result and current room configuration. Separate electrical availability from sampled acoustic/visual effectiveness, calculate which registered areas have valid evidence, and list unsampled or changed areas. Uncertainty and safe failure. Calibration, background noise, position, door state, test-signal difference, clock alignment and unobserved user needs. Return 'notification effectiveness not demonstrated for this area' and require competent sampling or inspection.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The alarm/voice-alarm system owner owns circuits and design; the competent test provider owns measurements and interpretation; Space/Estates owns room changes; the fire-safety lead owns operational response. Current impairment and signed competent test first; then supervised circuit state; then historic samples as qualified evidence only. Times, access and provenance. Detailed message content, secure zone mapping and raw recordings are restricted; tests should avoid recording speech or occupants. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live circuit state stale after one polling cycle; sampled evidence stale after relevant environmental/layout change or owner-defined test period. Conflict. A failed sample or competent report remains open despite normal supervision. Conflicting samples require repeat testing at defined positions. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not promise that any person will hear, understand or see an alarm, and do not generate or transmit test/emergency messages. Do not retain speech recordings, identify occupants or disclose restricted message/circuit design beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports a defensible test plan and prevents a healthy circuit from being misreported as effective notification in every space.

FS-015

FIRE-SAFETY PERSONNEL QUESTION

Which sprinkler or water-suppression supplies, valves, pumps, tanks or monitored pressures show an exception requiring specialist confirmation?

SENSORS AND TELEMETRY

Proposed evidence. Supervised valve/tamper switches, calibrated pressure transmitters at approved points, pump run/fault and power status, tank level where installed, heater/temperature status where relevant, and competent inspection/test records. Coverage and timing. Only assets in the current suppression register and hydraulic schematic, mapped to the supply, zone and protected area; unmonitored valves and gauges remain inspection-only. State changes within seconds; pressure/level at the approved sampling interval, typically 1-5 minutes; retain raw trends around tests/events and aggregate to current exception plus 24-hour range. Quality and limits. Retain instrument calibration, range, tap location, controller configuration, source clock/UTC offset, uptime, gaps, maintenance and provenance. Compare redundant gauges or test references for drift; flag frozen values and impossible transitions. Do not bridge gaps over 10 minutes or during a pump/test event. Pressure and level are location-specific and influenced by demand, testing and supply conditions; normal readings do not prove valve alignment, discharge capability, pipework integrity or design adequacy.

ANALYSIS REQUIRED

Operation and interval. Observation and comparison with owner-approved operating bands; recommendation to check; no hydraulic performance calculation or fault diagnosis unless separately authorised and engineered. Every minute for live exceptions, after test/pump events and in daily handover. Minimum evidence and method. Current asset/schematic mapping, healthy calibrated channel, current impairment/test context, expected operating state/band and recent local or controller evidence. Validate state and trend at each approved point, distinguish planned test/run from unexplained exception, and rank by the protected scope affected. Uncertainty and safe failure. Calibration drift, tap blockage, local demand, clock gaps, missing unmonitored valves and configuration changes. Return 'suppression supply/state not demonstrated' and direct an authorised local or specialist check.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The suppression system owner and specialist contractor own schematics, limits and test interpretation; Estates/utilities own supplies and power; the fire-safety lead owns impairment status. Fire-service command and signed impairment/test findings first; then local controller/gauge observations; then healthy telemetry. A closed or unverified valve report overrides normal pressure. Times, access and provenance. Hydraulic topology, valve locations and detailed vulnerabilities are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Two minutes for discrete valve/pump state; 10 minutes for pressure/level unless the owner specifies tighter; immediately stale during unrepresented testing or supply change. Conflict. Show each source and location. Any disagreement with a local gauge, impairment or competent finding requires local confirmation; never average away a conflict. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer discharge performance, design sufficiency, valve alignment or system availability from pressure/level alone, and do not start/stop pumps or move valves. Do not expose detailed valve topology or security-sensitive supply information beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This lets fire-safety staff escalate the affected zone and evidence gap quickly while giving the specialist the exact point, trend and test context to check.

FS-016

FIRE-SAFETY PERSONNEL QUESTION

Which specialist suppression system or protected enclosure lacks current evidence of authorised availability, agent condition, release readiness or enclosure integrity?

SENSORS AND TELEMETRY

Proposed evidence. Suppression panel state, cylinder pressure or manufacturer-approved weight/level telemetry, release-circuit supervision, room-door/vent position only where designed, environmental limits and current enclosure-integrity/test records. Coverage and timing. Only registered systems and protected enclosures, mapped to hazards, zones and interfaces; no inference for systems without manufacturer-supported monitoring. Discrete states on change; pressure/weight/environment at manufacturer-approved intervals; test/inspection and integrity results event-driven; aggregate to evidence status, not a single 'ready' score. Quality and limits. Retain instrument calibration/functional checks, cylinder/reference details, configuration, source clock/UTC offset, uptime, gaps, maintenance and provenance. Apply temperature compensation only from the approved method; compare with competent service readings for drift. Never interpolate across maintenance or disconnection. Pressure or weight does not prove agent quantity, distribution, pipework, nozzle condition, release logic or enclosure hold time; door/vent contacts do not prove enclosure integrity.

ANALYSIS REQUIRED

Operation and interval. Lookup, observation and evidence-completeness comparison; no estimate of extinguishing effectiveness and no diagnosis of agent loss cause. On every state change, daily evidence assessment and before/after work affecting the enclosure. Minimum evidence and method. Current system/enclosure record, current use/hazard, valid specialist service and integrity evidence, complete panel state, calibrated agent evidence if used, and impairment status. Separate authorisation, agent condition, release supervision and enclosure evidence; classify each as supported, qualified or unavailable without collapsing them into readiness. Uncertainty and safe failure. Temperature compensation, sensor drift, inaccessible cylinders, unobserved envelope changes, clock gaps and configuration mismatch. Return 'specialist-system availability not demonstrated' and require specialist/owner confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The specialist system owner and competent service provider own design, agent criteria, integrity testing and release logic; the hazard/process owner owns enclosure use; the fire-safety lead owns impairment coordination. Current signed impairment, specialist service result and enclosure-integrity record first; then panel/local indicators; then healthy telemetry. Times, access and provenance. Hazard, agent, enclosure and release details are tightly role-restricted; no public disclosure of vulnerabilities. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live panel state stale after one polling cycle; pressure/environment after the owner-set interval; integrity evidence stale after any envelope or ventilation change. Conflict. Any active work, open defect or failed integrity/service result overrides a normal panel or pressure reading. Conflicts remain unavailable until specialist resolution. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not declare the enclosure protected, estimate suppression success, arm/disarm/reset the system or authorise occupancy/work. Do not disclose hazard, agent quantity, release logic or secure enclosure details beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This prevents a normal panel or pressure value from masking missing integrity, service or authorisation evidence and supports a precise specialist call-out.

FS-017

FIRE-SAFETY PERSONNEL QUESTION

Which smoke-control fans, dampers, vents or automatic opening vents lack a complete, witnessed response sequence for the scenario being tested?

SENSORS AND TELEMETRY

Proposed evidence. Controller states, fan run/fault and VFD feedback, motor current, damper/vent end switches, approved pressure-difference sensors at specified points, power supplies and witnessed cause-and-effect tests. Coverage and timing. Only registered smoke-control components and monitored pressure points linked to current zones, scenarios and cause-and-effect; uninstrumented flow paths remain outside the answer. Discrete states on change; pressure at 1-5 seconds during authorised tests and slower approved intervals otherwise; retain full event trace for each test and aggregate to component/sequence exceptions. Quality and limits. Retain sensor calibration and tap locations, end-switch checks, controller configuration, source clock/UTC offset, uptime, gaps and provenance. Compare pressure sensors with calibrated references and test baselines for drift; flag configuration changes. Never reconstruct missing transitions. Run feedback, current and pressure are indirect; they do not prove airflow distribution, smoke containment, door forces, leakage or performance under fire conditions.

ANALYSIS REQUIRED

Operation and interval. Event-sequence observation and comparison; limited localisation of missing command/feedback; no diagnosis of smoke performance. At each authorised test, on every fault/state change, and in daily exception assessment. Minimum evidence and method. Approved scenario and cause-and-effect, mapped components, aligned clocks, complete command/feedback trace, calibrated measurements where required, witnessed result and current impairment status. Reconstruct scenario steps, verify required endpoints and timing, compare measured values only at approved points, and separate electrical, mechanical and performance evidence. Uncertainty and safe failure. Clock offset, sensor/tap drift, variable door/vent conditions, hidden mechanical defects, incomplete scenario coverage and gaps. Return 'smoke-control response not demonstrated' and require specialist testing or inspection.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The smoke-control system owner and competent specialist own design scenarios, thresholds and tests; Estates owns power/HVAC dependencies; the fire-safety lead owns operational impairment. Fire-service command and signed impairment/test result first; then controller/local indication; then complete healthy telemetry. Times, access and provenance. Control logic, pressure points and secure service topology are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live state stale after two minutes; test evidence valid only for the tested configuration, environmental condition and owner-defined period. Conflict. A witnessed failure or active impairment overrides normal telemetry. Any sequence or clock mismatch is incomplete evidence requiring competent assessment. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer smoke clearance, tenability, safe route conditions or future fire performance; do not command fans, dampers or vents. Do not expose cause-and-effect, secure plant locations or raw fault topology to unauthorised users. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the specialist a precise missing step or measurement and helps the fire-safety lead define the affected scenario without overstating performance.

FS-018

FIRE-SAFETY PERSONNEL QUESTION

Which fire-alarm interfaces to lifts, access control, ventilation, plant or door releases lacked destination-system confirmation in the latest authorised test?

SENSORS AND TELEMETRY

Proposed evidence. Read-only fire-alarm interface events plus destination-system endpoint state: lift controller/test result, access release relay/door state, HVAC shutdown feedback, plant interlock status and competent observer confirmation. Coverage and timing. Only interfaces in the current approved cause-and-effect matrix, mapped end-to-end from source event to destination endpoint; intermediate-only evidence is not treated as completion. Native event resolution during authorised tests, with pre-test clock synchronisation or measured offsets; retain complete trace and observer time for each endpoint. Quality and limits. Retain interface commissioning, configuration/version, functional checks, source clocks/UTC offsets, uptime, gaps and cross-system provenance. Measure and store clock offset before the test; flag dropped sequence numbers or timing drift. Do not infer an endpoint state from an intermediate relay if endpoint evidence is absent. A command or relay state does not prove the destination action, physical outcome or future reliability; systems may apply normal delays and safety interlocks.

ANALYSIS REQUIRED

Operation and interval. Cross-system event-sequence comparison; recommendation to retest or inspect; no causal diagnosis beyond the evidenced missing stage. At every integrated test, after relevant change and on demand for unresolved interfaces. Minimum evidence and method. Approved cause-and-effect, mapped source and endpoint IDs, measured clock offsets, complete logs, destination observation and current impairment/change status. Build a source-command-destination sequence, adjust only by documented clock offsets, verify endpoint evidence and classify source, transmission and endpoint stages separately. Uncertainty and safe failure. Clock uncertainty, vendor logging resolution, normal endpoint delay, missing intermediate logs, manual observation timing and configuration change. Return 'interface endpoint not demonstrated' and require the relevant system owners to confirm or repeat the test.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-alarm system owner owns the source cause-and-effect; each destination-system owner owns endpoint evidence; the competent test coordinator owns the witnessed integrated result; the fire-safety lead owns impairment action. Current impairment and signed integrated test result first; then destination-system logs/observation; then source-panel command. Source command alone has lowest evidential value for endpoint completion. Times, access and provenance. Cross-system logs and access/lift topology are restricted to named test, fire-safety and system-owner roles. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Valid only for the tested configuration and until any interface, software, network or endpoint change; live endpoint states stale after two minutes. Conflict. Do not reconcile conflicts by timestamp alone. Missing endpoint confirmation or unexplained sequence order remains an incomplete test requiring owner-led retest. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer safe lift, door, HVAC or plant behaviour from a fire-panel output, and do not trigger or reset interfaces. Do not disclose access-control events, lift topology or control logic outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This isolates exactly which destination owner must provide evidence and avoids closing an integrated test on the strength of a source relay alone.

FS-019

FIRE-SAFETY PERSONNEL QUESTION

Is the remote alarm-signalling path available now, and when was alarm receipt last demonstrated end to end?

SENSORS AND TELEMETRY

Proposed evidence. Alarm transmission unit heartbeat, primary/secondary path state, power/battery status, communication fault events, receiving-centre acknowledgement from an authorised test, and attended local panel/ATU observation. Coverage and timing. Only the commissioned transmission path(s) and receiving service in the current contract/system record; external network segments not instrumented by the service are explicit unknowns. Heartbeat and state at the provider-approved interval; event-level timestamps for faults and tests; present current path state plus last successful end-to-end receipt, not a derived availability promise. Quality and limits. Retain ATU/interface commissioning, configuration, source clocks/UTC offsets, heartbeat success, uptime, gaps, maintenance and provider provenance. Confirm clock offsets at test; detect prolonged stale heartbeat and path flapping. Never fill communication gaps. Heartbeat does not prove alarm receipt; a previous receipt does not guarantee current delivery; provider data may lag or abstract external failures.

ANALYSIS REQUIRED

Operation and interval. Observation and lookup; comparison of local and receiving-centre evidence; no forecast or guarantee of future transmission. At each heartbeat/event, at daily handover and after every authorised end-to-end test. Minimum evidence and method. Current ATU identity/configuration, recent healthy heartbeat, no unresolved fault/maintenance, provider state and dated receiving-centre receipt for any end-to-end claim. Separate local equipment, bearer path and receiving-centre receipt; report the last proven end-to-end test and current evidence gaps. Uncertainty and safe failure. External network visibility, provider latency, clock alignment, missed events, maintenance and test conditions. Return 'remote signalling not demonstrated' and direct authorised staff to the current provider/receiving-centre check and impairment process.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The alarm transmission/system owner owns local equipment; the receiving centre/provider owns receipt evidence and service state; the Responsible Person/fire-safety lead owns impairment and escalation. Fire-service/receiving-centre instruction and signed impairment first; then current provider/local indication; last test receipt is historic corroboration only. Times, access and provenance. Provider identifiers, routing details and contact information are restricted to authorised roles. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Current path state stale after one missed approved heartbeat interval; test receipt is valid only as dated and until relevant change/fault. Conflict. A local fault, provider incident or missing receipt cannot be overruled by a normal heartbeat. Show source-specific status and require direct confirmation where needed. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not promise alarm delivery, initiate a test call, contact emergency services or treat path status as evidence that building alarms or response are effective. Do not expose account numbers, routing, credentials or responder contacts to unauthorised users. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports prompt impairment escalation and gives handover staff a precise statement of what is live, what was last tested and what remains unproved.

FS-020

FIRE-SAFETY PERSONNEL QUESTION

For the briefing being prepared, which alarm, detection and suppression statements are supported, qualified or unavailable, and why?

SENSORS AND TELEMETRY

Proposed evidence. No additional sensing is implied. The service queries only verified, permitted sources already described: panel/loop states, suppression telemetry, specialist tests, current impairments and attended observations. Coverage and timing. Only assets, zones and time windows explicitly within the user's authorised question; every omitted, unmapped or unhealthy source is shown as a coverage gap. Use native event times and latest accepted records; aggregate by claim and dependency while preserving source age and the underlying evidence links. Quality and limits. Require commissioned source mapping, current configuration, calibration/functional checks where applicable, clock/UTC offset, uptime and gaps. Preserve source-to-claim provenance and flag drift or stale feeds inherited from each source. No imputation across missing evidence. Combining weak sources does not create strong evidence; a normal status cannot establish physical condition, design adequacy or compliance.

ANALYSIS REQUIRED

Operation and interval. Claim decomposition and evidence comparison. Observation, lookup, calculation or estimate are labelled separately; forecast and causal diagnosis are not used. Recommendation is only to obtain missing evidence or escalate. On demand for a defined decision and time; refresh whenever a critical source changes. Minimum evidence and method. A precise claim, current decision time/area, required dependencies, authoritative owner criteria, current evidence with provenance and no unresolved critical conflict. Break the claim into testable subclaims, apply source precedence, classify each as supported/qualified/unavailable, and state the narrowest defensible conclusion. Uncertainty and safe failure. Coverage, source quality, clock alignment, staleness, mapping and unresolved owner disagreement. Return 'claim not supportable' with the exact missing dependency and required owner-led check; do not synthesize a verdict.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person and relevant system/asset owners define the decision and authoritative records; competent maintainers/assessors own specialist findings; the fire-safety lead controls access and interpretation. Fire-service command/current emergency plan first during incidents; then signed owner/competent decisions and impairments; then current system records; then healthy telemetry; then derived analysis. Times, access and provenance. Claim-specific, least-privilege access; sensitive topology and raw evidence are withheld unless required and authorised. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Use each source's declared staleness; if any critical dependency is stale, the combined claim cannot be 'supported'. Conflict. Display conflicts and lower the claim status; never average, vote or select a source to obtain a reassuring answer. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not convert evidence status into certification, operational permission, a building-wide safety statement or an emergency instruction. Do not disclose more system detail than the authorised decision requires. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the Responsible Person a transparent briefing that separates current observation from records and assumptions, making gaps actionable rather than hidden.

Evacuation routes, exits, refuges and assembly points

FS-021

FIRE-SAFETY PERSONNEL QUESTION

Which final exits or protected-route segments need an immediate local check because they may be closed, locked, obstructed or otherwise unavailable?

SENSORS AND TELEMETRY

Proposed evidence. Door-position contacts, approved emergency-release status, lock/bolt feedback only where designed, selected clear-zone time-of-flight sensing, route/closure reports and attended observations. Coverage and timing. Only current final exits and route segments mapped to owner-approved evacuation information; unmonitored approaches, discharge paths and external sections are explicit gaps. State changes within 5-10 seconds; clear-zone exceptions per minute; closure and observation events at source time; retain the last known evidence age for every segment. Quality and limits. Retain contact/release functional checks, range calibration for clear-zone sensing, source clock/UTC offset, configuration, uptime, gaps and provenance. Recheck after door or layout work; detect stuck states and drift. Do not bridge live-state gaps over two minutes. A closed contact does not prove unlocked/openable status, panic hardware, opening effort or clear onward travel; clear-zone sensing may miss low or screened obstructions.

ANALYSIS REQUIRED

Operation and interval. Observation and comparison; recommendation limited to where an authorised physical check is needed, not which route occupants should use. Every 30 seconds during occupied periods and on demand before opening, events or planned route changes. Minimum evidence and method. A current route/exit ID, expected state, healthy recent evidence, current closure/impairment record and at least one attended source where openability is material. Reconcile door, release, obstruction and closure evidence separately; rank only by affected route dependency and evidence age. Uncertainty and safe failure. Unmonitored sections, lock/contact mismatch, clock offset, transient crowding, external conditions and hidden hardware defects. Return 'exit/route availability not demonstrated' and direct an authorised local inspection.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead owns route definitions; Estates/Security own doors, locks and closures; event/work owners hold temporary restrictions; competent inspectors own physical findings. Live fire-service command and current signed closure/impairment first; then owner-approved route records and competent observations; then healthy telemetry. Times, access and provenance. Detailed secure exits, lock states and access events are restricted to authorised operational roles. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Two minutes for live door/clear-zone state; immediately stale after a reported closure, crowd or layout change. Conflict. Any current closure or physical report overrides apparently available telemetry. Unexplained disagreement requires a local check. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer safe passage, legal adequacy, opening force or complete route availability from door/clearance signals, and do not unlock or release a door. Do not expose secure exit geometry, individual access events or restricted lock information outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This enables a focused pre-occupancy sweep and prompt escalation of a closed or uncertain exit without presenting telemetry as evacuation permission.

FS-022

FIRE-SAFETY PERSONNEL QUESTION

For each occupied evacuation zone, which approved egress options have current end-to-end evidence and which still need a physical route confirmation?

SENSORS AND TELEMETRY

Proposed evidence. Current route/closure records, mapped door and selected route telemetry, anonymous aggregate occupancy by zone, and attended route observations; no person tracking or route recommendation engine. Coverage and timing. Only zones and egress options in the current approved emergency information, with each physical segment's monitored/unmonitored status shown. Route states at their native intervals, occupancy aggregates every 5-15 minutes, and manual observations at source time; combine only for a stated decision time. Quality and limits. Retain mapping versions, sensor calibration/functional checks, source clocks/UTC offsets, uptime, gaps and end-to-end provenance. Reconcile counter drift at known empty/reset periods. Do not infer route state during gaps or propagate a segment result to unobserved sections. Occupancy estimates and partial route telemetry cannot establish suitability, capacity, tenability or availability during an incident.

ANALYSIS REQUIRED

Operation and interval. Lookup and claim comparison; limited calculation of evidence coverage. No forecast, optimisation or recommendation of an evacuation route. On demand before an activity, handover or planned closure; not as a substitute for live incident command. Minimum evidence and method. A current approved zone-to-egress mapping, current segment states/closures, source health and a clear decision time; occupancy is required only if the question concerns demand. Decompose each option into segments, apply evidence precedence and classify supporting evidence as current, qualified or missing without selecting a 'best' route. Uncertainty and safe failure. Unmonitored segments, changing occupancy, clock offset, external discharge conditions and emergency conditions not represented in routine data. Return 'egress option not established' and name the segment or owner confirmation required.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety lead owns approved egress information; route, door and occupancy system owners own current evidence; fire-service command controls live incident use. Fire-service command/current emergency plan first; then signed route closures and owner-approved plans; then competent observations; then healthy telemetry and aggregate estimates. Times, access and provenance. Zone-level evidence is role-based; do not expose sensitive route geometry or individual movement/access data. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Use the strictest staleness among required route and occupancy inputs; any critical stale segment makes the option qualified or unavailable. Conflict. Retain contradictory route states and defer to the current closure/competent source. Never use model output to overrule an approved restriction. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not tell occupants which route to take, certify any option, model smoke/tenability, or treat routine route evidence as live incident permission. Do not disclose personal movement, access histories or restricted emergency-route detail. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives fire-safety personnel a bounded pre-event evidence check and identifies exactly where a physical route confirmation or approved plan update is needed.

FS-023

FIRE-SAFETY PERSONNEL QUESTION

Which refuge communication points need action because power, line supervision, call, acknowledgement, location or intelligibility evidence is missing?

SENSORS AND TELEMETRY

Proposed evidence. Controller/line supervision, device heartbeat where supported, call initiation and acknowledgement logs from authorised tests, local power/battery state, and competent two-way functional/intelligibility observations. Coverage and timing. Every registered refuge communication point, master station and network segment, mapped to current refuge location and accessible approach; unintegrated points remain inspection-only. Supervised states on change; test calls at source event resolution; daily heartbeat summary; no recording of routine speech or call content. Quality and limits. Retain device and line functional checks, battery-test date, configuration, source clocks/UTC offsets, uptime, gaps and provenance. Measure cross-end clock offset during tests; flag repeated missed heartbeats, call-sequence gaps and drift in battery test values. Never infer availability through a lost link. Line supervision may not prove microphone, loudspeaker, call quality, physical access or user reach; a successful test does not guarantee future use.

ANALYSIS REQUIRED

Operation and interval. Event-sequence observation and comparison; recommendation to inspect or retest; no diagnosis of communication quality from supervision alone. On every fault, at each authorised test and in daily exception assessment. Minimum evidence and method. Current device/location register, complete line/power state, aligned test-call sequence, competent endpoint observation, and current defect/impairment status. Separate line availability, call initiation, master receipt, two-way communication and physical access; report the first unproved stage. Uncertainty and safe failure. Clock resolution, intermittent faults, acoustic conditions, battery condition, unobserved physical damage and incomplete test coverage. Return 'refuge communication not demonstrated' and require a competent local/end-to-end test.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The refuge communication system owner owns the register and technical state; the competent maintainer owns tests; the fire-safety lead owns refuge procedures and impairment action. Current incident command and signed impairment/test result first; then master/local observation; then complete healthy telemetry. Times, access and provenance. Device locations and test logs are role-restricted; do not retain speech, caller identity or personal assistance information. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live state stale after one approved heartbeat/poll interval; test evidence stale after fault, configuration or location change. Conflict. A failed test or local report overrides a normal line status. Missing acknowledgement or endpoint evidence is an incomplete test. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not initiate emergency calls, record conversations, infer a person's needs or promise assistance response. Do not expose refuge-user identities, call content or sensitive location details outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps arrange targeted remedial work or temporary authorised provision and prevents line supervision from being mistaken for a usable refuge communication point.

FS-024

FIRE-SAFETY PERSONNEL QUESTION

Which final-exit discharge areas or external onward paths need a local clearance or access check before the next occupied period?

SENSORS AND TELEMETRY

Proposed evidence. Selected outdoor gate/door contacts, fixed time-of-flight or radar presence sensing at tightly defined pinch points where privacy assessment permits, weather/works feeds, inspection reports and authorised closure records. Coverage and timing. Only mapped discharge areas and onward-path pinch points on the premises; public highway, remote paths and uninstrumented sections remain outside the service's observation. Door/gate states within 10 seconds; pinch-point occupancy/obstruction flag per minute; weather and works at source interval; manual observations retain validity. Quality and limits. Retain functional checks, sensing-zone geometry, calibration against reference distances, source clock/UTC offset, uptime, gaps and provenance. Revalidate after landscaping, hoarding or sensor movement; flag weather-related noise and drift. Do not bridge gaps over five minutes. Outdoor sensors are affected by rain, vegetation, vehicles and crowds and cannot identify an obstruction, assess surface condition, lighting, gradients or complete onward safety.

ANALYSIS REQUIRED

Operation and interval. Observation and comparison; recommendation to inspect or coordinate, not to declare an onward path safe. Every minute before and during planned high-occupancy use, and on demand after weather, works or security changes. Minimum evidence and method. Current discharge/path map, current closure/works/weather context, healthy recent evidence at all monitored points and attended confirmation for material conditions. Identify persistent monitored exceptions, overlay known closures and explicitly stop at the premises coverage boundary. Uncertainty and safe failure. Weather noise, vegetation, unmonitored sections, surface condition, public-realm changes, clock offset and crowd dynamics. Return 'external discharge condition not established' and request an attended premises sweep or external-owner confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates/Security own premises access and closures; the fire-safety lead owns discharge-path requirements; project/event owners hold temporary restrictions; external authorities own public-realm conditions. Fire-service command and current closure/works records first; then attended inspection; then healthy on-premises telemetry. External-authority information cannot be inferred from premises sensors. Times, access and provenance. Detailed gates and access routes are role-restricted; use aggregate presence only, with no images or identifiers. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Five minutes for live outdoor state, immediately stale after weather/works change; external data uses its declared update time. Conflict. Any current closure or attended defect overrides normal telemetry. Missing external evidence is shown as outside coverage. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer public-road availability, surface safety, accessibility, capacity or route suitability, and do not open gates or direct evacuees. Do not retain vehicle/person identities or disclose secure access arrangements without permission. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports a timely check of gates, stored items, hoardings or weather effects and identifies when coordination with an external owner is required.

FS-025

FIRE-SAFETY PERSONNEL QUESTION

Which emergency-lighting circuits, luminaires or route areas lack current functional, duration or physical-condition evidence?

SENSORS AND TELEMETRY

Proposed evidence. Central-battery or self-test luminaire status where supported, circuit faults, battery/charger state, test-controller logs, and competent functional, duration and physical inspection results including sampled illuminance only where required. Coverage and timing. Every registered circuit/luminaire plus the exact route areas and sample points tested; non-self-test fittings and physical conditions remain inspection-dependent. Fault states on change; self-test summaries at manufacturer interval; full event trace for authorised tests; inspection and measurements at source time. Quality and limits. Retain controller/luminaire functional checks, battery-test history, meter calibration and sample geometry, configuration, source clock/UTC offset, uptime, gaps and provenance. Compare repeated duration/capacity results for degradation only within the same method; flag drift and replaced components. Do not infer through missed tests. A self-test or central status does not prove light output, orientation, obstruction, cleanliness, coverage, duration under real conditions or route suitability.

ANALYSIS REQUIRED

Operation and interval. Lookup, observation and evidence-completeness comparison; calculation of tested coverage; no prediction of emergency performance from a health flag. On every fault, after each test and in 7/30/90-day maintenance planning. Minimum evidence and method. Current luminaire/circuit register, valid test scope, complete self-test data where used, competent physical inspection and current route configuration. Separate electrical status, functional operation, duration and physical/coverage evidence; flag assets/areas without each required evidence type. Uncertainty and safe failure. Battery ageing, self-test limitations, meter calibration, sample location, clock gaps and changes after test. Return 'emergency-lighting evidence incomplete' and require competent test or inspection.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The emergency-lighting system owner owns the register and test programme; competent maintainers own test results; Estates owns changes to routes/ceilings; the fire-safety lead owns operational impairment. Current signed competent test and impairment first; then self-test/controller evidence; then historic measurements as qualified evidence only. Times, access and provenance. Detailed circuit topology and secure route locations are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live fault state stale after one approved reporting interval; test evidence stale at due date or after layout, luminaire, battery or circuit change. Conflict. A physical failure or failed duration test remains open despite normal self-test. Missing sample coverage is not inferred from nearby luminaires. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not declare a route adequately lit, calculate legal compliance from sparse samples, or initiate tests or switch circuits. Do not expose secure circuit topology or restricted route details outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This creates a precise test and remedial work list and stops a self-test 'pass' from being used as proof of physical route illumination.

FS-026

FIRE-SAFETY PERSONNEL QUESTION

Which escape-route doors need an attended test of panic hardware, emergency release, opening effort or final openability?

SENSORS AND TELEMETRY

Proposed evidence. Door-position and approved emergency-release feedback, lock/bolt state where designed, authorised test events, calibrated force-gauge measurements where a competent method requires them, and witnessed physical results. Coverage and timing. Only doors in the current escape-door register, mapped to direction of travel, hardware type and dependent route; telemetry cannot cover uninstrumented mechanics or external conditions. Discrete states on change; full trace during authorised test; force/physical measurements at source time and defined test conditions. Quality and limits. Retain hardware and release functional checks, force-gauge calibration, configuration, source clocks/UTC offsets, uptime, gaps and provenance. Compare only like-for-like tests and flag post-maintenance baseline change. Do not infer openability between tests or across missing events. Release and contact feedback do not prove panic hardware operation, acceptable force, door swing, leaf condition or unobstructed onward travel.

ANALYSIS REQUIRED

Operation and interval. Event-sequence observation and comparison; measurement when competent testing is provided; no diagnosis of hardware failure cause. At scheduled tests, after hardware/access change, and before reopening an affected route. Minimum evidence and method. Current door/hardware record, approved test method, calibrated equipment if used, complete release/open/close trace, witnessed outcome and current defect/closure status. Separate command, release, manual actuation, openability and return state; compare only with owner-approved criteria and test conditions. Uncertainty and safe failure. Human technique, door pressure, weather, floor condition, instrument calibration, clock offset and hidden mechanical defects. Return 'escape-door openability not demonstrated' and require an attended competent test.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead owns escape-door requirements; Estates/Security own hardware and access interfaces; the competent maintainer/inspector owns physical test and force interpretation. Current impairment/closure and competent test first; then local interface state; then healthy telemetry. Times, access and provenance. Lock, release and secure-route details are role-restricted; do not expose individual access events. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live state stale after two minutes; physical test valid only for the tested condition and owner-defined period. Conflict. A failed physical test overrides normal electronic feedback. Any difference between door, lock and release states is unverified pending attendance. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not unlock, release or test a door, infer legal compliance from a contact, or advise occupants to use a door during an incident. Do not disclose secure lock states, access events or restricted route geometry. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps schedule the right hardware or access-control specialist and records exactly which part of an end-to-end openability test is missing.

Evacuation routes, exits, refuges and assembly points

FS-027

FIRE-SAFETY PERSONNEL QUESTION

Which dependencies of any designated evacuation lift are unavailable, impaired or not demonstrated by current integrated testing?

SENSORS AND TELEMETRY

Proposed evidence. Lift controller/fire mode status, power and standby supply, landing communication/alarm, relevant smoke-control or lobby states, water ingress alarms where installed, door/position telemetry and competent integrated-test results. Coverage and timing. Only lifts explicitly designated in the current emergency plan and their owner-defined dependencies; ordinary passenger lifts are not reclassified by the service. Discrete states on change; controller heartbeat at approved interval; full aligned trace during authorised integrated tests; aggregate by dependency with raw events retained. Quality and limits. Retain controller/interface functional checks, power-monitor calibration where applicable, configuration, source clocks/UTC offsets, uptime, gaps and provenance. Measure inter-system clock offset before tests; flag missing sequence events and changed dependencies. Do not reconstruct a test across any gap. Controller/fire-mode signals do not prove safe operation, capacity, lobby conditions, communication, staffing or suitability for any person or incident.

ANALYSIS REQUIRED

Operation and interval. Dependency lookup and cross-system sequence comparison; no recommendation to use a lift and no forecast of availability in fire conditions. At handover, after any dependency fault/change and at each authorised integrated test. Minimum evidence and method. Current designation/procedure, dependency map, complete live states, current impairment, aligned integrated-test evidence and competent owner acceptance. Evaluate each dependency separately, apply precedence and report supported/qualified/unavailable status without collapsing to a single 'safe' indicator. Uncertainty and safe failure. Hidden mechanical condition, inter-system clock uncertainty, untested scenarios, staffing, lobby conditions and changes since test. Return 'evacuation-lift dependency not demonstrated' and direct owner-led confirmation under the current procedure.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety lead owns the approved evacuation-lift procedure; the lift owner and competent lift contractor own lift evidence; power, alarm, smoke-control and communication owners own dependencies. Fire-service command/current emergency plan first; then current impairment and signed integrated test; then local controller/owner records; then healthy telemetry. Times, access and provenance. Lift logic, locations and vulnerability details are restricted to authorised operational and specialist roles. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live states stale after one approved heartbeat interval; integrated test valid only for the tested configuration and until any dependency changes. Conflict. Any owner-declared impairment or failed dependency overrides normal lift telemetry. Missing endpoint evidence makes the lift dependency unavailable, not assumed. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not designate a lift, advise anyone to use it, infer suitability for a person, operate/test it or override fire-service command. Do not disclose personal evacuation needs or restricted lift/control topology. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the fire-safety lead a dependency-specific impairment view for planning, briefing and specialist coordination without issuing use advice.

FS-028

FIRE-SAFETY PERSONNEL QUESTION

Does any assembly point or approach path need an authorised change because works, weather, access constraints or planned occupancy undermine the current arrangement?

SENSORS AND TELEMETRY

Proposed evidence. Current assembly-point and event records, premises gate/route states, authorised works/closure feeds, weather alerts, optional aggregate people counts at defined outdoor zones and attended observations. Coverage and timing. Only owner-approved assembly points and on-premises approaches; public-realm capacity, crowd behaviour and off-site alternatives remain outside coverage. Works/events by effective interval; weather at provider interval; aggregate counts every 5-15 minutes where permitted; observations at source time. Quality and limits. Retain counter calibration/zone geometry, source clocks/UTC offsets, uptime, gaps and provenance; reconcile counter drift at empty periods. Preserve weather/provider version. Do not extrapolate counts beyond the sensing zone or bridge gaps over 15 minutes. Counts do not establish capacity, accountability, accessibility, surface condition or suitability; forecasts and weather feeds may be uncertain.

ANALYSIS REQUIRED

Operation and interval. Lookup, observation, estimate of aggregate use and forecast of planned demand, each labelled separately; recommendation only for owner decision. Before major events, daily during works, and on demand after weather/security changes. Minimum evidence and method. Current assembly-point approval, approach map, current works/access/weather context, planned occupancy and recent local observation for material decisions. Overlay effective restrictions and planned demand, identify evidence gaps and present scenarios rather than declaring suitability. Uncertainty and safe failure. Forecast error, weather change, counter drift, external conditions, accessibility needs and crowd dynamics. Return 'assembly-point suitability not established' and require the Responsible Person or relevant owner to confirm arrangements.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead/Responsible Person owns assembly arrangements; Estates/Security own premises access; event/work owners own planned impacts; external authorities own public-realm constraints. Fire-service command/current emergency plan first; then signed closure/change and current approved assembly information; then attended observation; then aggregate telemetry/forecast. Times, access and provenance. Share only aggregate occupancy and approved operational information; no person identities or movement histories. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live access/count evidence stale after 15 minutes; weather/works use source-declared validity and become stale after any unrecorded change. Conflict. Any current closure or owner report overrides apparent access. If external conditions are unknown, show them as outside coverage. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not designate a new assembly point, direct evacuees, claim capacity/accessibility or expose individual location data. Do not publish secure access arrangements or sensitive event information beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports early relocation or briefing through the authorised plan process before a closure, event or weather condition makes the current arrangement unreliable.

Evacuation routes, exits, refuges and assembly points

FS-029

FIRE-SAFETY PERSONNEL QUESTION

For the next event or closure, where could evacuation demand depend too heavily on one route, create counterflow or exceed the movement evidence available?

SENSORS AND TELEMETRY

Proposed evidence. Current approved route graph, event/booking schedules, anonymous doorway/zone counts, selected temporary overhead time-of-flight counters during authorised trials, and drill/manual observations. Coverage and timing. Only planned scenarios, mapped zones and observed route segments; no prediction of individual behaviour, smoke conditions or unmonitored bottlenecks. Historical counts aggregated to 5-15 minute bands; planned occupancy by event interval; temporary trial data at 1-5 seconds reduced to flow bands; preserve dates and conditions. Quality and limits. Retain counter calibration, direction-validation checks, zone geometry, clocks/UTC offsets, uptime, gaps and provenance. Reconcile entry/exit drift at known resets and compare temporary counters with manual counts. Exclude periods with layout changes or gaps over five minutes. Routine flows and drills do not reproduce an emergency; counterflow detection can be ambiguous; capacity and movement depend on behaviour, accessibility, door state and incident conditions.

ANALYSIS REQUIRED

Operation and interval. Forecast and scenario comparison using aggregate estimates; no causal diagnosis and no operational route optimisation. During event planning, after any closure/layout change, and on the day with current inputs. Minimum evidence and method. Approved route graph, event scenario, current route restrictions, aggregate demand inputs, representative observed flow data with known coverage and owner-set assessment criteria. Estimate zone-to-route demand ranges, identify single dependencies and counterflow points, and compare with observed - not certified - flow bands under similar conditions. Uncertainty and safe failure. Demand uncertainty, non-emergency comparability, sensor drift, route-choice assumptions, accessibility needs, unobserved sections and configuration mismatch. Return 'scenario evidence insufficient' and require competent assessment or an attended trial; do not issue a route recommendation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety lead owns planning criteria and approved routes; event/space owners own planned numbers; Estates owns route configuration; competent assessors own any formal analysis. Current emergency plan and approved event/closure conditions first; then competent assessment and current route data; then qualified historic observations and model calculations. Times, access and provenance. Use aggregate data only; raw trajectories, device identities and event attendee details are prohibited. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Forecast stale when event numbers, layout, closures or route state change; historic data must match the stated configuration. Conflict. Conflicting schedules, counts or route records are shown as scenarios and escalated; do not choose the largest or smallest silently. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not calculate or certify evacuation time/capacity, select routes, predict individual behaviour or issue instructions for a live incident. Do not retain or disclose attendee identities, device traces or individual movement. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This highlights fragile route dependencies early enough to change the event plan, staffing, furniture or closure timing through the authorised approval process.

FS-030

FIRE-SAFETY PERSONNEL QUESTION

Which temporary route, exit, refuge or assembly-point changes have not yet been reflected consistently in the approved plan, signs, briefings and operational systems?

SENSORS AND TELEMETRY

Proposed evidence. Change-control records, temporary works/closure permits, digital plan and signage versions, briefing acknowledgements by role, asset/route register updates and time-stamped site observations; no continuous sensing is necessary. Coverage and timing. Every approved or emergency temporary change affecting egress, refuge, assembly or associated systems, including start/end, rollback and affected audiences. Event-driven at decision, implementation, briefing, inspection and withdrawal; retain effective intervals and all versions; aggregate to propagation status by change. Quality and limits. Require signed source, stable change ID, file hashes, source clock/UTC offset, version/supersession links, acknowledgement provenance and missing-step gaps. Calibration is not applicable except to supporting measurements. Flag overlapping effective dates, stale copies and orphaned temporary signs. An acknowledgement does not prove understanding or physical implementation; records may miss informal changes; signs may be moved after inspection.

ANALYSIS REQUIRED

Operation and interval. Workflow lookup and version/effective-time comparison; recommendation to complete or withdraw a propagation step, not to approve the change. At each change decision, before it becomes effective, at every shift/event handover and at expiry. Minimum evidence and method. Signed change ID, effective interval, current approved plan/instruction, implementation evidence, affected-system versions, briefing scope and withdrawal/rollback owner. Check the required propagation chain, verify exact version and effective time at each target, and list missing or conflicting steps. Uncertainty and safe failure. Unrecorded copies, late physical changes, acknowledgement without understanding, clock errors and incomplete audience coverage. Return 'temporary change not fully propagated' and direct the named owners to verify before reliance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety lead owns approval and emergency information; change/work owners implement; Estates/Space/Security own systems and locations; Communications/training owners hold briefing records. Fire-service command/current emergency plan and signed change decision first; then competent site verification; then system/version and briefing records. Times, access and provenance. Role-level acknowledgement only; do not expose personal training or attendance detail beyond authorised need. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale when the change, route, timing or affected population changes; every temporary change must carry a clear expiry/assessment time. Conflict. Any mismatch in effective date, plan, sign, system or observed implementation remains unresolved; do not treat distribution as implementation. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not approve the temporary arrangement, generate emergency instructions, infer comprehension or treat system consistency as fire-safety acceptance. Do not expose personal briefing records or restricted plan detail to unauthorised users. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This provides a concrete close-out list so plans, signs, systems and staff do not describe different evacuation arrangements.

FS-031

FIRE-SAFETY PERSONNEL QUESTION

For each evacuation zone, what aggregate occupancy range is supportable now, and where is the evidence too weak for staffing, drill or event planning?

SENSORS AND TELEMETRY

Proposed evidence. Anonymous bidirectional people counters at selected entrances, turnstile/access totals aggregated by zone where lawfully permitted, room-booking/teaching schedules and time-stamped manual headcounts. No Wi-Fi/Bluetooth device tracking or identity-level access analysis. Coverage and timing. Only zones with a documented counting boundary and current topology; internal movement, tailgating, open doors and unmonitored entrances are explicit coverage gaps. Doorway counts at 1-5 second events, reduced to 5-minute zone totals or bands; schedules by effective interval; manual counts at source time. No person-level histories are retained. Quality and limits. Retain counter calibration and direction-validation results, mounting geometry, source clock/UTC offset, uptime, gaps, resets and raw-to-aggregate provenance. Reconcile drift at verified empty periods or attended counts; never silently force totals to booking numbers or interpolate across gaps longer than 15 minutes. Counts are estimates, not identities or accountability; boundary crossings, tailgating, multiple entries, open-plan movement and sensor occlusion create error.

ANALYSIS REQUIRED

Operation and interval. Observation and calculation of an aggregate estimate; comparison with owner-set planning values. No identity inference, accountability claim or forecast unless explicitly requested. Every 5-15 minutes for routine planning and on demand before events or out-of-hours handover. Minimum evidence and method. A current zone boundary, healthy counters on all material entrances or a documented coverage fraction, current schedules/access aggregates, reset status and known uncertainty. Apply directional counts, reconcile only at verified reset points, combine independent sources as ranges, and label direct observation versus schedule-derived estimate. Uncertainty and safe failure. Counter error, unmonitored entrances, internal migration, clock offset, sensor occlusion, manual-count age and unknown starting balance. Return 'aggregate occupancy not established' with a range only if defensible; otherwise require an authorised local count.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Space Management/Timetabling owns planned use; Security or the access-system owner owns permitted aggregate access data; the fire-safety lead/Responsible Person owns the operational question and zone definitions. Fire-service command and current emergency plan first during incidents; then attended/accountability records; then current aggregate counters; schedules and bookings are planning context only. Times, access and provenance. Only minimum necessary zone bands or ranges may be shared; individual access records, device identifiers and named presence are excluded. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live estimate stale after 15 minutes or any unobserved boundary opening/reset; schedules stale when activity changes. Conflict. Show each source and range. Do not average conflicting counts into a precise total; widen uncertainty and request a local count if the decision is material. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not identify or track people, claim a roll call, infer individual location, or use an occupancy estimate to certify evacuation capacity or issue instructions. Do not expose personal access, booking or attendance data; use zone-level bands under role-based permission. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This tells fire-safety personnel where occupancy evidence is adequate for routine staffing or drill planning and where a local count or conservative owner decision is needed.

FS-032

FIRE-SAFETY PERSONNEL QUESTION

Which occupied spaces may exceed the current owner-approved planning value and therefore need a local count or authorised operational action?

SENSORS AND TELEMETRY

Proposed evidence. Anonymous doorway/zone counts, permitted aggregate access totals, current room booking/timetable numbers and an attended local count where escalation is considered. Coverage and timing. Only spaces with a current owner-approved planning value, stable counting boundary and known sensor coverage; open-plan or shared-boundary areas are treated as estimates with wider uncertainty. Counts reduced to 5-minute ranges; booking values by event interval; manual counts at source time. Alert only after a configurable persistence period to avoid transient doorway queues. Quality and limits. Retain counter calibration, direction tests, geometry, clocks/UTC offsets, uptime, gaps, resets and provenance. Validate against attended counts under representative conditions; track cumulative drift and reset only with recorded evidence. Do not interpolate across a boundary or feed outage. Counts cannot establish lawful capacity, occupant characteristics, seating layout, route availability or fire-risk acceptability; bookings may include no-shows or unrecorded attendees.

ANALYSIS REQUIRED

Operation and interval. Observation, estimate and comparison; recommendation limited to confirm, pause admission through authorised staff, or escalate under existing procedures. No capacity certification. Every five minutes during relevant occupancy and at event start/changeover. Minimum evidence and method. A current approved value with scope/effective time, current space configuration, healthy aggregate evidence with known start balance, and current route/closure context. Calculate a conservative occupancy range, compare its lower and upper bounds with the approved value, and classify confirmed, possible or unsupported exceedance. Uncertainty and safe failure. Counter drift, open doors, internal movement, booking variance, changing layout, clock gaps and unknown starting occupancy. Return 'possible exceedance - local count required' unless the minimum defensible estimate itself exceeds the approved value; do not automate access control.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Space/event owners own planned numbers and room configuration; the Responsible Person/fire-safety lead owns any approved planning value and response; Security owns permitted aggregate access evidence. Current signed operational restriction/approved value first; then attended count; then healthy aggregate estimate; booking is forecast/context only. Times, access and provenance. Only aggregate range and exception status may be shown; no attendee lists or individual access histories. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live estimate stale after 10 minutes or any unobserved entrance/layout change; planning value stale after configuration or risk-assessment change. Conflict. If sources disagree, show the widest plausible range and require local confirmation before action; never choose the lower estimate to clear an exception. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not set or reinterpret capacity, infer personal attributes, stop entry, lock doors, direct occupants or claim the space is unsafe/safe. Do not disclose attendee identity, access logs or detailed schedules beyond authorised need. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports a prompt local count, event-owner contact or use of existing admission controls before an uncertain crowd estimate becomes a hidden planning assumption.

Occupancy, visitors and inclusive evacuation information

FS-033

FIRE-SAFETY PERSONNEL QUESTION

What visitor and contractor totals are currently recorded by zone, and which hosts, permits or check-out records leave accountability incomplete?

SENSORS AND TELEMETRY

Proposed evidence. Visitor-management and contractor permit records aggregated by zone/status; badge issue/return totals; approved gate or reception counts; authorised manual muster/accountability updates. No continuous person-location tracking. Coverage and timing. Only people formally registered in the relevant visitor/contractor systems and zones those systems are authorised to represent; hosts, informal visitors and internal movement may be unobserved. Record status on check-in/out or permit event; aggregate to current zone/site totals and unmatched records. Manual accountability has explicit source and observed time. Quality and limits. Retain system/version, badge/permit status, source clock/UTC offset, uptime, gaps, duplicate/merge history and record provenance. Validate timezone and site boundary; flag orphaned check-ins, expired permits and impossible sequences. Do not infer checkout from access silence. Registration does not prove current presence or exact zone; badge sharing, missed checkout, host movement and emergency relocation create gaps.

ANALYSIS REQUIRED

Operation and interval. Lookup and aggregation; comparison of registered, checked-out and unaccounted statuses. No location inference or prediction. At handover, before out-of-hours work/events, and when authorised command requests an update. Minimum evidence and method. Current system health, defined site/zone scope, valid registrations/permits, known check-in/out status and current manual accountability source where material. Aggregate records by status and permitted zone, identify orphaned or expired entries, and state that totals are administrative records rather than a roll call. Uncertainty and safe failure. Late updates, missed checkout, badge sharing, zone ambiguity, system outages, clock error and informal visitors. Return 'formal accountability incomplete' and identify the responsible host, reception or permit process without exposing unnecessary personal detail.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Security/reception owns visitor records; contractor/permit owners own contractor status; hosts/work owners own local accountability; the Responsible Person/fire-safety lead defines required aggregate briefing. Fire-service command and attended accountability during an incident first; then current visitor/permit records; access aggregates may corroborate but not identify location. Times, access and provenance. Share only authorised aggregate totals and explicitly requested named information through established command channels; routine catalogue answers must not expose identities. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Current status stale after the owner-set check-out/permit interval or 15 minutes after a system outage; incident accountability is live only at its observed time. Conflict. Do not reconcile an unmatched visitor by guessing. Show registration and accountability sources separately and escalate missing check-out/host confirmation. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not claim exact presence/location, generate a roll call, infer identity from sensors, or publish personal emergency/visitor information. Do not expose names, contact details or access histories except through authorised emergency/accountability channels. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This lets fire-safety personnel focus hosts, reception and permit owners on unresolved records before a drill, out-of-hours period or incident briefing.

FS-034

FIRE-SAFETY PERSONNEL QUESTION

Where are teaching, events, meetings, deliveries or shift changes likely to create material movement peaks during the next two hours?

SENSORS AND TELEMETRY

Proposed evidence. Current timetable/booking and event schedules, anonymised doorway/zone counts, approved access-flow aggregates and known closures. No device tracking, camera identification or content inspection. Coverage and timing. Only scheduled activities and monitored route zones with current mappings; spontaneous activity and unmonitored routes are explicit uncertainty. Historical counts in 5-15 minute bands matched to comparable activity windows; schedules by effective time; forecast refreshed every 15 minutes and after changes. Quality and limits. Retain counter calibration, geometry, clocks/UTC offsets, uptime, gaps, resets and provenance; validate historical direction counts against manual samples. Version schedules and route topology. Exclude periods with sensor drift, abnormal closures or data gaps over 15 minutes. Schedule attendance and historic movement do not predict individual behaviour or emergency flow; peaks may shift with lateness, weather, room changes and cancellations.

ANALYSIS REQUIRED

Operation and interval. Forecast from aggregate schedules and observed baselines, with comparison to current route conditions; no causal diagnosis or emergency evacuation-time prediction. Every 15 minutes over a two-hour horizon and on demand for event planning. Minimum evidence and method. Current schedule/version, mapped origin/destination zones at coarse level, recent healthy counts, representative historic periods and current closures. Estimate time-banded movement ranges, adjust only for observed current attendance bands, and identify overlapping activity windows and monitored pinch points. Uncertainty and safe failure. Attendance variance, route choice, cancellations, counter drift, clock alignment, spontaneous activity and non-emergency comparability. Return 'movement peak forecast too uncertain' and use only confirmed schedules/closures for planning.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Timetabling/Space/Event owners own planned activity; Security/Estates own route closures; the fire-safety lead defines the planning question and acceptable uncertainty. Current closures and confirmed event changes first; then current schedule; then representative aggregate observations. Forecast never outranks live observation. Times, access and provenance. Use only aggregate event sizes and route bands; no attendee identity, class lists or individual movement. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Forecast expires after 15 minutes or any schedule/route change; historical comparators must match current topology. Conflict. Present multiple scenarios when schedule and live counts disagree; do not force a single peak time. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not predict individuals, claim evacuation capacity, alter access, issue crowd-control instructions or present routine movement as emergency behaviour. Do not expose class/event attendee data or individual access histories. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps schedule a route sweep, warden positioning, delivery pause or event briefing before predictable changeover pressure occurs.

Occupancy, visitors and inclusive evacuation information

FIRE-SAFETY PERSONNEL QUESTION

FS-035

What privacy-preserving aggregate evacuation-assistance demand has been authorised for today, and are approved resources and roles aligned?

SENSORS AND TELEMETRY

Proposed evidence. Authorised, privacy-preserving aggregate counts or demand bands supplied by the PEEP/emergency-planning owner; role rosters; approved equipment availability; refuge/lift dependency status. No extraction or generation of individual plans. Coverage and timing. Only the time, zones and assistance categories explicitly released in aggregate by the authorised owner; absence of data is not treated as no need. Event-driven when authorised aggregates, rosters or resource states change; show effective interval and retrieved time. No continuous personal sensing. Quality and limits. Retain source signature/version, purpose and permission, zone granularity, source clock/UTC offset, expiry, access log and transformation provenance. Calibration is not applicable. Flag suppression-rule changes, small-cell disclosure risk and missing effective periods; never derive demand from identity, disability or behaviour data. Aggregate demand may omit late changes and cannot specify individual needs, compatibility, consent, assistance method or whether a person is present.

ANALYSIS REQUIRED

Operation and interval. Lookup and aggregate resource comparison; recommendation limited to authorised owner decision. No estimation, prediction or inference of personal need. At daily or event-specific planning and after any authorised update; not as a live substitute for personal procedures. Minimum evidence and method. A currently authorised aggregate with scope/expiry, current role/resource/dependency status, owner-approved matching rules and valid access permission. Compare demand bands with approved resource categories by zone/time, report shortages as ranges, and preserve privacy suppression. Uncertainty and safe failure. Late updates, unknown current presence, resource compatibility, role availability, dependency state and deliberate aggregation. Return 'authorised aggregate insufficient for resource check' and direct the designated owner to assessment; never infer missing demand.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The designated PEEP/inclusive-evacuation owner controls personal plans and any aggregate release; the Responsible Person/fire-safety lead owns resource planning; HR/student/support services may own source records under strict permissions. Current authorised aggregate and emergency plan first; then current role/equipment/dependency records. Sensor or occupancy data must never infer assistance need. Times, access and provenance. Apply strict role-based access, purpose limitation, minimum cell sizes and suppression. Personal plans and identities are never generated, summarised or exposed by the catalogue. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Stale at the aggregate's explicit expiry, after a known plan/resource/occupancy change, or when permission is withdrawn. Conflict. If authorised aggregates and resource records disagree, show the mismatch without revealing individuals and refer to the designated owner. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not create, expose, summarise or modify any personal emergency evacuation plan; do not infer disability, health, identity, location or assistance requirement. Do not reveal small-cell aggregates, names or plan details, even to otherwise authorised building users without the specific emergency-planning permission. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This enables resource and role checks while preserving personal-plan confidentiality and keeping the accountable owner in control of inclusive evacuation arrangements.

FIRE-SAFETY PERSONNEL QUESTION

FS-036

Which approved evacuation chairs or other assistance devices need attention because presence, access, service status or trained-role coverage is not demonstrated?

SENSORS AND TELEMETRY

Proposed evidence. Asset-tag inspection scans, optional cabinet/tamper state, manufacturer-supported battery/charge status for powered devices, access/obstruction observations, maintenance records and aggregate trained-role rosters. Coverage and timing. Only approved assistance devices in the current register and their storage/access points; the service does not associate devices with individuals or personal plans. Inspection/service events at source time; optional cabinet/battery states at approved intervals; role coverage by effective shift/event interval. Quality and limits. Retain tag history, device serial/version, battery-monitor calibration/functional check where applicable, source clock/UTC offset, uptime, gaps and provenance. Validate location after moves and compare charge indicators with competent tests; flag stale cabinet states. Do not infer condition between inspections. Presence or charge does not prove serviceability, suitability, access, correct setup, trained operation or compatibility with any person or route.

ANALYSIS REQUIRED

Operation and interval. Lookup, observation and comparison of asset, access, service and aggregate role evidence; no suitability recommendation for an individual. Daily exception assessment, before events/out-of-hours use, and after relocation or maintenance. Minimum evidence and method. Current asset/location register, latest competent service/inspection, access observation, current defect state and authorised aggregate trained-role coverage. Check each device's presence, access, service and role dependency separately; flag any missing element rather than producing a single readiness score. Uncertainty and safe failure. Unrecorded relocation, cabinet sensor error, battery indicator limits, access obstruction, role absence and changes after inspection. Return 'assistance-device provision not demonstrated' and request owner-led inspection or roster confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The inclusive-evacuation/equipment owner owns approved devices and use procedures; the competent service provider owns tests; the fire-safety lead owns operational planning; training owner holds role-level competence. Current defect/withdrawal and competent service result first; then register/inspection; then optional telemetry. Personal plans do not flow into this answer. Times, access and provenance. Only asset-level and aggregate role status; no names unless a separate authorised operational roster process requires them. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live optional status stale after 24 hours; service/training at due date; location immediately stale after a reported move. Conflict. A competent defect or missing physical check overrides a normal presence/charge signal. Conflicting locations require attendance. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not assign a device to a person, infer personal need, judge suitability, train users, or expose personal evacuation information. Do not disclose named competence or individual schedules beyond authorised roster management. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports targeted equipment, access, servicing and role checks without linking assistance provision to identifiable people.

Occupancy, visitors and inclusive evacuation information

FS-037

FIRE-SAFETY PERSONNEL QUESTION

Which zones depend on visual, voice, tactile or other inclusive alarm modes, and where is current technical or sampled-test evidence incomplete?

SENSORS AND TELEMETRY

Proposed evidence. Current inclusive-alarm design/zone records, panel/circuit supervision, sampled competent tests of visual/voice/tactile devices, approved portable-device inventory and role-level deployment records. No inference from occupant characteristics. Coverage and timing. Only provisions explicitly recorded for current zones and use conditions; unrecorded personal arrangements or device assignments are excluded. Technical states on change; tests and deployment events at source time; aggregate to provision/evidence status by zone and mode. Quality and limits. Retain device/circuit functional checks, meter calibration where used, configuration, source clock/UTC offset, uptime, gaps and provenance. Revalidate after layout, use, alarm or device changes; flag stale portable-device assignments. Do not extrapolate sample results to untested zones or modes. Circuit health does not prove perception or understanding; portable-device presence does not prove receipt, charge, assignment or suitability; individual needs cannot be inferred.

ANALYSIS REQUIRED

Operation and interval. Lookup and evidence-completeness comparison by notification mode; no estimate of individual perception or recommendation for a person. At daily/event planning, after changes and in scheduled assurance checks. Minimum evidence and method. Current zone/use record, approved provision by mode, current technical/test evidence, device deployment status where applicable and valid permission. Map provision dependencies, distinguish design record, technical availability and sampled effectiveness, and report unsupported zones/modes. Uncertainty and safe failure. Unrecorded use changes, sample limits, battery state, personal arrangements, clock gaps and incomplete zone mapping. Return 'inclusive notification evidence incomplete' and require owner/competent assessment.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety and inclusive-evacuation owners own the approved provision; alarm/device owners and competent maintainers own technical evidence; designated support services control personal arrangements. Current approved plan and signed impairment/test results first; then system/asset records; then healthy telemetry. Personal-plan data is not used. Times, access and provenance. Zone-level provision only under role-based access. Never reveal identities, protected characteristics, diagnoses or personal-plan details. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Technical state uses source interval; test evidence stale after relevant change/due date; deployment records stale at the end of the approved period. Conflict. Any failed test, withdrawn device or owner-reported gap overrides normal circuit status. Conflicts go to the inclusive-evacuation owner. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer disability or need, expose personal plans, promise that anyone will perceive or understand an alarm, or prescribe an individual solution. Do not disclose sensitive personal or small-group information. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the inclusive-evacuation owner a zone-by-mode evidence gap list for testing, temporary provision or plan update without profiling occupants.

FS-038

FIRE-SAFETY PERSONNEL QUESTION

Where could aggregate refuge demand exceed the evidenced communication, space, trained-role or evacuation-lift provision?

SENSORS AND TELEMETRY

Proposed evidence. Only authorised aggregate assistance-demand bands; current refuge/communication and evacuation-lift dependency status; owner-approved refuge capacity/provision records; aggregate support-role rosters and planned occupancy. Coverage and timing. Only zones/refuges covered by current approved planning records and privacy-preserving aggregates; no individual allocation, route or plan is processed. Inputs by effective interval; technical states at source interval; forecast at 15-30 minute bands for planned use and refreshed after changes. Quality and limits. Retain source permissions, versions, clocks/UTC offsets, uptime/gaps for technical feeds and full provenance. Calibration applies to any occupancy counters, not personal data. Flag expired aggregates, small-cell risks and changed refuge/lift topology. Do not infer demand from occupancy demographics. Aggregate demand and nominal provision do not prove fit, accessibility, staffing, communication quality, lift availability in an incident or suitability for any person.

ANALYSIS REQUIRED

Operation and interval. Forecast and scenario comparison of aggregate demand against owner-recorded provision; no individual recommendation, capacity certification or live incident allocation. Before events/out-of-hours periods and when a relevant input changes. Minimum evidence and method. Valid authorised demand band, current refuge/provision records, technical dependency states, aggregate role coverage and owner-approved comparison rules. Compare lower/upper demand bands with each independent provision dependency and identify single points or possible shortfall scenarios. Uncertainty and safe failure. Privacy aggregation, unknown presence, role availability, compatibility, communication performance, lift dependency and forecast error. Return 'aggregate provision comparison not supportable' and require the designated owner to assessment.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The PEEP/inclusive-evacuation owner controls demand aggregates; the Responsible Person/fire-safety lead owns planning criteria; refuge, communication, lift and training owners hold provision evidence. Current emergency plan and authorised aggregate first; then current signed impairments and competent tests; then healthy technical states and planned occupancy. Times, access and provenance. Strict least privilege, minimum cell size and suppression; no individual plan, identity, disability or location. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Forecast stale after 15 minutes or any occupancy, role, device, refuge, lift or permission change. Conflict. Show ranges and source conflicts; never reduce a shortage by assuming missing demand is zero or by assigning people to resources. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not expose or generate personal plans, allocate assistance, infer need, certify refuge capacity or direct live evacuation. Do not reveal small-cell aggregates or named role availability outside authorised processes. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports early owner decision of a possible resource or dependency shortfall while preserving the confidentiality and individuality of evacuation arrangements.

FS-039

FIRE-SAFETY PERSONNEL QUESTION

What aggregate out-of-hours occupancy is evidenced, and where do lone-worker, visitor or contractor accountability arrangements have unresolved gaps?

SENSORS AND TELEMETRY

Proposed evidence. Anonymous entrance/exit counters, permitted aggregate access status, out-of-hours booking/permit records, visitor/contractor registers and authorised check-in/welfare-process status. No continuous location tracking. Coverage and timing. Only defined out-of-hours site/zone boundaries and registered processes; people who bypass or are outside those systems are an explicit unknown. Aggregate counts every 5-15 minutes; access/permit/check-in events at source time; present ranges and unmatched process records, not names. Quality and limits. Retain counter calibration/direction checks, source clocks/UTC offsets, uptime, gaps, reset history and provenance. Validate access-system timezone and duplicate events; reconcile only at verified empty periods. Do not infer departure from inactivity. Aggregate occupancy is not accountability or exact location; access and check-in records may be missed, shared or stale; lone-worker safety cannot be inferred.

ANALYSIS REQUIRED

Operation and interval. Observation, lookup and aggregate comparison; no identity/location inference or welfare diagnosis. At out-of-hours handover, every 15 minutes where justified, and when a permit/check-in becomes overdue. Minimum evidence and method. Defined boundary, healthy aggregate counts or known coverage, current permits/registrations/check-ins, source health and current time. Produce an occupancy range, list overdue/unmatched process records by owner and show system gaps separately. Uncertainty and safe failure. Unmonitored access, tailgating, missed checkout, clock error, internal movement, counter drift and informal presence. Return 'out-of-hours occupancy/accountability not established' and require Security or work owners to check under existing procedures.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Security owns access and out-of-hours control; permit/visitor/work owners own registered people; HR/safety process owners control lone-worker procedures; the fire-safety lead owns aggregate planning use. Fire-service command and attended accountability first; then current check-in/permit/visitor records; aggregate counters corroborate totals only. Times, access and provenance. Use aggregate bands and process-gap counts. Named data remains within authorised security/emergency processes. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live estimate stale after 15 minutes or a counter/system outage; check-in/permit status follows its declared expiry. Conflict. Conflicting totals widen the range and trigger local/process-owner confirmation; do not construct a person list from combined data. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not identify, track or contact individuals, infer lone-worker risk, perform welfare monitoring, or treat the result as a roll call. Do not expose names, access histories, phone numbers or personal schedules. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This lets fire-safety staff decide whether additional authorised coverage, a local sweep or process follow-up is needed for out-of-hours operation.

FS-040

FIRE-SAFETY PERSONNEL QUESTION

Can today's occupancy and inclusive-evacuation briefing be supported, or which exact input needs owner-led confirmation before it is used?

SENSORS AND TELEMETRY

Proposed evidence. No new sensing. Query only the permitted aggregate sources relevant to the defined decision: counters, schedules, visitor/permit status, authorised assistance aggregates, role rosters and dependency states. Coverage and timing. Only the specified zones, time window and decision; every source's spatial coverage and excluded population is displayed. Use each source's native observation/effective time and quality metadata; aggregate to a claim-by-claim quality table, not a blended confidence score. Quality and limits. Require calibration/functional status where applicable, current mapping, source clocks/UTC offsets, uptime, gaps, reset history, permission and provenance. Carry forward drift flags and missing intervals from source systems. No imputation or inference from identity/proxy variables. A data-quality classification does not establish actual occupancy, need, accountability or safe provision; multiple weak sources do not become reliable through combination.

ANALYSIS REQUIRED

Operation and interval. Evidence-quality lookup and comparison. Observation, estimate and forecast labels are preserved; no diagnosis or decision recommendation beyond obtaining evidence. On demand before each material decision and automatically when a critical source changes. Minimum evidence and method. A precisely stated decision, required inputs, current source metadata, permissions, owner criteria and conflict rules. Evaluate spatial coverage, temporal validity, calibration/health, provenance and conflict for each input; classify supported, qualified or unavailable. Uncertainty and safe failure. Unknown populations, privacy suppression, mapping, clock offset, sensor drift, administrative delay and source conflicts. Return 'decision evidence insufficient', name the exact input and responsible owner, and avoid producing the substantive decision answer.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Each data owner retains authority over meaning; the Responsible Person/fire-safety and inclusive-evacuation owners define the decision and minimum evidence; privacy owners control access. Current command/emergency plan and authorised owner records first; then direct aggregate observations; then schedules/forecasts. Personal-plan data remains outside the catalogue. Times, access and provenance. Apply least privilege, data minimisation and suppression; expose only the quality and aggregate information needed. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Use source-specific thresholds; if the decision time exceeds any critical source's validity, classify the claim as unavailable. Conflict. Retain conflicts and show owner/source/time. Do not average, infer missing values or choose a preferred source without the authorised precedence rule. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer personal need, identity, attendance or location; do not generate personal plans or convert data quality into fire-safety approval. Do not disclose underlying personal records or small-cell aggregates. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives fire-safety personnel an honest go/no-go evidence check before using occupancy or inclusion information in a briefing or planning decision.

FS-041

FIRE-SAFETY PERSONNEL QUESTION

Which occupied zones lack confirmed fire-warden or marshal coverage for the current operating period?

SENSORS AND TELEMETRY

Proposed evidence. Authorised role rosters, shift/event assignments, training-validity status, role check-in or explicit availability confirmation, current occupancy bands and zone/opening schedules. No continuous staff location tracking. Coverage and timing. Only zones and periods covered by the current warden plan and authorised roster process; temporary absences and movement within zones remain uncertain unless explicitly confirmed. Roster and assignment by effective interval; explicit availability events at source time; occupancy/opening status every 5-15 minutes. Present role coverage by zone, not individual movement. Quality and limits. Retain roster version/signature, role identifier, source clock/UTC offset, system uptime, gaps and provenance. Calibration is not applicable except to any occupancy counters. Flag expired training, overlapping assignments, stale confirmations and clock/time-zone errors; do not infer availability from access events or calendar presence. An assignment or check-in does not prove the person is present, able to act, familiar with current conditions or suitable for every duty; occupancy may change rapidly.

ANALYSIS REQUIRED

Operation and interval. Lookup and comparison of required versus evidenced role coverage; recommendation limited to contact the responsible manager or apply the current authorised contingency. No performance diagnosis. At each shift/event handover, after a zone opens/closes, and on demand before major activity. Minimum evidence and method. A current warden plan, open/occupied zone status, authorised roster, valid role competence and recent explicit availability confirmation where required. Map required roles to zones/time, validate training and availability, and classify coverage as confirmed, fragile, missing or unknown without tracking staff. Uncertainty and safe failure. Short-notice absence, stale confirmation, double assignment, changing occupancy, privacy-limited roster detail and unknown physical location. Return 'warden coverage not confirmed' and identify the responsible roster owner; do not nominate a person automatically.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety lead owns the warden plan and minimum coverage; training/HR owners hold competence records; line/event managers own assignments; Security/Space owners hold operating status. Current incident command and emergency plan first; then authorised live roster/availability and current role restrictions; then schedule and aggregate occupancy context. Times, access and provenance. Show named details only to authorised operational coordinators; routine output should use role/zone status and avoid personal schedules. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Availability stale at shift/event change, explicit withdrawal or owner-set check-in interval; training stale at expiry or role change. Conflict. Conflicting roster, training or availability records are shown separately and the zone remains unconfirmed until the role owner resolves them. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not track staff, infer attendance from access, appoint or reassign wardens, evaluate individual performance or issue emergency commands. Do not expose personal schedules, absence reasons or training detail beyond authorised need. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This lets fire-safety personnel resolve a genuine coverage gap before opening, an event or an out-of-hours period rather than discovering it during a drill or alarm.

FS-042

FIRE-SAFETY PERSONNEL QUESTION

Where does the current warden arrangement depend on a single person, communication path, device, route or other fragile point?

SENSORS AND TELEMETRY

Proposed evidence. Current warden-role and zone plan, authorised rosters, communication/device inventory, route and refuge dependencies, equipment status and recent drill observations. No personal movement sensing. Coverage and timing. All roles and dependencies explicitly documented in the current emergency plan; undocumented informal workarounds are outside the model and must be captured by owners. Plan/roster by effective interval; device and route states at source interval; drill evidence by event. Recompute after any dependency change. Quality and limits. Retain plan/version, source clocks/UTC offsets, system health, gaps and provenance. Calibration applies only to supporting sensors. Validate dependency IDs and supersession links; flag orphaned roles/devices and stale drill evidence. Do not infer redundancy from two records that share the same underlying dependency. Recorded redundancy may fail for common causes, simultaneous absence, inaccessible equipment or changed conditions; routine data cannot predict emergency behaviour.

ANALYSIS REQUIRED

Operation and interval. Dependency-graph lookup and comparison; calculation of single points and common dependencies; recommendation only for owner decision, not plan approval. When the plan is checked, before major events/out-of-hours operation and after any impairment/change. Minimum evidence and method. Current approved plan, role/dependency IDs, live availability/status where relevant, current impairments and owner-defined resilience criteria. Build a dependency graph, identify nodes whose loss removes required coverage, group common-cause dependencies and label assumptions and unverified links. Uncertainty and safe failure. Undocumented workarounds, correlated absences, hidden common infrastructure, changing occupancy, stale states and incomplete topology. Return 'resilience not demonstrated' for any incomplete dependency chain and request owner-led confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety lead owns the plan and resilience criteria; line/training owners own roles; Security/IT/Estates/system owners own communication, route and equipment dependencies. Current emergency plan and signed owner restrictions first; then current roster/device/route records; then drill observations and derived dependency analysis. Times, access and provenance. Role names may be visible only to authorised coordinators; wider output uses anonymised role/dependency descriptions. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale after plan, roster, route, device or communication change; drill evidence is historical and context-bound. Conflict. Conflicting dependency records are retained and treated as a fragility until the owner resolves them; analytics cannot declare redundancy. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not redesign or approve the emergency plan, appoint personnel, infer individual reliability or issue operational instructions. Do not expose personal availability, secure communication details or restricted route topology outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the Responsible Person a precise list of single points to address through authorised backup roles, devices, routes or procedures.

FS-043

FIRE-SAFETY PERSONNEL QUESTION

Which current fire-safety duties are assigned to people whose training, competence, induction or local briefing evidence is missing or out of date?

SENSORS AND TELEMETRY

Proposed evidence. Authorised training/competence records, role assignments, local induction and change-briefing acknowledgements, qualification/certificate references and current duty period. No biometric, productivity or behavioural monitoring. Coverage and timing. Only fire-safety duties and competence requirements defined by the responsible owner; external contractor evidence is included only where permission and verification exist. Event-driven on training, assignment, expiry, induction and briefing; evaluate at current duty effective time and upcoming 30/90-day windows. Quality and limits. Retain issuer, certificate/reference, verification status, source clock/UTC offset, version, expiry, access decision and provenance. Calibration is not applicable. Flag duplicate identities, mismatched role codes, revoked credentials, stale imported records and missing briefing versions; do not infer competence from job title or past assignment. A valid record does not prove current practical competence, physical presence, language understanding or performance; a missing record may reflect an integration gap.

ANALYSIS REQUIRED

Operation and interval. Lookup and exact requirement comparison; recommendation to verify, brief, reassign through authority or train. No assessment of personal competence. At each assignment/shift, before high-risk work/events and in 30/90-day planning. Minimum evidence and method. A current duty requirement, authorised assignment, verified person/role link, valid evidence with scope/expiry, and latest required local briefing. Compare each assigned duty with exact competence and briefing requirements, distinguish missing, expired, unverified and out-of-scope evidence. Uncertainty and safe failure. Record latency, identity matching, qualification scope, external verification, role changes and practical competence not represented. Return 'duty competence evidence not confirmed' and direct the authorised manager to verify or apply existing contingency.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The training/HR/contract owner owns records; the Responsible Person/fire-safety lead owns duty requirements and appointment; line/project managers own assignments; competent bodies own qualification validity. Current owner appointment/restriction and verified competence evidence first; then training systems; self-declaration or calendar data is insufficient. Times, access and provenance. Least-privilege access; routine output should show duty-level gap and responsible manager, not unnecessary personal details or reasons. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Stale at expiry, revocation, role/site change or when a new briefing/version becomes effective. Conflict. Any identity, course, scope or expiry conflict remains unresolved and the duty is unconfirmed; do not assume equivalence between courses. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not appoint, remove or judge a person, infer ability or protected characteristics, or disclose disciplinary/performance information. Do not expose personal training history beyond the authorised duty decision. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This lets managers correct a genuine assignment gap before a drill, event, test or incident rather than relying on a role title or stale certificate.

FS-044

FIRE-SAFETY PERSONNEL QUESTION

Which zones, operating periods, populations or critical procedures were not meaningfully exercised by the latest relevant fire drill?

SENSORS AND TELEMETRY

Proposed evidence. Approved drill plan and scope, alarm/panel event logs, anonymous zone/doorway counts, observer checklists, role participation records at aggregate level, refuge/lift/communication test records and authorised debrief evidence. Coverage and timing. Only the planned and actually observed drill areas, times, populations and procedures; excluded, simulated and unobserved elements are explicitly identified. Event logs at native resolution; aggregate movement in 5-15 second or one-minute bands; observations at source time; retain the drill scenario and configuration. Quality and limits. Retain counter calibration, clock/UTC offsets across alarm and observations, uptime, gaps, device/zone mapping and provenance. Validate clock offset before the drill and compare manual counts. Flag sensor relocation and missing observers; never infer participation from access or routine occupancy data. A drill is not a real fire; attendance, behaviour, staffing, doors and weather may differ; observed participation does not prove understanding or future performance.

ANALYSIS REQUIRED

Operation and interval. Lookup, coverage calculation and comparison with drill objectives; no causal diagnosis or prediction of incident performance. After each drill and during annual/programme planning. Minimum evidence and method. Approved objectives, scenario, included/excluded scope, current zone/procedure map, aligned logs, observer evidence and aggregate participation. Create an objective-to-evidence matrix, classify exercised, partially exercised, simulated, excluded or unobserved, and show gaps by zone/time/procedure. Uncertainty and safe failure. Clock alignment, observer coverage, counter drift, unknown participation, scenario realism and changed conditions. Return 'drill coverage not demonstrated' for any objective without direct evidence and require owner decision or future exercise planning.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety lead owns drill objectives and acceptance; Space/Event/line managers own affected populations; system owners and observers own evidence; privacy owner controls use. Signed drill plan/debrief and competent observations first; then complete system/aggregate telemetry; schedules or inferred presence are context only. Times, access and provenance. Use aggregate participation and role-level records; no individual behaviour scoring, tracking or exposure of personal evacuation arrangements. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Evidence applies only to the drill's date, scenario and configuration; stale after material plan, route, system or population change. Conflict. Conflicting attendance, observation or log evidence is shown as uncertainty; a scheduled zone is not counted as exercised without evidence. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not score individuals, expose personal plans, claim compliance or predict that people/systems will perform the same in an incident. Do not disclose identities, health/disability information or raw movement traces. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the fire-safety lead a defensible next-drill agenda instead of treating attendance or an alarm activation as proof that all critical procedures were tested.

FS-045

FIRE-SAFETY PERSONNEL QUESTION

What evidenced intervals were observed from alarm initiation through movement, final-exit flow and assembly stabilisation in the latest drill?

SENSORS AND TELEMETRY

Proposed evidence. Fire-alarm initiation/transmission events, anonymous doorway/zone counters, final-exit counts, observer timestamps and an authorised declaration of assembly stabilisation. No identity tracking or continuous video. Coverage and timing. Only instrumented/observed stages in the drill; unmonitored zones, personal assistance procedures and public-realm movement are explicit gaps. Native event times; movement counts at 1-5 seconds reduced to cumulative curves; observer events at source time. Store raw timestamps and calculated intervals separately. Quality and limits. Before the drill, measure source clock offsets to UTC and logging resolution; retain counter calibration/direction checks, uptime, gaps and provenance. Flag clock drift, resets and duplicate events. Do not calculate an interval across a missing stage or correct clocks without documented offsets. Intervals depend on scenario, participants, route choices, observer definitions and equipment; assembly 'stabilisation' is an operational observation, not accountability or completion.

ANALYSIS REQUIRED

Operation and interval. Observation and calculation of elapsed intervals; comparison with prior drills only under matched definitions. No evacuation-time certification or causal diagnosis. After each drill and during debrief; not for live command. Minimum evidence and method. A defined start/end event for each stage, measured clock offsets/resolution, complete logs, known coverage and observer confirmation. Align by documented offsets, calculate intervals with uncertainty bounds, and separate alarm, recognition/movement, exit-flow and assembly-observation stages. Uncertainty and safe failure. Clock offset/drift, event-definition ambiguity, missing zones, counter error, participant mix, route conditions and observer reaction time. Return 'interval not calculable' for any stage lacking aligned endpoints and preserve the raw evidence for assessment.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead owns interval definitions and drill interpretation; alarm/system owners and observers own source events; Space/Security owners hold aggregate counts. Signed drill record/observer declaration first; then aligned system/counter logs; derived intervals are lowest-precedence analysis. Times, access and provenance. Use aggregate times only; no individual journey times, identities or movement traces. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Valid only for the stated drill/scenario and until evidence is corrected; not a live performance forecast. Conflict. Where timestamps disagree beyond measured uncertainty, report an interval range or mark the stage unavailable; do not force sequence order. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not claim total evacuation time, individual performance, legal adequacy, accountability or expected incident performance. Do not expose individual movement or named delays. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports a technically defensible debrief and shows whether delay evidence lies in alarm/system timing, observed movement, exit flow or the assembly process.

FS-046

FIRE-SAFETY PERSONNEL QUESTION

Which specific doors, junctions or route segments showed repeatable congestion, counterflow, hesitation or delayed release during the latest drill?

SENSORS AND TELEMETRY

Proposed evidence. Temporary or permanent anonymous time-of-flight/thermal counters at approved route points, door/release event logs, aggregate flow curves and observer checklists. Use no facial recognition, device tracking or general CCTV analytics. Coverage and timing. Only monitored drill pinch points, doors and observer zones; full-route and individual behaviour remain unobserved. Samples at 1-5 seconds during the drill, reduced to flow, occupancy band and persistent queue/hesitation indicators; retain aligned door events and observer times. Quality and limits. Retain sensor calibration/geometry, direction validation, source clocks/UTC offsets, uptime, gaps and provenance. Measure offsets before the drill; compare counts with manual samples and flag drift/occlusion. Do not classify congestion through a gap longer than 10 seconds. Depth/thermal data can confuse groups, mobility aids and direction; a pause may reflect instruction, door use or normal behaviour; no causal interpretation is automatic.

ANALYSIS REQUIRED

Operation and interval. Observation, calculation and comparison of aggregate movement indicators; recommendation to inspect/test/assessment. No causal diagnosis or capacity claim. After the drill, with preliminary indicators available for debrief and confirmed analysis later. Minimum evidence and method. Current route/door map, calibrated sensors or observer evidence, aligned timestamps, complete drill interval and defined indicator thresholds. Detect persistent low flow with growing local count, opposite-direction crossings, repeated door-state delays and observer-marked hesitation, then rank for human assessment. Uncertainty and safe failure. Occlusion, counter drift, clock error, route-choice variation, participant mix, observer bias and non-emergency realism. Return 'movement issue not established' where evidence is incomplete; require attended assessment of the location and drill context.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead owns drill objectives and assessment; Estates/Security own door/release records; observers and privacy owner govern temporary sensing; competent assessors own any formal movement analysis. Competent observer/debrief findings first; then complete aligned aggregate telemetry. A model indicator is only a prompt. Times, access and provenance. Keep only aggregate derived metrics and short-lived raw sensor data if authorised; no identities or individual trajectories. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Valid only for the drill configuration and monitored conditions; stale after route/door/layout change. Conflict. Where telemetry and observers disagree, retain both and mark the location for assessment; do not label participant behaviour. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not identify or score participants, diagnose behaviour, certify capacity, or use drill analytics to direct a live evacuation. Do not retain raw identifiable imagery, audio or device traces. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This focuses debrief and physical checks on specific doors or pinch points while avoiding unsupported claims about why people moved as they did.

FS-047

FIRE-SAFETY PERSONNEL QUESTION

Which alarm zones or message modes lack separate evidence of technical delivery and participant understanding under the tested conditions?

SENSORS AND TELEMETRY

Proposed evidence. Panel/circuit supervision, calibrated sampled sound/visual tests, voice-alarm message delivery logs, observer checklists and anonymous post-drill feedback designed for comprehension. No continuous microphones or speech recording. Coverage and timing. Only zones, positions, conditions, languages/modes and participant groups actually sampled; untested spaces or needs remain unverified. Technical events at native resolution; measurements and observations at source time; anonymous feedback within the defined post-drill window and aggregated above privacy thresholds. Quality and limits. Retain meter calibration and test position, message/version, source clocks/UTC offsets, uptime, gaps and provenance. Validate survey version and minimum response count. Recheck after acoustic/layout/system change; do not extrapolate between positions or treat non-response as understanding. Signal delivery does not prove perception or comprehension; feedback may be biased or sparse; tested conditions may not match normal noise, doors or future participants.

ANALYSIS REQUIRED

Operation and interval. Observation and comparison of delivery, sampled perception and aggregate comprehension evidence; no individual assessment or guarantee. After each relevant drill/test and after message/system changes. Minimum evidence and method. Current approved message/mode, documented test conditions/positions, calibrated measurements, complete delivery logs, adequate anonymous feedback and owner criteria. Separate command/delivery, sampled audibility/visibility and reported understanding; report coverage and uncertainty for each. Uncertainty and safe failure. Calibration, background noise, sample location, response bias, language/context, clock alignment and limited group coverage. Return 'delivery/understanding not demonstrated for this condition' and require competent retest or communication assessment.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The alarm/voice-alarm owner owns technical delivery; the fire-safety lead owns messages and drill objectives; competent testers own measurements; Communications/inclusion owners govern wording and feedback. Approved message and competent test/debrief first; then system logs and aggregate feedback. Circuit supervision alone cannot establish understanding. Times, access and provenance. Use anonymous aggregated feedback; do not record speech, identify respondents or expose protected characteristics. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Evidence valid only for the tested message, configuration and conditions; feedback expires when wording, system or population context changes. Conflict. Conflicting technical, observer and feedback evidence remains qualified and triggers targeted retest; do not choose the most positive source. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not promise any person will perceive or understand a message, generate emergency wording, record conversations or infer personal capability. Do not expose respondent identity, health/disability data or small-cell feedback. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports targeted technical testing or message redesign without treating a delivered circuit event as proof of human understanding.

FS-048

FIRE-SAFETY PERSONNEL QUESTION

Which refuge, assisted-evacuation, evacuation-lift or inclusive communication procedures still lack a meaningful exercise?

SENSORS AND TELEMETRY

Proposed evidence. Approved drill objectives, aggregate role participation, refuge call and lift test logs, device/equipment test records, observer checklists and privacy-preserving scenario/debrief records. No personal-plan content. Coverage and timing. Only procedures explicitly authorised for the drill; simulated, excluded and non-participating personal arrangements are clearly separated. Events at native/system time; observations and role participation at source time; aggregate to procedure stages and evidence status. Quality and limits. Retain drill plan/version, source clocks/UTC offsets, system health, gaps and provenance. Validate clock offsets for integrated tests; verify that scenario labels match the observed evidence. Calibration applies to any test instruments. Do not infer an exercised step from mere device presence or role attendance. A drill may simulate rather than perform steps; participation does not prove competence, suitability or future response; personal arrangements remain individual and confidential.

ANALYSIS REQUIRED

Operation and interval. Coverage lookup and event-sequence comparison; no individual suitability judgement or incident-performance prediction. After each drill and during future exercise planning. Minimum evidence and method. Approved objectives, explicit simulation rules, current procedure stages, aligned system logs, competent observations and aggregate role participation. Map each procedure stage to direct evidence, distinguish performed, simulated, discussed, excluded and unobserved, and record dependencies not tested. Uncertainty and safe failure. Observer coverage, privacy limits, simulation realism, clock alignment, role absence and changed conditions. Return 'procedure not meaningfully exercised' where direct evidence is absent and require owner-led planning.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety and inclusive-evacuation owners own procedure and drill scope; system/equipment owners and competent observers own evidence; the PEEP owner controls personal information. Signed drill scope/debrief and competent observations first; then complete system logs; aggregate participation is corroboration only. Times, access and provenance. No personal plans, identities, diagnoses or individual performance. Use procedure-level and role-level evidence with privacy suppression. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Valid only for the drill scenario/configuration and stale after procedure, equipment, route, role or system change. Conflict. If the plan says a step was included but no direct evidence exists, classify it unobserved. Conflicts go to the drill owner. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not expose personal emergency evacuation plans, assign assistance, judge individuals or present the drill as approval of the procedure. Do not reveal identities or sensitive assistance information. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This creates a safe, specific future-drill agenda for inclusive and assisted procedures without turning personal arrangements into catalogue data.

FS-049

FIRE-SAFETY PERSONNEL QUESTION

Which actions from drills, tests or briefings are overdue, blocked or still unsupported by competent completion evidence?

SENSORS AND TELEMETRY

Proposed evidence. Action-management records, linked defects/work orders, owner decisions, due dates, dependencies, restricted completion evidence and current asset/system state where relevant. No productivity monitoring. Coverage and timing. All fire-safety actions accepted into the current action register, linked to the originating drill/test/briefing and affected asset, zone or procedure. Event-driven on assignment, status, evidence upload, due-date change and acceptance; daily and 7/30/90-day summaries. Quality and limits. Retain signed action version, owner, effective dates, source clock/UTC offset, file hashes, supersession, access and provenance. Calibration applies only to supporting measurements. Flag status reversals, closed actions without acceptance, broken links and long evidence gaps; do not infer completion from inactivity or a closed work order. A status label does not prove implementation or effectiveness; delays may have valid authorised reasons; recurrence does not identify individual fault.

ANALYSIS REQUIRED

Operation and interval. Lookup, evidence-completeness comparison and due-date calculation; recommendation to escalate under current governance. No blame or causal diagnosis. Daily, at governance meetings and before related impairments/work are closed. Minimum evidence and method. Current action record, origin, accountable owner, due/approved extension, required evidence, linked asset/procedure and explicit competent acceptance where closure is claimed. Calculate due status, validate evidence against closure criteria, identify blocked dependencies and separate implementation from acceptance/effectiveness. Uncertainty and safe failure. Record latency, unclear criteria, changed scope, inaccessible evidence, dependency status and unobserved effectiveness. Return 'action completion not demonstrated' and name the missing acceptance/evidence or blocking owner.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead/Responsible Person owns action acceptance and closure criteria; action owners implement; competent inspectors/system owners provide evidence; governance owner controls escalation. Current signed acceptance/rejection and impairment decision first; then competent evidence; then work-order status and telemetry. 'Closed' in a task system cannot overrule missing acceptance. Times, access and provenance. Role-based access; do not expose personal performance, disciplinary context or unnecessary contractor detail. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Stale immediately after scope, due date, asset condition or acceptance criteria change; open actions refreshed daily. Conflict. Retain conflicting status and evidence; classify as not accepted until the accountable owner resolves it. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not close, approve, reassign or judge an action owner, infer negligence, or treat a task-system status as fire-safety assurance. Do not disclose personal performance or sensitive incident details beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This lets governance focus on actions that are genuinely overdue or evidentially incomplete rather than those merely labelled closed.

FS-050

FIRE-SAFETY PERSONNEL QUESTION

Which emergency contacts, radios, public-address or approved message routes are unavailable, untested or dependent on a single point of failure?

SENSORS AND TELEMETRY

Proposed evidence. Approved contact directory versions, role/on-call availability, radio network/repeater health, handset battery/functional-test records, public-address supervision, messaging-platform service state and authorised test acknowledgements. No message-content inspection. Coverage and timing. Only communication routes in the current emergency plan and authorised contact process; personal devices and external networks beyond contracted status are explicit dependencies. Technical health at approved heartbeat intervals; role/contact availability by effective period; tests at source event time; aggregate to route/dependency status. Quality and limits. Retain directory version/owner, device functional checks, battery test, source clocks/UTC offsets, uptime, gaps and provenance. Measure cross-system clock offsets during tests; flag stale contacts, shared infrastructure and repeated missed acknowledgements. Calibration applies only to test instruments. Do not infer availability from recent normal use. A heartbeat or delivered test does not prove receipt, comprehension, network performance under incident load or a person's availability.

ANALYSIS REQUIRED

Operation and interval. Dependency lookup, observation and comparison; calculation of single points; no guarantee of delivery or recommendation to send an emergency message. At handover, after changes/outages and during scheduled communication tests. Minimum evidence and method. Current emergency-plan route list, current directory/roster, live service health, valid functional-test evidence, and known shared dependencies. Map source-to-recipient/channel dependencies, verify technical and organisational stages, and flag routes with no independent alternative. Uncertainty and safe failure. External network status, battery condition, human availability, clock alignment, message-load effects and hidden shared infrastructure. Return 'communication route not demonstrated' and direct authorised testing or owner confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/fire-safety lead owns approved communication routes and messages; Security/IT/telecom/system owners own technical services; line managers own on-call roles; Communications owns public wording. Fire-service command/current emergency plan first; then current owner-approved directory and signed test/impairment; then technical health and historical delivery evidence. Times, access and provenance. Contact details, radio topology and message logs are tightly role-restricted; catalogue answers should use role/channel status rather than personal details. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Technical state stale after one missed heartbeat; contact availability at roster change; test evidence only as dated and until relevant change. Conflict. A failed test or known outage overrides normal health. Unacknowledged tests remain unproved; do not infer why. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not send messages, call emergency services, record content, infer individual availability or expose contact details. Do not reveal phone numbers, radio channels, credentials or message content outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports targeted testing, battery/device replacement, contact correction or authorised contingency planning before an emergency communication route is assumed available.

FS-051

FIRE-SAFETY PERSONNEL QUESTION

Which fire-safety inspections, tests and maintenance tasks fall due within 7, 30 or 90 days, and which are already overdue?

SENSORS AND TELEMETRY

Proposed evidence. Current asset/task registers, planned-maintenance system, competent test certificates, work orders, impairment records and asset telemetry only where it changes the authorised task trigger. No sensing is required to calculate calendar due dates. Coverage and timing. All in-scope fire-safety assets and procedures with a current owner, task rule, last accepted completion and location/asset ID; missing-register items are reported separately. Event-driven on task, certificate, defect or asset change; compute daily 7/30/90-day windows using effective dates and timezone. Quality and limits. Retain schedule version, signed completion/acceptance, source clock/UTC offset, work-order history, asset mapping and provenance. Calibration/functional health is inherited from any telemetry used as a condition trigger. Flag duplicate tasks, date-format/time-zone errors, suspended assets and missing periods; do not infer completion from an auto-closed job. Due-date calculation does not prove work scope, competence, physical completion, asset condition or whether the stated interval is appropriate.

ANALYSIS REQUIRED

Operation and interval. Lookup and calendar calculation; comparison of due, completed and accepted evidence; recommendation to schedule or escalate, not to certify. Daily and at maintenance/governance planning. Minimum evidence and method. A current asset/task ID, owner-approved interval/trigger, last accepted completion, effective dates, current asset status and unresolved defect/impairment. Calculate next due date from the accepted rule, separate due from booked and completed-from-accepted, and group by 7/30/90-day window and dependency. Uncertainty and safe failure. Incorrect task rule, missing acceptance, asset replacement, suspended service, timezone and record latency. Return 'maintenance due status not established' and identify the missing owner rule or accepted completion.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Each asset/system owner owns the maintenance requirement and register; the competent service provider owns completion evidence; the Responsible Person/fire-safety lead owns oversight and prioritisation. Current legal/owner requirement and active impairment first; then accepted competent completion; then maintenance-system status. Telemetry cannot extend a due date without an authorised rule. Times, access and provenance. Role-based access; suppress commercial, personal and secure-location detail not needed for planning. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Recalculate daily and immediately after any task/asset/change update; source schedules are stale if owner/version is unknown. Conflict. Show conflicting due dates and requirements with owners. Use the most restrictive only when the approved precedence rule says so; otherwise escalate. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not create or alter statutory/owner intervals, approve deferral, close work, certify compliance or infer condition from due status. Do not expose personal technician data, prices or secure asset details beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives fire-safety personnel a credible forward workload and escalation list based on accepted evidence rather than optimistic job-system labels.

FS-052

FIRE-SAFETY PERSONNEL QUESTION

Which fire-door findings remain open, and which claimed close-outs still lack door-specific competent acceptance?

SENSORS AND TELEMETRY

Proposed evidence. Fire-door inspection findings, tagged photographs and measurements where authorised, work orders, remedial work records, door contacts as contextual state only, and competent reinspection/acceptance. Coverage and timing. Every finding linked to a current door/leaf ID, location, component and inspection date; unmatched legacy records are a separate data-quality issue. Event-driven at finding, remedial work, reinspection and acceptance; live contacts at source interval but never used as close-out proof. Quality and limits. Retain inspection instrument calibration where measurements are used, photo/file hashes, source clocks/UTC offsets, door/component mapping, version, uptime/gaps for telemetry and full provenance. Flag reused photographs, mismatched leaf IDs, post-remedial work changes and closed jobs without reinspection. Do not infer acceptance across related doors. A photograph, remedial work note, door contact or invoice does not prove the defect is rectified; concealed and performance conditions require competent examination.

ANALYSIS REQUIRED

Operation and interval. Lookup and evidence-completeness comparison; no diagnosis from telemetry and no automated close-out recommendation. Daily exception check and before handback, formal assurance or affected-route reliance. Minimum evidence and method. Current door/leaf record, original finding, remedial work scope, competent reinspection and explicit acceptance tied to the same component/location. Link finding-to-work-to-reinspection, verify exact ID/scope and classify open, remedied-awaiting-check, accepted or conflicting. Uncertainty and safe failure. Legacy IDs, incomplete component scope, image ambiguity, unrecorded damage after remedial work and record latency. Return 'fire-door close-out not demonstrated' and retain the finding until competent owner resolution.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-safety lead/door owner owns the register and acceptance criteria; competent inspectors own findings and reinspection; maintenance/project owners implement remedial actions. Current competent finding and explicit acceptance first; then remedial work/work-order evidence; telemetry is contextual only. Times, access and provenance. Inspection images and secure door locations are role-restricted; technician identity is minimised. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Open finding remains current until explicit accepted close-out or superseding competent inspection; any new damage makes prior close-out stale. Conflict. A work-order closure conflicting with an open finding remains open. Asset/leaf mismatches require attended reconciliation. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer door performance or close a finding from a job status, photograph, invoice or contact signal. Do not expose access histories, secure locations or personal performance data beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This stops defects disappearing between maintenance and inspection systems and gives teams a precise reinspection queue.

FS-053

FIRE-SAFETY PERSONNEL QUESTION

What did the latest alarm test actually cover across panels, zones, call points, notification circuits and interfaces, and what was not demonstrated?

SENSORS AND TELEMETRY

Proposed evidence. Approved test plan, panel/event logs, call-point activations, circuit supervision and sampled notification evidence, interface endpoint logs, observer checklist and signed test record. Coverage and timing. Only the exact devices, zones, circuits and interfaces in the current test scope; simulated, excluded or unlogged elements are separately marked. Native event timestamps during the test; observer/test records at source time; retain sequence and coverage matrix, not only a pass/fail summary. Quality and limits. Retain panel/configuration version, source clocks/UTC offsets, measured inter-system offsets, uptime, gaps, test-device IDs and provenance. Validate sequence numbers and observer coverage; calibration applies to sound/light test instruments. Do not infer an activation from a downstream event or bridge missing logs. A recorded alarm event does not prove physical operation, notification effectiveness, endpoint response or complete test coverage.

ANALYSIS REQUIRED

Operation and interval. Coverage lookup and event-sequence comparison; calculation of tested percentage is descriptive, not compliance certification. After every test and before acceptance or an assurance pack is assembled. Minimum evidence and method. Current approved scope, mapped IDs, aligned complete logs, observer evidence, endpoint confirmation and signed result. Build a scope-to-evidence matrix, verify each intended stage, and classify directly demonstrated, simulated, excluded, missing or failed. Uncertainty and safe failure. Clock resolution, configuration mismatch, logging gaps, observer coverage, sample limitations and post-test change. Return 'test coverage not demonstrated' for missing elements and require corrected evidence or retest.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire-alarm system owner owns test scope and cause-and-effect; competent maintainer/test coordinator owns execution and result; destination-system owners own endpoint evidence; the fire-safety lead owns acceptance/impairment. Signed competent test and current impairment first; then complete panel/destination logs and observer evidence. Test-management status alone is insufficient. Times, access and provenance. Detailed topology and cause-and-effect are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Valid only for the tested configuration and until relevant change/fault; missing raw logs cannot be replaced by a later summary. Conflict. Any device/scope mismatch, missing endpoint or observer disagreement remains untested/failed until competent resolution. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not acknowledge/reset the panel, initiate a test, infer physical success from logs or certify the alarm system. Do not disclose device addresses, secure topology or test credentials beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the fire-safety lead a defensible test-coverage statement and prevents a high-level pass label from hiding untested interfaces or areas.

FS-054

FIRE-SAFETY PERSONNEL QUESTION

Which emergency-lighting functional or duration tests are overdue, failed, incomplete or still missing physical confirmation?

SENSORS AND TELEMETRY

Proposed evidence. Self-test/central controller records where supported, test schedules, battery/charger data, competent functional/duration results, defect work orders and physical inspection records. Coverage and timing. Every registered circuit/luminaire and required test type, with clear distinction between self-test and attended physical evidence. Controller events at native interval; tests and inspections at source time; daily due/exception summary and event-level trace. Quality and limits. Retain controller configuration, source clock/UTC offset, uptime, gaps, battery/test instrument calibration where applicable, asset mapping and provenance. Detect missed scheduled tests, duplicate auto-results, flat-line battery values and clock changes. Do not infer a test during an outage. Self-test can miss orientation, obstruction, cleanliness, damage and actual route coverage; duration data may not represent future battery performance.

ANALYSIS REQUIRED

Operation and interval. Lookup, calendar calculation and evidence comparison; no forecast of battery duration or certification. Daily and after each test/fault. Minimum evidence and method. Current register, required test type/interval, complete controller data where applicable, competent physical evidence and defect/impairment status. Separate overdue, failed, incomplete, unobserved and accepted results for functional, duration and physical checks. Uncertainty and safe failure. Controller gaps, asset mapping, battery ageing, post-test change and physical conditions not sensed. Return 'emergency-lighting test evidence incomplete' and require owner-led test or inspection.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The emergency-lighting owner owns the register/schedule; competent maintainer owns tests and defects; Estates owns route/layout changes; fire-safety lead owns operational impairment. Current competent failed test/impairment first; then accepted test; then controller/self-test evidence. A normal status cannot close a physical defect. Times, access and provenance. Role-restricted circuit and location detail. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Due at owner-approved interval; test evidence stale after relevant change, battery/luminaire replacement or new defect. Conflict. Conflicting self-test and physical findings remain failed/unverified pending competent inspection. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not initiate tests, switch circuits, predict performance, declare routes lit or approve deferral. Do not expose secure circuit topology beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This creates an actionable test and remedial work list while making clear which self-test results still need physical confirmation.

FS-055

FIRE-SAFETY PERSONNEL QUESTION

Which portable firefighting, sprinkler, riser or specialist suppression assets need service, retest or evidence close-out?

SENSORS AND TELEMETRY

Proposed evidence. Current asset/service registers, competent certificates, defect and work-order records, optional manufacturer-supported condition telemetry, valve/pump/pressure states and attended inspection scans. Coverage and timing. All registered extinguishers, hose reels, sprinklers, risers and specialist systems, with asset-specific service rules and protected scope; unregistered discoveries are separate exceptions. Event-driven on service/test/defect; live telemetry at approved intervals; daily 7/30/90-day and open-close-out summary. Quality and limits. Retain instrument calibration/functional checks, certificate issuer/signature, asset IDs, source clock/UTC offset, uptime, gaps and provenance. Validate serial/zone matches, test-method version and supersession; flag closed work without accepted retest. Do not let normal telemetry extend a service date. Service records and telemetry do not prove design adequacy, full physical condition or future discharge; asset classes have different competent evidence requirements.

ANALYSIS REQUIRED

Operation and interval. Lookup, due-date calculation and evidence-completeness comparison; no cross-asset readiness score or certification. Daily, after service/test and before assurance or handback. Minimum evidence and method. Current asset/service rule, latest accepted competent evidence, current defect/impairment, linked remedial work and any required retest. Evaluate each asset class against its own evidence rule, distinguish service performed from accepted close-out, and group by protected dependency. Uncertainty and safe failure. Legacy IDs, certificate scope, unrecorded replacement, telemetry limitations, record delay and changed condition. Return 'service/retest status not demonstrated' and identify the competent evidence required.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Each system/equipment owner owns register and service requirement; competent service providers own tests/certificates; fire-safety lead owns impairment and assurance. Current competent defect/failed test and impairment first; then accepted service/retest; then work-order and telemetry context. Times, access and provenance. Detailed suppression topology, hazards and commercial records are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Service evidence stale at owner-set due date or after fault/change/use; live telemetry uses source-specific validity. Conflict. Any serial, location, scope or result mismatch remains open; a completed invoice/job cannot close it. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not certify availability, recommend firefighting use, move valves, start pumps or approve service deferral. Do not expose secure system topology, hazard details or technician personal data unnecessarily. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps coordinate the right specialist service and prevents assets being treated as available because a generic job or invoice is closed.

FS-056

FIRE-SAFETY PERSONNEL QUESTION

Which dampers, smoke-control systems, refuge communications, evacuation lifts or fire-service facilities lack current specialist evidence?

SENSORS AND TELEMETRY

Proposed evidence. Specialist asset registers, certificates and integrated-test records; controller/endpoint status where supported; defects, impairments, maintenance and competent observations. No generic sensor is treated as a substitute for specialist evidence. Coverage and timing. All specialist assets within current owner registers, mapped to systems, zones and dependencies; inaccessible or unmapped assets are explicit gaps. Event-driven on test, maintenance, defect and change; live states at approved intervals; daily evidence-age and dependency summary. Quality and limits. Retain certificate signature/scope, test instrument calibration, configuration, source clocks/UTC offsets, uptime, gaps, asset mapping and provenance. Validate specialist competence/source, exact asset IDs and test scenario. Flag expired certificates, missing appendices and configuration changes. Do not infer evidence across similar assets. Controller health or a certificate title may not prove full scope, physical condition, integration or current configuration.

ANALYSIS REQUIRED

Operation and interval. Lookup and exact-scope evidence comparison; no engineering diagnosis or certification. Daily evidence-status view and before formal assurance, handback or planned reliance. Minimum evidence and method. Current asset/dependency register, applicable specialist requirement, competent source, complete test scope/result, current configuration and impairment status. Match evidence by asset, scenario and configuration; report missing, expired, partial, failed or conflicting specialist evidence. Uncertainty and safe failure. Certificate metadata, inaccessible appendices, hidden configuration change, record delay and unmapped dependencies. Return 'specialist evidence not demonstrated' and name the required owner/competent source.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Relevant specialist system owners own registers/requirements; competent contractors/engineers own tests; Estates/fire-safety lead owns acceptance, impairment and access. Current signed specialist finding and impairment first; then complete test evidence; then live telemetry as context. Times, access and provenance. Secure system details, certificates and vulnerabilities are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Use owner-defined validity and invalidate after relevant change/fault/impairment. Conflict. Any asset/scope/configuration mismatch remains unsupported; do not choose a certificate because its date is recent. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not interpret specialist results beyond their scope, certify systems, approve restoration or operate/test assets. Do not disclose secure topology or sensitive vulnerabilities beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives fire-safety personnel a precise specialist-evidence gap list rather than a misleading building-wide 'maintenance complete' statement.

FS-057

FIRE-SAFETY PERSONNEL QUESTION

Which fire-safety records are unsigned, time-ambiguous, untraceable or linked to the wrong asset, location or competent source?

SENSORS AND TELEMETRY

Proposed evidence. Document and record metadata, digital signatures, file hashes, controlled-system controlled-system access logs, asset/location IDs, version/supersession links and restricted original evidence. No physical sensing is necessary. Coverage and timing. All records proposed as evidence for the current catalogue decision; archived material is included only with retention and authenticity status. Event-driven at creation, import, edit, signature and supersession; preserve original and retrieved times and immutable change history. Quality and limits. Retain signer/issuer verification, source-system ID, clock/UTC offset, file hash, version, access history and transformation provenance. Calibration is not applicable. Flag impossible dates, timezone ambiguity, duplicate files, broken links, post-signature edits and orphaned IDs; never rewrite originals. A complete metadata envelope does not prove factual correctness, competence, physical condition or that a signature is legally sufficient.

ANALYSIS REQUIRED

Operation and interval. Metadata validation, lookup and provenance comparison; no judgement of technical truth beyond explicit checks. At ingestion, before evidence use and during assurance preparation. Minimum evidence and method. A defined claim, original/controlled record, verifiable source/signature, unambiguous times, current asset/location linkage and access permission. Validate required fields, hashes, signatures, time semantics and asset/source links; classify traceable, qualified or unusable. Uncertainty and safe failure. Legacy scans, timezone, missing originals, migrations, broken IDs and unverified issuers. Return 'record not suitable as authoritative evidence' and identify the missing provenance or owner decision.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Record owner and system owner own authenticity and retention; asset/system owners own linkage; competent issuer owns technical content; fire-safety lead owns evidence use. Original signed/controlled record first; verified system copy next; derived extracts last. A newer upload does not automatically supersede an accepted record. Times, access and provenance. Role-based access and retention; sensitive originals are not exposed beyond the claim need. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Stale when superseded, permission/retention changes, asset mapping changes or authenticity cannot be verified. Conflict. Preserve all conflicting versions and require the owner to identify the controlling record; do not auto-merge. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not alter, backdate, sign, authenticate beyond available verification, or claim that valid metadata proves fire safety. Do not disclose restricted records, signatures or personal information unnecessarily. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This prevents weak records from silently entering an assurance pack and tells owners exactly what must be corrected or reissued.

FS-058

FIRE-SAFETY PERSONNEL QUESTION

Which asset identifiers, locations, controlled drawings or controller configurations conflict and could misdirect testing or maintenance?

SENSORS AND TELEMETRY

Proposed evidence. Current asset registers, controlled drawings/models, panel/controller exports, commissioning records, field inspection scans and change records. Supporting position measurements may be used only with documented accuracy. Coverage and timing. All mapped fire-safety assets and interfaces in the decision area; each source's coordinate system, floor/zone naming and version are recorded. Event-driven on register/drawing/configuration/change; field scans at source time; daily mismatch summary during live change. Quality and limits. Retain file/configuration hashes, source clocks/UTC offsets, version/supersession, coordinate reference, measurement calibration where used, uptime/gaps for controller feeds and provenance. Flag duplicate IDs, impossible relocations, stale exports and name aliases. Do not force coordinate matches beyond stated tolerance. Agreement between digital sources does not prove physical presence, correct installation or current condition; field scans may be mislocated.

ANALYSIS REQUIRED

Operation and interval. Version comparison, identifier matching and spatial join; recommendation to inspect/reconcile, not to certify configuration. Daily during change and before tests, maintenance, handback or incident-information updates. Minimum evidence and method. Current versions of each source, stable asset IDs, known coordinate/name mappings, field evidence where material and owner precedence. Normalise aliases without losing originals, compare asset/config/location fields, apply stated spatial tolerance and classify exact, probable, conflicting or unmapped. Uncertainty and safe failure. Coordinate error, legacy naming, partial exports, unrecorded moves, duplicate tags and clock/version ambiguity. Return 'asset/configuration mapping not established' and block derived claims that depend on the mismatch.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates/BIM/Space owners own current geometry; system/asset owners own registers/configuration; project/change owners own updates; fire-safety lead owns operational interpretation. Competent physical finding and signed change/commissioning decision first; then controlled registers/configuration; derived spatial match last. Times, access and provenance. Secure drawings, addresses and controller topology are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale after asset, configuration, naming or layout change; field observation valid only as dated. Conflict. Retain all mismatches and require owners to resolve the controlling ID/location; do not overwrite one system automatically. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not reconfigure controllers, update authoritative registers automatically, infer physical installation or expose secure topology. Do not disclose detailed drawings or controller addresses beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This reduces test and maintenance errors caused by staff working from different identifiers, locations or configuration versions.

FS-059

FIRE-SAFETY PERSONNEL QUESTION

Which recurring alarms, faults, door defects or maintenance findings justify a competent diagnostic investigation?

SENSORS AND TELEMETRY

Proposed evidence. Historical panel/fault events, accepted fire-door and maintenance findings, work orders, environmental/operational context at aggregate level and competent investigation outcomes. No person or productivity monitoring. Coverage and timing. Only comparable events tied to stable asset/system IDs and configurations; periods with known testing, works or data outage are labelled or excluded. Use native event times; aggregate counts, duration and recurrence by asset/zone in weekly/monthly windows while retaining raw event provenance. Quality and limits. Retain source clock/UTC offset, configuration, uptime, gaps, maintenance/test periods, asset mapping and provenance. Validate event-code semantics across upgrades; flag duplicated imports, reset-induced bursts and changed baselines. Calibration applies to supporting measurements. Do not count missing periods as zero. Recurrence does not establish cause, risk, severity or common mechanism; maintenance activity and reporting practice can change apparent trends.

ANALYSIS REQUIRED

Operation and interval. Historical comparison and anomaly screening; recommendation to competent investigation. Causal diagnosis is explicitly prohibited. Weekly/monthly and on demand for repeated active exceptions. Minimum evidence and method. A stable asset ID/configuration, adequate observed exposure time, consistent event definitions, known gaps/testing periods and current defect/maintenance context. Calculate event rate/duration with denominator and confidence, compare with the asset's own baseline or peer class only when comparable, and flag persistent recurrence. Uncertainty and safe failure. Reporting changes, configuration upgrades, data gaps, differing exposure, maintenance cycles and low event counts. Return 'recurrence evidence insufficient' when exposure or coding is unstable; otherwise present a assessment prompt, not a diagnosis.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. System/asset owners and competent maintainers own event meaning and investigation; fire-safety lead owns prioritisation; data owner controls permitted context. Current active defect/impairment and competent findings first; then traceable historical events; derived trend is a prompt only. Times, access and provenance. Use asset/zone aggregates; do not infer staff behaviour, blame or individual actions. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Trend valid only for stable configuration and sufficient coverage; recompute after mapping/code changes. Conflict. Show conflicting code mappings, missing periods and closed defects. Do not suppress recurrence because a job is closed. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not claim root cause, negligence, future failure, risk level or fire-safety non-compliance from recurrence alone. Do not expose individual operator/maintainer activity or sensitive incident details. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps direct scarce specialist investigation toward persistent, well-evidenced patterns while avoiding unsupported causal claims.

FS-060

FIRE-SAFETY PERSONNEL QUESTION

Can a claim-specific evidence pack be assembled for the Responsible Person or competent assessor without overstating what the records demonstrate?

SENSORS AND TELEMETRY

Proposed evidence. No new sensing. Retrieve authorised originals, signed records, current asset/source metadata, permitted telemetry extracts, quality flags, conflicts and a reproducible claim-to-evidence index. Coverage and timing. Only the exact claim, assets, locations and time interval requested; unrelated records and sensitive system detail are excluded. Use original observed/effective times and a pack generation/retrieval time; freeze a manifest and hashes so later changes are visible. Quality and limits. Verify source mapping, calibration/functional status where telemetry is included, clock/UTC offsets, uptime, gaps, versions, signatures, file hashes, permissions and full transformation provenance. Flag stale/derived evidence and preserve raw references. Do not edit originals or hide missing periods. An organised pack does not certify truth, compliance, completeness of the building or competence of an interpretation; absence from the pack may reflect scope or permission.

ANALYSIS REQUIRED

Operation and interval. Claim decomposition, provenance validation and evidence packaging; no new diagnosis, certification or substantive recommendation. On demand, with a stated cut-off time and version. Minimum evidence and method. A precise claim, authorised purpose/recipient, required evidence list, current source permissions, validated provenance and unresolved-conflict statement. Build a manifest mapping each subclaim to source, owner, times, quality, limitation and status; include reproducible extracts and a clear absent-evidence list. Uncertainty and safe failure. Scope error, hidden records, permission limits, stale sources, clock/mapping uncertainty and competent interpretation outside the service. Return 'pack cannot support the stated claim' if critical evidence or permission is missing, while still listing what is available.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/assessor defines the claim and pack purpose; each record/system owner retains authority; information governance controls access/retention; fire-safety lead curates but does not certify. Original signed competent/owner decisions first; controlled records next; telemetry and derived analysis last, clearly labelled. Conflicts remain visible. Times, access and provenance. Claim-based least privilege, redaction, retention and access logging; personal plans and unnecessary identities are excluded. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Pack becomes stale when any critical source changes, permission expires or the claim/time scope changes. Conflict. Include a conflict register and do not choose or harmonise sources without an owner decision. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not certify fire safety, sign on behalf of owners, rewrite conclusions, suppress conflicts or expose personal emergency evacuation plans. Do not include unrelated sensitive records, credentials, personal data or secure topology. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives decision-makers a bounded, reproducible pack that makes evidence strength and gaps visible instead of presenting a polished but misleading assurance bundle.

FS-061

FIRE-SAFETY PERSONNEL QUESTION

Which fire-safety impairments are active now, what systems or areas do they affect, who owns each one and when does the authorised period expire?

SENSORS AND TELEMETRY

Proposed evidence. Authorised impairment register, system disablement/fault states, permit/work-order status, affected asset/zone mappings, temporary-control records and attended confirmation. Telemetry is corroboration only. Coverage and timing. Every impairment declared by the Responsible Person or delegated owner, linked to affected systems, zones, dependencies, start/assessment/expiry and responsible contacts; undeclared faults remain separate exceptions. Event-driven at declaration, extension, status change and closure; live system states at approved intervals; present a current impairment view plus next 24/72-hour expiries. Quality and limits. Retain signed decision/version, source clocks/UTC offsets, system health, gaps, asset/zone mapping and provenance. Calibration/functional checks apply to supporting telemetry. Flag expired authorisations, overlapping records, stale disabled-state feeds and closure without acceptance. Do not infer an impairment solely from a fault code. The register may lag a physical fault, and a declared impairment does not quantify risk, adequacy of controls or permission to operate.

ANALYSIS REQUIRED

Operation and interval. Lookup and temporal/spatial comparison; recommendation limited to escalate expiry, confirm scope or verify closure. No approval or risk diagnosis. Continuously on change, at shift handover and in 24/72-hour planning. Minimum evidence and method. A current signed impairment record, affected asset/zone/dependencies, effective interval, current technical evidence, temporary-control reference and explicit closure criteria. Reconcile declared scope with live faults/disablements and work context, calculate expiry windows and list undeclared or mismatched exceptions separately. Uncertainty and safe failure. Record latency, incomplete dependency map, clock offset, hidden faults, temporary changes and stale technical feeds. Return 'impairment status/scope not established' and require the authorised owner to confirm; do not infer permission.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/delegated fire-safety lead owns impairment authorisation, scope and expiry; system/asset owners and competent maintainers own technical evidence; work/event owners own affected activity. Fire-service command/current emergency plan first; then current signed impairment decision; then competent system evidence and live telemetry. Times, access and provenance. Role-restricted because impairments expose vulnerabilities, contacts and operational restrictions. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Current until explicit expiry/closure; live corroborating state uses source validity and any missed update makes closure unverified. Conflict. If system state and register disagree, show both, keep the more restrictive operational status pending owner confirmation, and never auto-close or extend. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not create, approve, extend or close an impairment, set risk level, or treat a normal system state as authorisation to restore. Do not disclose vulnerabilities, contact details or secure system topology beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives fire-safety personnel a reliable handover and expiry list while preserving the Responsible Person's control of impairment decisions.

FS-062

FIRE-SAFETY PERSONNEL QUESTION

For each active impairment, which authorised temporary controls lack current evidence of placement, operation, coverage or handover?

SENSORS AND TELEMETRY

Proposed evidence. Temporary-control register, patrol/fire-watch logs, temporary detector/alarm heartbeat and test records, barrier/sign/route inspection scans, communication checks and shift handover acknowledgements. Coverage and timing. Only controls explicitly authorised for a named impairment, area and effective interval; informal measures and unrecorded patrol routes are outside the evidence set. Control states/logs at the authorised frequency; temporary device heartbeat at approved interval; handover and inspection events at source time; aggregate to missed/late/unverified control periods. Quality and limits. Retain device calibration/functional checks where applicable, patrol/log source, source clocks/UTC offsets, uptime, gaps, route/zone mapping and provenance. Flag copy-forward logs, impossible sequences, stale placements and clock drift. Do not infer a completed patrol or control from silence or a device heartbeat alone. Logs and device status do not prove quality, vigilance, physical coverage or adequacy of the temporary control; conditions may change between checks.

ANALYSIS REQUIRED

Operation and interval. Lookup, observation and compliance-to-authorised-plan comparison; recommendation limited to immediate owner check/escalation, not approval. At each required control interval, shift handover and impairment assessment. Minimum evidence and method. Current signed control plan, exact scope/frequency, responsible role, current placement/operation evidence, complete interval log and handover. Build a control-by-time matrix, verify each required evidence item and identify gaps, late checks, untested devices and incomplete handovers. Uncertainty and safe failure. Manual log accuracy, clock alignment, unobserved quality, device coverage, staff availability and changing conditions. Return 'temporary-control performance not demonstrated' and notify the authorised control owner through existing procedures.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The impairment authoriser/fire-safety lead owns the required controls and frequency; Security/work/system owners implement; competent maintainers own temporary devices; line managers own handover. Current signed impairment/control plan first; then attended/competent evidence; then healthy telemetry/logs. A missed required check remains a gap even if no alarm occurred. Times, access and provenance. Patrol routes, staff identities and vulnerability detail are tightly role-restricted; routine output uses role and time status. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Stale immediately after a missed required interval, location/scope change, shift change without handover or device fault. Conflict. Any conflict between plan, placement, log and telemetry is shown as an unverified control and escalated; do not backfill. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not approve controls, infer adequacy, monitor individual performance, issue fire-watch instructions or extend the impairment. Do not expose named patrols, routes or vulnerability information beyond authorised need. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This enables rapid correction of a missed patrol, failed temporary device or incomplete handover before an impairment continues unnoticed.

FS-063

FIRE-SAFETY PERSONNEL QUESTION

Which active or imminent hot-work permits overlap detection, suppression, escape routes, combustibles or other fire-safety dependencies?

SENSORS AND TELEMETRY

Proposed evidence. Authorised hot-work permits, geolocated work zones, schedule and status, detector/suppression isolations, nearby asset/route mappings, gas/energy controls, housekeeping inspections and attended permit checks. Environmental/thermal sensing may corroborate but cannot authorise work. Coverage and timing. All current and next-72-hour hot-work permits with verified work extents and vertical/adjacent effects; unknown coordinates or changed scope are explicit gaps. Event-driven on permit issue, start, suspension, extension and close-out; live system/route states at approved intervals; evaluate current plus 72-hour timeline. Quality and limits. Retain permit signature/version, work-zone coordinates, source clocks/UTC offsets, system health/gaps and provenance. Calibration applies to any gas/temperature instruments used by competent staff. Flag expired permits, overlapping effective times, moved work and clock/time-zone errors. Do not infer completion from permit end time. Spatial overlap does not establish hazard severity, adequacy of controls, sparks/heat travel, hidden combustibles or permission to proceed.

ANALYSIS REQUIRED

Operation and interval. Spatial/temporal lookup and comparison; recommendation to coordinate or inspect. No risk score, causal diagnosis or permit decision. On permit issue/change, before start, continuously for active dependency changes and over a 72-hour planning horizon. Minimum evidence and method. A current signed permit, verified work extent/time, required controls, current nearby fire-safety dependencies, active impairments and competent local confirmation where material. Intersect work volumes and time windows with assets/routes/isolations, include adjacent/vertical zones defined by the owner, and list exact conflicts and unknowns. Uncertainty and safe failure. Geolocation error, hidden services/combustibles, changing work, incomplete vertical mapping, sensor limits and record latency. Return 'hot-work dependency overlap not established' and require the permit issuer/fire-safety owner to check.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The permit issuer and work owner own hot-work authorisation/scope; the Responsible Person/fire-safety lead owns fire-safety conditions; system/route owners and competent testers own dependencies. Current signed permit conditions and impairment decisions first; then competent local inspection; then current system/route records and telemetry. Times, access and provenance. Work location, hazards, isolations and personnel detail are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale after work scope/location/time/control change or missed permit assessment; live dependencies use their source validity. Conflict. Any mismatch in location, isolation, route or permit status is an unresolved conflict requiring the permit issuer's assessment; analytics cannot approve continuation. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not issue, approve, suspend or close a permit, prescribe controls, infer safe separation or operate/isolate systems. Do not expose worker identities, hazardous-process detail or secure system topology beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives permit and fire-safety owners a concrete coordination list before work starts or conditions change, without turning spatial analytics into work authorisation.

FS-064

FIRE-SAFETY PERSONNEL QUESTION

Which temporary detectors, alarms, patrols or fire-watch arrangements lack current placement, heartbeat, test or handover evidence?

SENSORS AND TELEMETRY

Proposed evidence. Temporary detector/alarm controller heartbeat, battery/power state, device placement scans, authorised functional-test results, patrol/fire-watch logs and shift handover records. Coverage and timing. Only temporary provisions listed in the current impairment/control plan, mapped to exact zones, coverage assumptions and effective interval; ad hoc devices remain unverified. Heartbeat at manufacturer/plan-approved interval; test and placement events at source time; patrols at authorised frequency; present gap durations and last verified handover. Quality and limits. Retain device calibration/functional checks, battery test, configuration, source clocks/UTC offsets, uptime, gaps, placement history and provenance. Detect moved/offline devices, stale scans, copied logs and clock drift. Never infer coverage from heartbeat or interpolate across a missed patrol/test. A heartbeat does not prove detection coverage, siting, audibility or physical condition; patrol logs do not prove observation quality or adequacy.

ANALYSIS REQUIRED

Operation and interval. Observation and comparison with the authorised temporary-control plan; recommendation only for owner check/retest. At every heartbeat/control interval, handover and impairment assessment. Minimum evidence and method. Current plan, unique device/control IDs, verified placement, complete heartbeat/test/log evidence, responsible role and current handover. Check placement, power/communication, test, interval completion and handover separately; report the first missing evidence and affected scope. Uncertainty and safe failure. Battery/communications, moved devices, coverage assumptions, manual-log quality, clock gaps and changed conditions. Return 'temporary provision not demonstrated' and require immediate authorised owner confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The impairment/control-plan owner defines provision; competent temporary-system provider owns devices/tests; Security/work owners implement patrol/fire watch; fire-safety lead owns coordination. Current signed temporary-control plan and competent test first; then attended placement/handover; then healthy telemetry/logs. Times, access and provenance. Locations, patrol patterns and staff details are role-restricted because they expose vulnerabilities. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Stale after one missed heartbeat/control interval, placement change, shift without handover or any device fault. Conflict. Plan-device-log mismatches remain unavailable and are escalated; do not assume a nearby temporary device covers the gap. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not approve temporary coverage, infer detector performance, direct fire-watch activity, arm/reset devices or extend an impairment. Do not reveal patrol routes, staff identity or sensitive temporary-system locations outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This enables prompt correction of a dead, moved, untested or unhandled temporary provision before it is relied upon.

FS-065

FIRE-SAFETY PERSONNEL QUESTION

Which live work interfaces need competent inspection because compartment boundaries, fire doors, protected routes or asset access could be compromised?

SENSORS AND TELEMETRY

Proposed evidence. Current permits/work packages, verified work-zone geometry, access/ceiling/riser events, temporary barrier/route records, asset/door states, site observations and restricted close-out evidence. No sensor can verify concealed compromise. Coverage and timing. All live works intersecting current owner-approved compartments, protected routes or fire-safety asset access zones, including vertical/adjacent effects and material staging. Event-driven on work status/location/change; relevant door/route states at source interval; daily and pre-handover overlap summary. Quality and limits. Retain permit/work version, source clocks/UTC offsets, location/asset IDs, telemetry uptime/gaps and provenance. Calibration applies to any clearance/position sensors. Flag scope creep, moved boundaries, unrecorded access and stale close-out. Do not infer reinstatement between inspections or from a closed job. Overlap indicates need for assessment, not actual compromise; hidden work, dust, temporary opening, workmanship and internal asset access may be unobserved.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The work/project owner owns scope and controls; the Responsible Person/fire-safety lead or competent fire engineer owns fire-safety assessment; Estates/system owners own assets/routes; competent inspector owns physical findings. Current signed permit/impairment and competent observation first; then controlled work/change records; then healthy telemetry as context. Times, access and provenance. Work details, secure spaces and contractor information are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale after scope/location/sequence or site-condition change; live state uses source validity. Conflict. Any mismatch between permit, observed work and system/route state remains an unresolved work-interface issue requiring local check. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation and interval. Spatial/temporal lookup and comparison; recommendation to inspect/coordinate, not diagnosis or approval. Daily, before shifts/handback and on any work or dependency change. Minimum evidence and method. Current work extent/time, current compartment/route/asset map, active permits/impairments, recent local evidence and named owners. Intersect work with protected boundaries and access zones, classify direct, adjacent, vertical and unknown overlaps, and list evidence required for each. Uncertainty and safe failure. Drawing registration, hidden scope, temporary conditions, unrecorded materials, clock/data gaps and physical change after observation. Return 'work impact not established' and require an attended competent assessment.

ANSWER BOUNDARY

Do not infer damage or integrity, stop/approve work, authorise access or close out compartment/route effects. Do not expose restricted construction, service-route or contractor details beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This directs competent inspections and coordination to the exact work interfaces most likely to be missed during routine project reporting.

FS-066

FIRE-SAFETY PERSONNEL QUESTION

How have temporary partitions, hoardings, scaffolds, displays or furniture changed routes, detector context, signs or emergency-lighting coverage?

SENSORS AND TELEMETRY

Proposed evidence. Approved temporary-layout plans, geolocated installation records, mobile inspection scans, selected clearance/depth measurements, detector/sign/light locations and work/event schedules. No continuous imaging is required. Coverage and timing. Only temporary installations with current mapped extents and affected fire-safety dependencies; hidden, unrecorded or off-plan items remain unknown. Event-driven at installation/change/inspection/removal; selected clearance measurements at source time; daily summary while temporary layouts remain. Quality and limits. Retain plan/version, measurement instrument calibration, capture-device clock/UTC offset, coordinate reference, file hashes, source, gaps and provenance. Re-register after moved hoarding/furniture; flag outdated images and coordinate drift. Do not infer geometry from a single photograph without scale/reference. Digital geometry and spot measurements cannot establish full route adequacy, detector coverage, lighting, sign visibility, stability or fire performance.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The event/work/space owner owns temporary layout; Estates owns geometry and services; fire-alarm/lighting owners and competent designers own technical effects; fire-safety lead owns assessment. Current signed temporary-change approval and competent inspection/design first; then controlled layout records; then measurements/telemetry as supporting evidence. Times, access and provenance. Detailed event layouts, secure routes and images are role-restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale after any physical move, crowd/furniture change or works update. Conflict. Any mismatch between approved layout and observation remains unresolved and requires owner-led inspection; do not auto-update controlled plans. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANALYSIS REQUIRED

Operation and interval. Spatial comparison and evidence-gap recommendation; calculations are geometric estimates, not compliance or engineering assessment. Before installation/use, after changes and before removal/handback. Minimum evidence and method. Current approved temporary layout, current fire-safety asset/route map, verified installed extent, recent measurements/observation and relevant system-owner criteria. Overlay temporary extents with routes, detector/sign/light influence zones defined by owners, and flag direct obstruction, possible shadowing and unknown effects separately. Uncertainty and safe failure. Coordinate/measurement error, three-dimensional obstruction, occupancy, lighting conditions, unrecorded movement and owner-criteria uncertainty. Return 'temporary-layout impact not established' and require competent site assessment.

ANSWER BOUNDARY

Do not approve a layout, calculate compliant coverage, infer fire performance, move items or issue evacuation directions. Do not expose restricted layouts or images containing identifiable people. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports timely repositioning, additional testing or controlled plan/sign updates before a temporary installation goes live.

FIRE-SAFETY PERSONNEL QUESTION

FS-067

Where do combustible materials, waste, gas cylinders or charging equipment require a housekeeping or control check now?

SENSORS AND TELEMETRY

Proposed evidence. Authorised inspection reports, waste/service schedules, permit and storage records, cabinet/door or fill-level sensors where appropriate, manufacturer-supported charger/battery alarms, selected temperature sensing in approved fixed charging areas and attended observations. Coverage and timing. Only defined storage/waste/charging zones and monitored containers/equipment; the service does not inspect contents, chemistry or unregistered items. Discrete alarms on change; fill/temperature at owner-approved intervals, typically 1-5 minutes for temperature and 15-60 minutes for fill; inspections at source time. Aggregate to zone-level exceptions. Quality and limits. Retain sensor calibration/location, alarm configuration, source clock/UTC offset, uptime, gaps and provenance. Validate temperature sensors against a reference and fill sensors after container change; flag drift, relocation and flat-line readings. Do not bridge gaps or infer material type from sensor response. Fill, door or temperature data does not identify contents, assess segregation, cylinder security, battery condition, ignition risk or compliance; hotspots can have non-fire causes.

ANALYSIS REQUIRED

Operation and interval. Observation and comparison with owner-set housekeeping/control triggers; recommendation to inspect/remove through existing authority. No fire-risk diagnosis. At daily rounds, before high-risk work/events and on active alarms/exceptions. Minimum evidence and method. Current zone/equipment record, authorised storage/control criteria, healthy recent sensor evidence if used, current service/permit context and attended confirmation for material decisions. Identify persistent owner-defined exceptions, distinguish sensor, schedule and inspection evidence, and rank by access/route/system dependency. Uncertainty and safe failure. Sensor placement, unknown contents, transient heat, container change, hidden storage, clock gaps and owner-criteria limits. Return 'housekeeping/control condition not established' and require local inspection.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Estates/Cleaning/waste owners own housekeeping and collections; laboratory/work/equipment owners own materials, cylinders and charging; fire-safety lead owns inspection/escalation; competent specialists own hazardous-storage assessment. Current permit/storage restriction, alarm and competent observation first; then owner records; healthy telemetry is a prompt only. Times, access and provenance. Hazardous-material and secure-location detail is role-restricted; no person or activity profiling. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Temperature/live alarm stale after one reporting interval; fill after source interval; inspection stale after material movement or owner-defined period. Conflict. A competent concern or active alarm overrides normal telemetry. Unknown contents or location conflicts require attended owner check. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not identify substances, judge legal storage, diagnose battery/gas risk, command equipment or authorise continued use. Do not disclose sensitive material inventories or work activities beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This focuses housekeeping, waste collection or specialist checks on areas with credible evidence of a control gap without pretending sensors can identify the hazard.

FIRE-SAFETY PERSONNEL QUESTION

FS-068

Which changes in use, occupancy, equipment, work activity or stored material require competent fire-safety reassessment before operation?

SENSORS AND TELEMETRY

Proposed evidence. Approved change requests, space-use and occupancy records, equipment/utility commissioning, process permits, hazardous-material registers and time-stamped site observations. Sensors may corroborate changed patterns but cannot determine reassessment need alone. Coverage and timing. All proposed, temporary, emergency or discovered changes within current building zones and their fire-safety dependencies; informal/unreported changes are an explicit unknown. Event-driven at proposal, implementation and discovery; occupancy/equipment telemetry at owner-approved aggregation; daily change-screening summary. Quality and limits. Retain change/version/signature, source clocks/UTC offsets, equipment/zone IDs, telemetry calibration/health/gaps and provenance. Flag changes without owner, effective date or baseline; detect sustained pattern shifts only with adequate comparable data. Do not infer process or material from energy, occupancy or environmental signatures. Administrative or sensor change indicators do not establish materiality, risk, compliance or required control; hidden activities and transient patterns may be missed.

ANALYSIS REQUIRED

Operation and interval. Lookup, version comparison and change screening; recommendation only for competent reassessment. No risk diagnosis or approval. At each change proposal, before implementation, after discovery and in periodic assessment. Minimum evidence and method. Current accepted baseline, precise change description/effective time, affected fire-safety dependencies, relevant authoritative records and owner-defined screening criteria. Compare change attributes with owner-defined triggers, classify confirmed trigger, possible trigger or insufficient information, and list required competent evidence. Uncertainty and safe failure. Unreported activity, ambiguous scope, proxy-data error, baseline drift, temporary conditions and incomplete material/process data. Return 'reassessment need cannot be determined' and require the responsible owner/competent professional to assessment.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The change/use/process owner owns description; the Responsible Person/fire-safety lead or competent assessor owns whether reassessment is required; Space/Estates/system owners provide current records. Current signed change and competent assessment first; then authoritative use/equipment/material records; telemetry-derived change indicators last. Times, access and provenance. Process, research, material and occupancy details are restricted; no inspection of research content, code, speech or network content. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale after scope/use/material/equipment change; pattern indicators require a current comparable baseline. Conflict. Conflicting use or change records are shown and referred to owners; analytics cannot determine the controlling use. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not conduct or certify the fire-risk assessment, approve change/use, infer sensitive activities or materials, or authorise occupancy. Do not expose research/work content, personal data or hazardous-material detail beyond authorised need. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives change owners an early, explainable referral to competent fire-safety assessment before a new use or process becomes operational.

FS-069

FIRE-SAFETY PERSONNEL QUESTION

What evidence remains missing before the authorised owner may consider restoring an impaired system or returning an affected area to normal use?

SENSORS AND TELEMETRY

Proposed evidence. Current impairment and restoration checklist, competent test/inspection results, remedial work evidence, system/asset states, integrated endpoint confirmation, temporary-control close-out and owner acceptance record. Coverage and timing. Only the named impairment, affected assets/zones and owner-defined restoration criteria; no building-wide inference from a local close-out. Event-driven on remedial work, test, state change and evidence upload; present a current checklist with source age and unresolved conflicts. Quality and limits. Retain test instrument calibration, configuration, source clocks/UTC offsets, system uptime/gaps, file hashes, signatures and provenance. Verify asset/scenario match and sequence completeness; flag post-test changes and evidence captured before remedial work. Do not infer restoration through missing logs. Completed remedial work or normal telemetry does not prove full restoration, integrated operation, physical condition, control withdrawal or owner acceptance.

ANALYSIS REQUIRED

Operation and interval. Evidence-completeness comparison against an owner-approved restoration checklist; recommendation limited to obtain evidence or request decision, not restoration approval. Continuously during close-out and at formal handback. Minimum evidence and method. Current impairment, explicit restoration criteria, asset-specific remedial work evidence, complete competent tests/endpoints, temporary-control close-out and authorised owner acceptance. Map each criterion to current evidence, validate sequence and scope, and classify met, qualified, missing or conflicting. Uncertainty and safe failure. Hidden condition, incomplete integrated test, clock/configuration mismatch, post-test change and acceptance outside the system. Return 'restoration evidence incomplete' and list exact missing criteria; keep the impairment active in the service view until owner decision.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The impairment authoriser owns restoration decision and criteria; system/asset owners and competent maintainers own technical evidence; temporary-control owners own handback; Responsible Person/fire-safety lead coordinates. Current signed impairment/restoration criteria and competent test first; then complete system/local evidence; work-order status and telemetry are supporting only. Times, access and provenance. Role-restricted due to system vulnerabilities, work details and decision records. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Any evidence becomes stale after relevant fault, change or configuration update; checklist refreshes on every event. Conflict. Conflicting or missing criterion remains open; the service must not substitute a normal status or closed job for owner acceptance. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not approve restoration, close the impairment, withdraw controls, reset systems or declare an area safe for use. Do not disclose secure technical detail beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the authorised owner a precise, traceable close-out checklist while preventing premature restoration based on a normal dashboard.

FS-070

FIRE-SAFETY PERSONNEL QUESTION

Which conflicts among impairments, permits, bookings, closures, events and isolations need coordination during the next 72 hours?

SENSORS AND TELEMETRY

Proposed evidence. Current signed impairments, work/hot-work permits, bookings/events, route/area closures, system isolations, maintenance windows and approved temporary controls. Live telemetry is used only to confirm current state, not future permission. Coverage and timing. All records with spatial or dependency overlap in the next 72 hours, including setup/clearance buffers and vertical/adjacent zones defined by owners. Event-driven updates; evaluate timeline in 15-minute or owner-defined intervals and refresh after any source change. Quality and limits. Retain versions, effective times/timezones, source clocks/UTC offsets, asset/zone IDs, permissions, uptime/gaps for live feeds and provenance. Flag open-ended records, conflicting time zones, stale copies and missing buffers. Calibration applies only to supporting sensors. Do not infer cancellation from no activity. A schedule overlap does not establish unsafe work or inadequate controls; unrecorded changes, duration uncertainty and owner discretion remain.

ANALYSIS REQUIRED

Operation and interval. Temporal/spatial/dependency comparison and scenario screening; recommendation limited to convene coordination or verify a record. Continuous on change, with 24/48/72-hour handover views. Minimum evidence and method. Current versioned records, effective intervals, mapped zones/assets/dependencies, setup/clearance buffers and owner precedence. Intersect time, space and dependencies, identify compound conflicts such as impairment plus event plus route closure, and present owner-specific actions/questions. Uncertainty and safe failure. Late changes, uncertain duration, hidden dependencies, mapping error, time-zone/clock issues and missing records. Return 'coordination screen incomplete' when any critical source is stale/unmapped and require owner confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Each permit/event/system/space owner controls its record; the Responsible Person/fire-safety lead coordinates fire-safety dependencies; competent specialists own technical constraints. Fire-service command/current emergency plan first; then current signed impairment/permit/closure decisions; then system and booking records; derived conflict screen last. Times, access and provenance. Role-based because schedules, hazards, vulnerabilities and contacts may be sensitive. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Recompute immediately on change; future records are valid only to their owner-set assessment/expiry and current version. Conflict. Show all conflicting source records and responsible owners. Do not cancel, reschedule or choose precedence beyond the approved rule. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not approve, cancel or reschedule work/events, authorise an isolation/impairment, or imply the remaining plan is safe. Do not expose sensitive schedules, hazards or system vulnerabilities beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the fire-safety lead an early, concrete coordination agenda before independent plans combine into an unmanaged dependency.

Incidents, fire-service liaison and post-event evidence

FS-071

FIRE-SAFETY PERSONNEL QUESTION

What incident identifier, command status and authoritative information source are confirmed for this event?

SENSORS AND TELEMETRY

Proposed evidence. Read-only incident log, fire-alarm event identifier, control-room log, authorised command/liaison update and current emergency-plan version. No automatic incident declaration or command channel is created. Coverage and timing. Only the named event and authorised operational systems; unrelated alarms, exercises and personal communications are excluded. Event-driven with original source timestamps; show last confirmed command update and retrieval time, not a continuously inferred status. Quality and limits. Retain source identity, signature/role, incident ID, source clock/UTC offset, logging resolution, uptime, gaps and provenance. Calibration is not applicable. Flag duplicate incident IDs, exercise/live ambiguity, clock reversal and unverified relays. Never infer command status during a communications gap. A system alarm does not establish an incident command structure, fire-service attendance, event severity or required occupant action.

ANALYSIS REQUIRED

Operation and interval. Lookup and source-authentication comparison; no diagnosis, forecast or recommendation beyond confirming the authorised channel. On event creation and each authorised command update; during live incidents, minimise queries and defer to command. Minimum evidence and method. A unique event ID, current authorised command source, current emergency-plan reference, source times and verified liaison/role. Validate identity and source precedence, separate system events from command statements, and present only the latest confirmed status with its age. Uncertainty and safe failure. Clock ambiguity, duplicate events, communications delay, exercise/live confusion and incomplete liaison updates. Return 'incident command/source not confirmed' and direct users to the current approved liaison process.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire service controls incident command when in attendance; the Responsible Person/control room owns internal incident records and emergency-plan activation; the fire-safety liaison owns authorised updates. Current fire-service command first; then the current emergency plan and signed/authorised internal incident record; system alarms are factual events only. Times, access and provenance. Strict incident-role access. Do not expose personal caller details, responder contacts, radio channels or sensitive building information outside authorised command. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Command information is stale immediately after a newer authorised update or loss of the command channel; system state uses source-specific validity. Conflict. If incident IDs or command sources conflict, display the conflict and require direct confirmation through the approved liaison; do not merge events. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not declare an incident, activate plans, issue evacuation or re-entry instructions, contact emergency services, or infer command from alarm telemetry. Do not disclose caller identity, personal data, secure communications or sensitive incident detail beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This prevents staff acting on an unverified dashboard or duplicated event and anchors every subsequent evidence request to the correct command source.

FS-072

FIRE-SAFETY PERSONNEL QUESTION

After command authorisation, what is the best-supported sequence from the first relevant detection or call-point signal through alarm transmission and recorded system responses?

SENSORS AND TELEMETRY

Proposed evidence. Read-only fire-alarm/panel events, call-point/device identifiers, remote-signalling events, destination-system logs, control-room observations and authorised incident-log entries. Coverage and timing. Only the named incident, mapped systems and time window; missing panels, external networks and unintegrated endpoints are explicit gaps. Native event resolution with original timestamps; preserve raw sequences and produce a post-event aligned timeline only after clock offsets and log completeness are assessed. Quality and limits. Retain configuration, source clocks/UTC offsets, logging resolution, uptime, gaps, sequence numbers and provenance. Estimate inter-system offsets from pre-event sync or documented tests; flag drift, resets and imported summaries. Do not invent order within timestamp uncertainty or bridge missing logs. Logged sequence does not prove physical outcome, cause, fire growth, human action or message receipt; device address may not identify the ignition source.

ANALYSIS REQUIRED

Operation and interval. Post-event observation and temporal comparison; calculation of bounded intervals. No causal diagnosis, fire-origin inference or live recommendation. After the live command phase and only with authorised scope; refresh on new evidence. Minimum evidence and method. Confirmed incident ID, frozen original logs, system/configuration versions, known clock offsets/resolution, gap assessment and authorised incident observations. Align by documented offsets, order only where uncertainty permits, link source-to-destination stages and distinguish observed, calculated and unknown events. Uncertainty and safe failure. Clock drift, coarse logging, dropped events, device mapping, manual observation delay, unintegrated systems and post-event configuration change. Return 'sequence not establishable for this interval' and preserve competing evidence without filling gaps.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire service/incident commander controls live interpretation; system owners own logs; the Responsible Person/fire-safety lead owns post-event evidence use; competent investigators determine cause. Fire-service/authorised incident record first; then original system logs and competent observations; derived aligned timeline last. Times, access and provenance. Incident and system detail is restricted; personal caller/operator identity is minimised and disclosed only through authorised investigation. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Timeline is provisional until all critical logs are frozen and verified; stale if new authoritative evidence is added. Conflict. Retain conflicting timestamps and present ranges/order uncertainty. A narrative cannot overrule original logs or command findings. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not infer cause, origin, intent, culpability or physical fire development; do not use the timeline to issue live commands or reset systems. Do not expose identities, call content, secure device topology or investigation-sensitive information outside authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives authorised investigators a reproducible chronology and makes missing or uncertain stages visible before maintenance, investigation or governance decisions are made.

FS-073

FIRE-SAFETY PERSONNEL QUESTION

Which doors, lifts, ventilation, smoke-control, suppression, access or power systems changed state during the event, and which physical outcomes remain unproved?

SENSORS AND TELEMETRY

Proposed evidence. Read-only event/state logs from each relevant system, local controller observations, power-quality/UPS events where commissioned, competent post-event inspections and authorised incident records. Coverage and timing. Only mapped systems linked to the incident area/time; unintegrated endpoints and physical conditions without inspection are explicit unknowns. Native event timestamps; retain pre-event baseline, event transitions and restoration states. Aggregate by dependency while preserving raw source events. Quality and limits. Retain configuration, functional/calibration status, source clocks/UTC offsets, logging resolution, uptime, gaps and provenance. Measure inter-system offset where possible; flag resets, stale gateways and impossible transitions. Do not treat a command as an endpoint or reconstruct across gaps. System state does not prove door opening, lift movement, airflow, smoke control, suppression discharge, access release, power quality or physical condition unless directly observed by an appropriate source.

ANALYSIS REQUIRED

Operation and interval. Event-sequence observation and cross-system comparison; limited localisation of missing evidence, no causal diagnosis or safe-state declaration. During live incident only if requested by authorised command; fuller analysis post-event. Minimum evidence and method. Confirmed incident/time window, current configuration, complete logs, measured clock offsets, endpoint evidence and competent inspections for any physical outcome claim. Build per-system state chains, map interfaces, distinguish commanded, reported and physically verified outcomes, and label unknown endpoints. Uncertainty and safe failure. Clock/gap uncertainty, vendor semantics, hidden mechanical condition, unmonitored endpoints and post-event changes. Return 'physical outcome not demonstrated' and identify the owner/inspection required; do not infer restoration.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The incident commander/fire service controls live interpretation; each system owner and competent maintainer owns technical evidence; the Responsible Person/fire-safety lead owns post-event coordination. Fire-service command and competent physical findings first; then local controller/system logs; derived cross-system comparison last. Times, access and provenance. Detailed system topology and vulnerabilities are incident-role restricted. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live evidence is stale after a communication gap; post-event status requires a current physical/competent check before handback. Conflict. Show command, feedback and physical observation separately. Conflicts remain unproved and are referred to the relevant owner. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not operate/reset systems, declare safe state, authorise re-entry/handback, or infer causal failure from temporal association. Do not expose sensitive control logic, security events or identities beyond authorised incident roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps command and post-event teams separate what systems reported from what actually happened physically, supporting targeted inspection and evidence preservation.

FS-074

FIRE-SAFETY PERSONNEL QUESTION

What aggregate occupancy, visitor and contractor information can be shared with authorised command, with its time, coverage and uncertainty explicit?

SENSORS AND TELEMETRY

Proposed evidence. Authorised aggregate people-count ranges, visitor/contractor registration totals, manual accountability updates and zone/opening status. Personal identities are accessed only through established emergency-accountability channels, not the catalogue. Coverage and timing. Only zones and populations requested by authorised command and covered by current systems; unregistered people, internal movement and unmonitored boundaries are explicit gaps. Use latest source observations; present ranges/bands and observed time. Do not continuously track or create person-level movement histories. Quality and limits. Retain counter calibration/direction checks, source clocks/UTC offsets, uptime, gaps, resets, registration-system health, permissions and provenance. Reconcile only at verified reset/count points. Flag drift and stale check-ins; do not infer location from access or device data. Aggregate counts are not a roll call, exact location or proof of evacuation; registration may not equal presence, and manual updates are time-bound.

ANALYSIS REQUIRED

Operation and interval. Observation, aggregation and uncertainty reporting; no identity inference, location prediction or evacuation-completion claim. Only on authorised command request during the event, with explicit observation time. Minimum evidence and method. Confirmed incident and authorised requester, defined zone/population, current source health, known coverage/reset and latest manual accountability source. Report administrative totals and sensor estimates separately, provide defensible ranges and list unobserved groups/boundaries. Uncertainty and safe failure. Counter drift, tailgating, internal movement, missed check-out, unregistered presence, clock/gap and manual-update delay. Return 'aggregate occupancy/accountability not established' and refer to the authorised manual/accountability process.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. Authorised incident command defines need; Security/reception/permit owners own administrative records; the Responsible Person/fire-safety liaison controls release; privacy owner governs minimum disclosure. Fire-service/incident-command requests and attended accountability first; then current administrative totals; aggregate sensor estimates last. Times, access and provenance. Share minimum necessary aggregate information. Named data, when lawfully required, follows the established command channel and is not generated or exposed by the question service. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale after a newer manual accountability update, a counter/system gap or material movement not observed. Conflict. Show source-specific ranges and gaps. Do not merge conflicting totals into a precise number or infer missing people. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not identify, track or locate individuals, produce a roll call, expose personal emergency plans or declare everyone evacuated. Do not reveal names, contact details or access histories through routine catalogue output. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives command a truthful, time-stamped aggregate picture while preventing false precision or inappropriate disclosure during a pressured incident.

FS-075

FIRE-SAFETY PERSONNEL QUESTION

Which fire-service access routes, standing areas, gates, information boxes, hydrants or riser inlets require immediate physical confirmation?

SENSORS AND TELEMETRY

Proposed evidence. Gate/door states, authorised closure and works records, selected outdoor clearance sensing, access-box tamper/status where designed, hydrant/riser inspection records and direct observations by authorised staff or fire service. Coverage and timing. Only on-premises facilities in the current fire-service information and asset register; public roads, traffic and external hydrants are outside coverage unless an authoritative external source is available. Gate/door states within 10 seconds; outdoor exception flags per minute; inspection/closure at source time. During an incident, updates are only on command request. Quality and limits. Retain functional checks, clearance-sensor calibration/geometry, source clocks/UTC offsets, uptime, gaps and provenance. Revalidate after works/weather/layout change; flag outdoor drift and stale box status. Do not bridge gaps or infer contents/access from a tamper switch. A normal gate/contact or box state does not prove access width, key/content availability, hydrant/riser serviceability, traffic clearance or fire-service suitability.

ANALYSIS REQUIRED

Operation and interval. Observation and comparison; recommendation limited to local confirmation or owner contact, not operational deployment advice. During an incident only when requested by command; otherwise at handover and planned access checks. Minimum evidence and method. Confirmed incident/request or routine task, current facility map, current closures/works, healthy recent evidence and authorised local observation for material claims. Separate route/gate, standing area, information box, hydrant and inlet evidence; report each as supported, qualified or unknown. Uncertainty and safe failure. Outdoor sensing, traffic, weather, unmonitored public realm, hidden contents, physical serviceability and clock gaps. Return 'fire-service facility/access not demonstrated' and direct authorised physical confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire service/incident commander controls operational access during an event; Estates/Security owns premises routes/gates; fire-safety and asset owners hold information boxes, hydrants and risers. Fire-service observation/command first; then current signed closure/defect and competent inspection; then healthy telemetry. Times, access and provenance. Access layouts, keys, information-box contents and vulnerabilities are tightly restricted to authorised responders and owners. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live state stale after two minutes or any reported obstruction/weather/works change; inspection evidence follows owner validity. Conflict. Any command/local report of obstruction overrides normal telemetry. Missing external information is shown as outside coverage. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not direct appliances, open gates/boxes, disclose keys or contents, certify hydrants/risers or override fire-service decisions. Do not publish sensitive responder-access information beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the liaison a concise list of facilities needing immediate physical confirmation without substituting building telemetry for fire-service judgement.

FS-076

FIRE-SAFETY PERSONNEL QUESTION

Which current hazards, isolations, hazardous materials, structural changes or system impairments must be passed to the fire service through the authorised liaison?

SENSORS AND TELEMETRY

Proposed evidence. Current hazard/material registers, permits, isolations, impairment and structural/change records, authorised incident updates and relevant system states. Environmental sensors may corroborate a measured condition but must not infer a material or hazard. Coverage and timing. Only information within the incident/request scope and the current premises; exact sensitive detail follows approved fire-service information procedures. Event-driven on owner/command updates; show effective/observed/retrieved times and a last-confirmed timestamp. No autonomous hazard discovery. Quality and limits. Retain signed source/version, asset/zone IDs, source clock/UTC offset, system uptime/gaps, permissions and provenance. Calibration applies to any included measurement. Flag superseded permits, unresolved material IDs and stale isolations. Do not infer chemicals, structural stability or hazard severity from proxy sensors. Registers may be incomplete and conditions may change; a listed material or isolation does not establish current release, exposure, stability or operational consequence.

ANALYSIS REQUIRED

Operation and interval. Lookup, filtering and evidence-quality comparison; recommendation only to contact the authorised owner/liaison. No hazard diagnosis or prioritisation beyond owner rules. On authorised command/liaison request and whenever a critical source changes. Minimum evidence and method. Confirmed incident/request, current signed records, precise location/scope, source owner, effective time, known uncertainty and permission to share. Create a source-attributed briefing list, separate confirmed, reported and unknown conditions, and preserve owner wording without unsupported reinterpretation. Uncertainty and safe failure. Unrecorded changes, material/location ambiguity, stale permits, measurement limitations and rapidly evolving incident conditions. Return 'hazard/isolation information not confirmed' and identify the authoritative owner; do not invent or omit the uncertainty.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire service/incident commander determines information need; the Responsible Person/fire-safety liaison controls transmission; hazard/process/Estates/system owners own current records; competent specialists own interpretation. Fire-service command and current authorised incident briefing first; then signed hazard/isolation/impairment records and competent observations; telemetry is supporting only. Times, access and provenance. Strict need-to-know incident access; sensitive materials, research, infrastructure and personal information are not broadly exposed. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Immediately stale after any owner/command update, system change or unverified lapse in the hazard/isolation record. Conflict. Conflicting hazard, location or isolation records are passed as a conflict with source and time, not resolved by the service. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not identify unknown materials, judge structural safety, issue tactical advice, alter isolations or bypass the authorised fire-service liaison. Do not expose sensitive material inventories, research content or infrastructure details beyond authorised incident need. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps the liaison pass concise, current and source-attributed information while making unresolved conflicts visible to command.

FS-077

FIRE-SAFETY PERSONNEL QUESTION

Which approved incident messages were sent, by which channel and at what time, and what delivery or acknowledgement evidence exists?

SENSORS AND TELEMETRY

Proposed evidence. Authorised messaging-system event logs, public-address/voice-alarm command logs, radio/control-room log entries and approved manual communication records. Message content is referenced by controlled version; no speech recording or content generation. Coverage and timing. Only channels and audiences in the current emergency plan and incident command scope; external recipient devices and human understanding are not directly observed. Native send/delivery/acknowledgement events with source times; preserve message version and channel, and report latest event age. Quality and limits. Retain system configuration, controlled message hash/version, source clocks/UTC offsets, logging resolution, uptime, gaps and provenance. Measure cross-channel offsets where possible; flag retries, duplicates, missing receipts and clock reversals. Do not infer delivery from send or understanding from acknowledgement. A sent or delivered event does not prove perception, comprehension, action or receipt by every intended person; radio/manual logs may be incomplete.

ANALYSIS REQUIRED

Operation and interval. Lookup and event-sequence comparison; no message drafting, comprehension inference or live recommendation. During incident only on authorised command request; post-event after evidence preservation. Minimum evidence and method. Confirmed incident, approved message/version, authorised sender/channel, complete event logs, clock metadata and known coverage. Link approve-send-deliver-acknowledge stages by channel, distinguish system receipt from human acknowledgement, and report gaps. Uncertainty and safe failure. Clock resolution, retries, external-device status, manual communications, channel outages and unavailable acknowledgements. Return 'message delivery/acknowledgement not demonstrated' for the missing stage and preserve the evidence trail.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The incident commander/Responsible Person controls approved messages; Communications and system owners own templates/channels/logs; authorised operators own manual entries. Fire-service/incident-command record first; then controlled system controlled-system access logs; operator notes and derived summaries last. Times, access and provenance. Message content, recipient identity, radio detail and operator data are incident-role restricted. Use aggregate delivery status where possible. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Live status stale after channel outage or newer command; post-event record remains provisional until logs are frozen. Conflict. Conflicting send times, versions or channels are retained; do not select a preferred narrative or infer unlogged messages. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not compose, send, repeat or modify incident messages, issue evacuation commands, infer understanding or contact recipients. Do not expose recipient identities, phone numbers, radio channels or message content beyond authorised roles. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This provides a defensible communication record and shows which channels or stages need follow-up without claiming that people understood or acted.

FS-078

FIRE-SAFETY PERSONNEL QUESTION

Which digital and physical records must be preserved for this event, and are provenance, access, integrity, custody and retention controls in place?

SENSORS AND TELEMETRY

Proposed evidence. Incident log, original system/event exports, configuration snapshots, authorised communications logs, photographs/observations, permits/impairments, maintenance records and physical evidence references. The service records hashes and custody metadata; it does not seize or alter evidence. Coverage and timing. Only records within the authorised incident/investigation preservation scope; personal, unrelated and legally restricted material is excluded or handled by the proper owner. Freeze/capture at authorised times; retain original observed/effective times plus collection time, chain-of-custody events and retention hold. Quality and limits. Retain source system/device, collector, clock/UTC offset, file hash, signature, format/version, access log, transformation and storage provenance. Calibration applies to any measurement evidence. Verify read-only copies and clock uncertainty; flag overwritten buffers, export gaps and post-capture changes. Never modify originals. A hash or custody record does not prove truth, completeness, admissibility, causal relevance or correct interpretation; volatile evidence may already be lost.

ANALYSIS REQUIRED

Operation and interval. Evidence inventory, integrity verification and provenance comparison; no determination of admissibility, cause or blame. Immediately after authorisation and throughout evidence handling. Minimum evidence and method. Confirmed preservation authority/scope, source inventory, original exports/physical references, clock metadata, hashes, custody records, permissions and retention instruction. Create an evidence manifest, verify integrity/custody, identify volatile sources and gaps, and separate original, working copy and derived product. Uncertainty and safe failure. Overwrite windows, proprietary exports, clock error, incomplete custody, access restrictions and missing originals. Return 'preservation controls incomplete' and identify the source owner or governance action needed; do not improvise retention.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The incident commander/investigation lead and information-governance/legal owners define preservation scope; system/record owners produce originals; fire-safety lead coordinates operational evidence. Fire-service/investigation/legal direction first; then original source evidence with custody; working copies and derived timelines last. Times, access and provenance. Strict role, purpose, retention and legal controls; personal emergency plans and unrelated personal data are excluded unless specifically lawfully required by the authorised investigation. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Preservation status changes with legal/investigation direction; volatile sources require immediate owner action. Conflict. Do not overwrite, merge or discard conflicting versions. Record gaps and custody breaks explicitly and refer to the investigation owner. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not alter, delete, seize, backdate or certify evidence, decide legal admissibility, assign blame or expose restricted records. Do not disclose personal or investigation-sensitive material beyond authorised scope. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This helps protect fragile records early and gives investigators a transparent manifest of what exists, what changed and what may be missing.

FS-079

FIRE-SAFETY PERSONNEL QUESTION

Which systems, routes or areas remain restricted, and what competent evidence is still required before the authorised owner considers handback?

SENSORS AND TELEMETRY

Proposed evidence. Current command/closure and impairment records, post-event inspections, system test results, structural/hazard findings, route/door states, remedial work and owner handback checklist. Telemetry is supporting evidence only. Coverage and timing. Only restrictions linked to the named event and current affected zones/dependencies; unaffected areas are not inferred from silence. Event-driven on command, inspection, test, work and owner decision; show current restriction, source and age. Quality and limits. Retain signed decision/version, test instrument calibration, configuration, source clocks/UTC offsets, uptime/gaps, file hashes and provenance. Validate that evidence post-dates the event/remedy and matches the exact area/system. Flag post-test changes and closure without owner acceptance. Do not infer handback from normal telemetry. A normal state, completed remedial work or passed component test does not establish structural safety, environmental safety, route condition, full system integration or permission to reoccupy.

ANALYSIS REQUIRED

Operation and interval. Evidence-completeness comparison against an owner-approved handback checklist; no handback recommendation or safety diagnosis. Continuously on change and at formal handback assessment. Minimum evidence and method. Current restriction, explicit handback criteria, competent post-event findings, complete remedial/test evidence, current system/route state and authorised owner decision. Map each restriction to criteria and evidence, validate timing/scope, and classify met, qualified, missing or conflicting. Uncertainty and safe failure. Hidden damage, evolving hazard, incomplete inspection, clock/configuration mismatch, post-test change and owner judgement. Return 'handback evidence incomplete' and retain the restriction in the service view until authorised decision.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The fire service/incident commander controls restrictions during the event; the Responsible Person/Estates owner controls handback after command release; competent structural, system, hazard and fire-safety professionals own findings. Current fire-service/owner restriction first; then competent findings and accepted tests; then system/local telemetry. Derived analysis cannot lift a restriction. Times, access and provenance. Restriction and vulnerability detail is role-restricted; public communication uses only approved owner/command wording. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Current until explicitly superseded; evidence stale after any relevant change or new finding. Conflict. Any conflict keeps the restriction unresolved until the authorised owner decides. Do not select a favourable result or auto-close. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not authorise re-entry, occupancy or handback, declare an area safe, remove barriers, reset systems or override fire-service/owner restrictions. Do not publish sensitive restrictions or investigation findings beyond authorised communication. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This gives the authorised owner a precise evidence checklist while preventing an area or system being reopened because a dashboard has returned to normal.

FS-080

FIRE-SAFETY PERSONNEL QUESTION

Which evidence-supported follow-up actions arise from the event without assigning blame or claiming a cause that has not been established?

SENSORS AND TELEMETRY

Proposed evidence. Authorised incident/debrief findings, preserved event evidence, competent inspection/test results, open defects/impairments, existing action register and owner decisions. No behavioural surveillance or automated blame analysis. Coverage and timing. Only the named event, validated findings and affected assets/procedures; hypotheses and unverified reports are labelled and kept separate. Post-event after command authorisation and initial evidence preservation; update when competent findings or owner decisions change. Quality and limits. Retain source signatures/versions, clocks/UTC offsets, hashes, access, evidence links and provenance. Calibration/health metadata follows any measurement/telemetry evidence. Flag unsupported narratives, missing evidence and actions created before findings. Do not infer causation from sequence, recurrence or correlation. A follow-up action may address a gap without proving cause, fault, responsibility, effectiveness or legal duty; some decisions require formal investigation or competent assessment.

ANALYSIS REQUIRED

Operation and interval. Evidence-to-action mapping and recommendation for owner consideration; no causal diagnosis, blame, risk certification or automatic action creation. At debrief, governance assessment and when new competent evidence is accepted. Minimum evidence and method. Confirmed event/scope, preserved evidence, accepted findings or explicit gaps, current defects/impairments, accountable owner and existing action criteria. Link each proposed action to a specific supported finding or evidence gap, distinguish contain/inspect/remedial work/test/assessment/learn, and state owner, evidence needed and uncertainty. Uncertainty and safe failure. Unfinished investigation, missing records, competing explanations, action effectiveness and changes after the event. Return 'no defensible action recommendation from current evidence' where linkage is weak; refer hypotheses to the authorised investigation.

AUTHORITATIVE SUPPORTING DATA

Owners and precedence. The Responsible Person/governance owner accepts actions; fire service/investigation owners control findings; system/asset owners and competent professionals define technical work; fire-safety lead coordinates. Authorised investigation/competent findings and current owner decisions first; then traceable evidence; derived suggestions last and clearly non-authoritative. Times, access and provenance. Role-based access; personal performance, disciplinary and sensitive incident information is excluded unless lawfully required by the authorised process. Retain effective, observed and retrieved times, source ID, version or signature, and the transformation used. Action basis stale when findings, scope or owner decision changes; regenerate with versioned evidence links. Conflict. Conflicting findings are retained and actions tied only to the agreed evidence gap, not to a disputed cause. Spatial claims require current owner-approved mapping; the May 2021 floor plan is historical reference only.

ANSWER BOUNDARY

Do not assign blame, identify a cause, judge individual conduct, create/close actions automatically, certify corrective effectiveness or issue operational commands. Do not expose personal or investigation-sensitive information beyond authorised governance. Do not disclose personal emergency evacuation plans, identities, individual movement or security-sensitive detail beyond authorised need. Do not certify fire safety, authorise work or impairments, operate or reset life-safety systems, issue emergency instructions, or override the Responsible Person, fire service or current emergency plan.

WHY THIS IS USEFUL

This supports proportionate, traceable follow-up focused on evidenced gaps while protecting the integrity of investigation and avoiding premature blame.

Operational decision map

The catalogue shortens the path from a question to the correct owner-led action. It never converts incomplete evidence into a safety verdict.

01

Compartmentation, doors, routes and assets

Prioritise attended checks, competent inspection and evidence close-out. Safe failure: physical condition or integrity is not demonstrated.

02

Alarms, detection, suppression and monitoring

Separate panel/controller state, endpoint evidence and competent test. Safe failure: system or coverage state is not established.

03

Routes, exits, refuges and assembly points

Identify local checks and missing dependencies without selecting an evacuation route. Safe failure: availability or suitability is not demonstrated.

04

Occupancy, visitors and inclusive information

Use aggregate ranges and authorised inclusion data only. Safe failure: occupancy, accountability or provision is not established.

05

Drills, training, wardens and communications

Measure exercise coverage and role evidence without scoring people. Safe failure: coverage, competence or communication is not demonstrated.

06

Inspection, testing, maintenance and records

Distinguish due, performed, evidenced and accepted work. Safe failure: completion, test scope or record suitability is not demonstrated.

07

Impairments, hot work, changes and controls

Expose overlaps, expiry and missing control evidence without approving work. Safe failure: authorised control or restoration evidence is incomplete.

08

Incidents, liaison and post-event evidence

Anchor information to command, preserve provenance and separate system reports from physical outcomes. Safe failure: sequence, outcome or handback is not established.

COMMON DECISION RULE

Use the narrowest claim the evidence supports. State observation, estimate, forecast and comparison separately. When a critical source is stale, unmapped, conflicting or unauthorised, stop and require the named competent or owner-led check.

Permissions and answer rules

01 Verify deployment

Treat every sensor, integration and analytic as proposed until mapping, commissioning, permission and current health are independently verified.

03 Preserve time

Keep effective, observed and retrieved times, timezone and source-clock offset; never invent event order within clock uncertainty.

05 Preserve provenance

Keep source ID, version, signature or hash, transformations and access decisions so every claim can be reproduced.

07 Expose conflict

Show incompatible sources, owners and times. Do not average, vote, merge or choose the reassuring record.

09 Separate operation types

Label lookup, observation, calculation, estimate, forecast, comparison, diagnosis and recommendation; do not present one as another.

02 State coverage

Name the asset, zone, side, height or path observed; show unmonitored areas and never generalise from sparse points.

04 Control measurement quality

Retain calibration or functional checks, configuration, uptime, gaps, maintenance periods and drift tests; do not fill critical gaps.

06 Respect authority

Fire-service command, the current emergency plan, signed Responsible Person decisions and competent findings outrank telemetry and derived analysis.

08 Minimise personal data

Use aggregate, purpose-limited evidence. No tracking, identity inference, personal-plan generation, speech recording or movement profiling.

10 Fail safely

Return unknown, qualified or not demonstrated, state the missing evidence and name the responsible local or owner-led check.

INCIDENT BOUNDARY

During an incident, follow the current emergency plan and authorised command. Talking Abacws may provide source-attributed evidence only when requested and permitted; it must never issue evacuation or re-entry instructions, contact emergency services, or operate any fire-safety system.